

Deformation Theory of Lie Algebraic Structures

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Preface

This thesis develops the deformation theory of three closely related algebraic structures: Lie algebras, Lie algebra homomorphisms, and Lie subalgebras, within a single, unified framework. Its aim is to give a detailed and self-contained explanation of how each of these objects behaves under small perturbations, and to make explicit the differential graded Lie algebra theory behind the scenes.

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Abstract

A classical question in algebra asks how a given structure behaves under small perturbations: if we deform it slightly, do we obtain a genuinely new structure, or merely one isomorphic to the original? This thesis studies that question for three objects: Lie algebras, Lie algebra homomorphisms and Lie subalgebras, following the unified viewpoint of Crainic, Schätz and Struchiner, and building on the foundational work of Nijenhuis and Richardson.

For each object we define what it means to deform it and when two deformations should be regarded as equivalent. We show that the infinitesimal deformations of a given structure are classified by a first cohomology group, that equivalent deformations lead to the same cohomology class, and that the obstructions to integrating an infinitesimal deformation to a genuine one are detected by an associated Kuranishi map. These results are then applied to answer the central questions of rigidity, local smoothness of the space of structures, and stability.

A unifying theme runs through the three cases: each deformation problem is governed by a differential graded Lie algebra, in the sense that the objects of interest arise as its Maurer-Cartan elements and the cohomology controlling deformations is the cohomology of this controlling algebra. We make this structure explicit and show that the three concrete deformation theories are instances of a single abstract one.

For Lie subalgebras our treatment differs from the existing literature: we work under the additional assumption that the subalgebra admits a complement that is itself a Lie subalgebra. Under this hypothesis the deformation problem is again governed by a genuine differential graded Lie algebra, which allows us to develop the theory in complete analogy with the other two cases and to avoid the L_∞ -machinery required in the general setting. The resulting statements follow from the general theory, but are obtained here by elementary and self-contained means.

List of Symbols

$\mathbb{N}$	Nonnegative integers
V or W	(Finite dimensional $\mathbb{R}$ -)vector space
$\wedge^k V^* \otimes W$	Set of k -linear skew symmetric maps from V to W
μ	Lie bracket
$(\mathfrak{g}, \mu)$	Lie algebra
$\mathcal{J}$	Jacobiator
ρ	Lie algebra homomorphism
μ_t, ρ_t or $\mathfrak{h}_t$	Deformation of respectively μ, ρ or $\mathfrak{h}$
M	Differentiable manifold
E or F	Vector bundle
G	Lie group
d or δ	Differential
$\llbracket \cdot, \cdot \rrbracket$	Graded Lie bracket
$(C^\bullet, \llbracket \cdot, \cdot \rrbracket)$	Graded Lie algebra
$(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$	Differential graded Lie algebra (DGLA)
d^x	Differential d twisted by Maurer-Cartan element x
$\text{Gr}_k(V)$	set of k -dimensional subspaces of V
$\text{Tr}_k(V, W)$	set of k -dimensional subspaces of V , transverse to W

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Chapter 1

Introduction

A Lie algebra is a vector space endowed with a Lie bracket, which is a bilinear skew-symmetric map that satisfies the Jacobi identity. Lie algebras arise naturally in areas of mathematics and physics, where they encode infinitesimal symmetries of algebraic and geometric structures. A natural and classical question is how such a structure behaves under small perturbations. When studying a Lie algebra, one asks: If we deform the Lie bracket slightly, do we obtain a genuinely new Lie algebra or merely one isomorphic to the original? Consider, for instance, the Lie algebra of infinitesimal rotations of a three-dimensional object, equipped with the commutator bracket. This Lie algebra does not deform: every nearby Lie algebra is isomorphic to it. This stands in contrast to the two-dimensional abelian Lie algebra, for which holds that any nearby Lie algebra is non-isomorphic to it. The same questions can be posed for Lie algebra homomorphisms and for Lie subalgebras, and their systematic study forms the deformation theory of these objects.

Deformation theory has been studied for roughly 60 years, and in particular the deformation theory of Lie algebras, Lie algebra homomorphisms and Lie subalgebras was first developed by Nijenhuis and Richardson in [NR67b], [NR67a],[NR66] and [Ric69] among other papers, after Gerstenhaber developed the deformation theory of associative algebras in [Ger64]. Here, deformed structures were represented by the set of solutions of a ‘deformation equation’ in graded Lie algebras. These early observations have led to the formalism, generally attributed to Deligne [Del86], that these deformation problems are sometimes governed by a differential graded Lie algebra (DGLA). Here, the objects of interest are the Maurer-Cartan elements of the controlling DGLA.

The main goal of this thesis is to develop this deformation theory of Lie algebras and Lie algebra homomorphisms in detail according to the viewpoint of Crainic, Schätz and Struchiner in [CSS14], and to establish several fundamental results. For the deformation theory of Lie subalgebras, our approach differs from the one in [CSS14]: we will follow the same structure but make the extra assumption that the given Lie subalgebra has a complement that is also a Lie subalgebra.

A classical notion in deformation theory is rigidity. Let X be an element of the

set A of algebraic structures under study. In our case A will consist of Lie algebras, Lie algebra homomorphisms or Lie subalgebras. An object X is called rigid, when every $Y \in A$ 'close to X ' is isomorphic to X . One can also impose the extra condition that these isomorphisms have to depend smoothly on Y .

Problem 1 (Rigidity). a) (Rigidity of Lie algebras). Given a Lie algebra $(\mathfrak{g}, \mu)$:

- When is every Lie bracket μ' on $\mathfrak{g}$ close to μ isomorphic to μ ?
- When can we find these isomorphisms depending smoothly on μ' ?

b) (Rigidity of Lie algebra homomorphisms). Given a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$:

- When is every Lie algebra homomorphism $\rho' : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ close to ρ of the form

$$\rho' = \text{Ad}_g \circ \rho$$

for a g in the simply connected Lie group G integrating $\mathfrak{g}$?

- When can we find these $g \in G$ depending smoothly on ρ' ?

c) (Rigidity of Lie subalgebras). Given a Lie algebra $(\mathfrak{g}, \mu)$ and a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ which has a complement $\mathfrak{h}^c \subset \mathfrak{g}$ that is also a Lie subalgebra:

- When is every Lie subalgebra $\mathfrak{h}'$ transverse to $\mathfrak{h}^c$ close to $\mathfrak{h}$ of the form

$$\mathfrak{h}' = \text{Ad}_l(\mathfrak{h})$$

for an element l in a connected Lie group K integrating $\mathfrak{h}^c$?

- When can we find $l \in K$ depending smoothly on $\mathfrak{h}'$?

Let X be an element of the set A of algebraic structures we consider. One can also wonder when A has a smooth structure around X .

Problem 2. a) Given a Lie algebra $(\mathfrak{g}, \mu)$, when is the space of Lie brackets on $\mathfrak{g}$ locally smooth around μ ?

b) Given a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, when is the space of Lie algebra homomorphisms $(\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ locally smooth around ρ ?

c) Given a Lie algebra $(\mathfrak{g}, \mu)$ and a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ which has a complement $\mathfrak{h}^c \subset \mathfrak{g}$ that is also a Lie subalgebra, when is the space of Lie subalgebras transverse to $\mathfrak{h}^c$ locally smooth?

Lastly, we have a problem regarding stability:

Problem 3 (Stability of Lie algebra homomorphisms). Given a Lie algebra $(\mathfrak{g}, \mu)$ and a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, when is true that for every Lie bracket μ' close to μ , there exists a Lie algebra homomorphism $\rho' : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu')$ close to ρ ?

We remark that the problem of stability can also be studied for Lie subalgebras. However, this would involve theory about L_∞ -algebras, which is beyond the scope of this thesis.

The problems concerning Lie algebras and Lie algebra homomorphisms: problems 1a, 1b, 2a, 2b and 3, correspond to the problems stated in chapter 1 of [CSS14], and were already solved by Nijenhuis and Richardson in the 1960s. In 2014, Crainic, Schätz and Struchiner gave in [CSS14] simple and unified proofs of these rigidity and stability results for Lie algebras, Lie algebra homomorphisms and Lie subalgebras, without introducing graded Lie algebras or DGLAs, and additionally they formulated a condition under which a Lie algebra homomorphism is stable under *all* automorphisms of its codomain. This thesis works out their unified approach in full detail, supplying the proofs and intermediate results that the original sources often leave out, while also recognizing the DGLA theory behind the deformation theories. For Lie subalgebras we diverge from [CSS14]: we make the additional assumption that the Lie subalgebra $\mathfrak{h}$ admits a complement $\mathfrak{h}^c$ that is itself a Lie subalgebra.

This thesis is organized as follows. Chapter 2 introduces Lie algebras, Lie algebra homomorphisms and Lie subalgebras, together with the relevant Lie group actions and Chevalley–Eilenberg complexes, laying the groundwork needed to formulate and approach the problems above. Chapter 3 develops the formalism of differential graded Lie algebras and shows how each of the three objects arises as a Maurer–Cartan element of an appropriate DGLA. Chapter 4 turns to the deformations of these objects, and in section 4.4 identifies the controlling DGLA in each case. Chapter 5 establishes three abstract theorems, which chapter 6 then applies to the three objects of interest to answer the problems stated above.

This thesis is, for the most part, expository in nature. We develop the deformation theory of Lie algebras, Lie algebra homomorphisms, and Lie subalgebras in a single, uniform framework, following the viewpoint of Crainic, Schätz and Struchiner [CSS14] and drawing on the earlier work of Nijenhuis and Richardson. None of the results concerning Lie algebras and Lie algebra homomorphisms are new. Our aim there has been to give a detailed and self-contained account. Concretely, for the results in the following parts, we supply full proofs together with the intermediate results needed to follow them, which the original sources often only sketch or omit: in section 3.1 and subsections 3.2.1 and 3.2.2 based on [Tor22], [NR67b] and [NR67a], and in chapter 5 and sections 6.1 and 6.2, which are derived from [CSS14]. We regard this complete and unified presentation of the deformation theories of Lie algebras, Lie algebra homomorphisms and Lie subalgebras in terms of DGLAs, as the main contribution of the thesis.

Our more specific contributions concern Lie subalgebras. Throughout, we make the additional assumption that the Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ under consideration admits a complement $\mathfrak{h}^c \subset \mathfrak{g}$ that is itself a Lie subalgebra; the presentation of section 2.3 is organized around this assumption. Under this assumption, we develop the deformation theory of Lie subalgebras in complete analogy with the cases of Lie algebras and Lie algebra homomorphisms: in subsection 3.2.3 we derive the differential graded Lie algebra controlling the deformations of a Lie subalgebra transverse to a fixed Lie subalgebra, and in sections 4.3 and 6.4 we develop this deformation theory and obtain the corresponding rigidity and smoothness results. These results are not new. They are implied by the general deformation theory of Lie subalgebras. However, the extra

assumption keeps the problem governed by a genuine differential graded Lie algebra, so that our treatment is self-contained and avoids the L_∞ -machinery required in the general case. Hence, we give a new way of proving the stated results. Finally, in section 4.4 and section 6.3 we make explicit the differential graded Lie algebra underlying the constructions of the preceding sections, connecting the concrete computations to the abstract formalism of chapter 3. Although the results in 4.4 are classical, the presentation that connects the three deformation theories to the general DGLA theory, is our own. Section 6.3 is completely our own contribution. Here, we recognize that the theorems in chapter 5 can be applied to the DGLA deformation theory we built in 4.4.

Chapter 2

Lie Algebras and Cohomology

This chapter develops the foundational material needed throughout the thesis. We introduce the three central objects of study: Lie algebras, Lie algebra homomorphisms and Lie subalgebras, together with their basic properties and examples. For each of these objects, we wish to understand when two such structures should be considered equivalent. This leads naturally to the introduction of appropriate Lie group actions, whose orbits give the equivalence classes. The fundamental definitions are classical and also appear in [NR67b] and [NR67a] respectively, but we provide additional context and detail. For Lie subalgebras we take a different route than the original sources: rather than treating Lie subalgebras in general, we restrict to subalgebras transverse to a fixed Lie subalgebra, under the assumption that the subalgebra of interest admits a complement that is itself a Lie subalgebra.

The last part of the chapter introduces Lie algebra cohomology via the Chevalley-Eilenberg complex. To each of the three objects above, we associate a cochain complex which lays the foundation for the corresponding DGLA we will construct in chapter 3. Additionally, these complexes yield cohomology groups which will play a central role in the deformation theory developed in chapters 4 and 6: they govern infinitesimal deformations, obstructions to integrability, rigidity and stability. This part about Lie algebra cohomology follows chapter 2 in [CSS14].

2.1 Lie algebras

Definition 2.1.1. Let $\mathfrak{g}$ be a finite-dimensional vector space over $\mathbb{R}$. We say that $\mathfrak{g}$ is a **Lie algebra** if it is equipped with a **Lie bracket**, which is a map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ that satisfies the following conditions:

- Bilinearity: For all $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{R}$

$$[ax + by, z] = a[x, z] + b[y, z] \tag{2.1}$$

$$[x, ay + bz] = a[x, y] + b[x, z] \tag{2.2}$$

- Skew symmetry: $[x, y] = -[y, x]$ for all $x, y \in \mathfrak{g}$

- The Jacobi identity: For all $x, y, z \in \mathfrak{g}$

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \quad (2.3)$$

The first two conditions can be merged to say that the bracket is an element of the vector space $\text{Hom}(\wedge^2 \mathfrak{g}, \mathfrak{g}) = \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, which is exactly the vector space of bilinear skew-symmetric maps $\eta : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$. More generally, for any vector space V ,

$$\begin{aligned} \wedge^k \mathfrak{g}^* \otimes V &= \text{Hom}(\wedge^k \mathfrak{g}, V) \\ &= \{\text{skew-symmetric multilinear maps } c : \underbrace{\mathfrak{g} \times \cdots \times \mathfrak{g}}_{k \text{ times}} \rightarrow V\}. \end{aligned} \quad (2.4)$$

If we denote by $n = \dim(\mathfrak{g})$ and $m = \dim(V)$, then $\wedge^k \mathfrak{g}^* \otimes V$ is a vector space of dimension $\dim(\wedge^k \mathfrak{g}^* \otimes V) = \binom{n}{k} m$.

Remark 2.1.2. Instead of writing a Lie bracket as brackets $[\cdot, \cdot]$, we often use a letter, such as μ to denote it.

Remark 2.1.3. We sometimes use the notation ‘+ c.p.’ to indicate that we have to sum over all cyclic permutations, meaning

$$f(x, y, z) + \text{c.p.} := f(x, y, z) + f(y, z, x) + f(z, x, y) \quad (2.5)$$

Using this notation, we can simplify the Jacobi identity: For all $x, y, z \in \mathfrak{g}$:

$$[x, [y, z]] + \text{c.p.} = 0 \quad (2.6)$$

Example 2.1.4. Letting $[\cdot, \cdot] = 0$ turns any vector space $\mathfrak{g}$ into a Lie algebra. Such a Lie algebra is called an abelian Lie algebra.

Example 2.1.5. The vector space $\mathbb{R}^{n \times n}$ of square matrices with the commutator bracket $[X, Y] := XY - YX$ is a Lie algebra and is denoted by $\mathfrak{gl}(\mathbb{R}^n)$.

Example 2.1.6. More generally, we can define $\mathfrak{gl}(V)$ as the vector space of all linear endomorphisms of a vector space V equipped with the commutator bracket $[f, g](v) = f \circ g(v) - g \circ f(v)$ for $f, g \in \mathfrak{gl}(V)$ and $v \in V$.

Example 2.1.7. The n -by- n matrices that have zero trace together with the commutator bracket $[X, Y] := XY - YX$ form a Lie algebra, denoted by $\mathfrak{sl}(\mathbb{R}^n)$.

Example 2.1.8. We denote by $\mathfrak{so}(\mathbb{R}^n)$ the Lie algebra of skew-symmetric n -by- n matrices, endowed with the commutator bracket $[X, Y] := XY - YX$.

Example 2.1.9. The three-dimensional Lie algebra with basis x, y, z on which the bracket is defined by

$$[x, y] = z, \quad [y, z] = 0, \quad [z, x] = 0$$

is called the Heisenberg algebra $\mathfrak{h}_3(\mathbb{R})$. It can be viewed as matrix Lie algebra where

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and the bracket is just the commutator bracket.

An important class of Lie algebras when studying rigidity of Lie algebras is the class of semisimple Lie algebras.

Definition 2.1.10. A **simple Lie algebra** is a non-abelian Lie algebra that has no nonzero proper ideals. A Lie algebra that is a direct sum of simple Lie algebras is called a **semisimple Lie algebra**.

Example 2.1.11. For all $n \geq 2$, the Lie algebra $\mathfrak{sl}(\mathbb{R}^n)$ is semisimple.

Example 2.1.12. For $n \geq 3$, the Lie algebra $\mathfrak{so}(\mathbb{R}^n)$ is semisimple.

Definition 2.1.13. For Lie algebras $(\mathfrak{h}, [\cdot, \cdot]_{\mathfrak{h}})$ and $(\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}})$, a linear map $\phi : (\mathfrak{h}, [\cdot, \cdot]_{\mathfrak{h}}) \rightarrow (\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}})$ is called a **Lie algebra homomorphism** if it preserves the bracket:

$$\phi([u, v]_{\mathfrak{h}}) = [\phi(u), \phi(v)]_{\mathfrak{g}} \text{ for } u, v \in \mathfrak{h}. \quad (2.7)$$

Additionally, if ϕ is a bijection, then ϕ is a **Lie algebra isomorphism**.

Definition 2.1.14. Given a vector space V , a **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, where $\mathfrak{gl}(V)$ denotes the Lie algebra of endomorphisms of V , equipped with the commutator bracket $[f, g](v) = f \circ g(v) - g \circ f(v)$, as in example 2.1.6.

Example 2.1.15. For any Lie algebra $\mathfrak{g}$ and vector space V , the map $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V) : u \mapsto 0$ is a representation of $\mathfrak{g}$, and is called the trivial representation of $\mathfrak{g}$.

Example 2.1.16. If we have a Lie algebra $(\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}})$ and let $V = \mathfrak{g}$, we obtain a representation by letting $r(u) = [u, \cdot]_{\mathfrak{g}} : v \mapsto [u, v]_{\mathfrak{g}}$. Indeed, as $[\cdot, \cdot]_{\mathfrak{g}}$ is a Lie bracket, r is a linear map from $\mathfrak{g}$ to $\mathfrak{g}$. Furthermore, we have that

$$r([u, v]_{\mathfrak{g}})(w) = [[u, v]_{\mathfrak{g}}, w]_{\mathfrak{g}} = -[w, [u, v]_{\mathfrak{g}}]_{\mathfrak{g}}$$

which is, by the Jacobi identity, equal to

$$\begin{aligned} &= [u, [v, w]_{\mathfrak{g}}]_{\mathfrak{g}} + [v, [w, u]_{\mathfrak{g}}]_{\mathfrak{g}} = [u, [v, w]_{\mathfrak{g}}]_{\mathfrak{g}} - [v, [u, w]_{\mathfrak{g}}]_{\mathfrak{g}} \\ &= [u, r(v)(w)]_{\mathfrak{g}} - [v, r(u)(w)]_{\mathfrak{g}} = r(u)(r(v)(w)) - r(v)(r(u)(w)) \\ &= [r(u), r(v)]_{\mathfrak{gl}(\mathfrak{g})}(w) \end{aligned}$$

showing that r is a Lie algebra homomorphism. This representation is called the adjoint representation of $\mathfrak{g}$, and is denoted by $\text{ad}_{\mathfrak{g}}$.

Example 2.1.17. Let $(\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}})$ and $(\mathfrak{h}, [\cdot, \cdot]_{\mathfrak{h}})$ be Lie algebras together with a Lie algebra homomorphism $\rho : \mathfrak{h} \rightarrow \mathfrak{g}$. Then $r = \text{ad}_{\mathfrak{g}} \circ \rho$ defines a representation on $\mathfrak{g}$. Indeed, we have the composition

$$(\mathfrak{h}, [\cdot, \cdot]_{\mathfrak{h}}) \xrightarrow{\rho} (\mathfrak{g}, [\cdot, \cdot]_{\mathfrak{g}}) \xrightarrow{\text{ad}_{\mathfrak{g}}} (\mathfrak{gl}(\mathfrak{g}), [\cdot, \cdot]_{\mathfrak{gl}(\mathfrak{g})})$$

where ρ is a Lie algebra homomorphism and $\text{ad}_{\mathfrak{g}}$ as well, by example 2.1.16. As a composition of Lie algebra homomorphisms is again a Lie algebra homomorphism, we proved that $\text{ad}_{\mathfrak{g}} \circ \rho$ is a representation.

Definition 2.1.18. A **Lie group** G is a group that is also a manifold, where the multiplication

$$G \times G \rightarrow G : (g, h) \mapsto gh$$

and inversion

$$G \rightarrow G : g \mapsto g^{-1}$$

are smooth maps.

Example 2.1.19. For any finite dimensional vector space V , the group

$$\mathrm{GL}(V) := \{\text{invertible linear maps } A : V \rightarrow V\}$$

is a Lie group, with Lie algebra $\mathfrak{gl}(V)$ from example 2.1.6.

Example 2.1.20. The $n \times n$ real orthogonal matrices, meaning matrices for which $A^{-1} = A^T$, form a Lie group called the special orthogonal group. This group is denoted by $\mathrm{SO}(n)$ and it has Lie algebra $\mathfrak{so}(\mathbb{R}^n)$ from example 2.1.8.

Definition 2.1.21. A group action by a Lie group G on a manifold M is called a **Lie group action** if the group action

$$G \times M \rightarrow M : (g, p) \mapsto g \cdot p$$

is a smooth map.

If we have a vector space $\mathfrak{g}$, we can let the Lie group $\mathrm{GL}(\mathfrak{g})$ act on $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ in the following way: If we take $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ and define

$$(A \cdot \eta)(u, v) := A\eta(A^{-1}u, A^{-1}v) \quad (2.8)$$

for all $u, v \in \mathfrak{g}$, then $A \cdot \eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ too. As A is linear, and μ is bilinear and skew-symmetric, $A \cdot \mu$ is bilinear and skew-symmetric too. Indeed, by bilinearity of η follows for all $u, v, w \in \mathfrak{g}$ that

$$\begin{aligned} (A \cdot \eta)(au + bv, w) &= A\eta(A^{-1}(au + bv), A^{-1}w) \\ &= A\eta(aA^{-1}u, A^{-1}w) + A\eta(bA^{-1}v, A^{-1}w) \\ &= a(A \cdot \eta)(u, w) + b(A \cdot \eta)(v, w) \end{aligned}$$

which can be computed analogously for the second component. From the skew-symmetry of η follows

$$(A \cdot \eta)(u, v) = A\eta(A^{-1}u, A^{-1}v) = -A\eta(A^{-1}v, A^{-1}u)$$

This already proves that $\mathrm{GL}(\mathfrak{g})$ acts on $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$.

However, if η also satisfies the Jacobi identity, then $A \cdot \eta$ does so too: For $u, v, w \in \mathfrak{g}$, we get that

$$\begin{aligned} &(A \cdot \eta)(u, v), w) + \text{c.p.} \\ &= A\eta(A^{-1}A\eta(A^{-1}u, A^{-1}v), A^{-1}w) + \text{c.p.} \\ &= A\eta(\eta(A^{-1}u, A^{-1}v), A^{-1}w) + \text{c.p.} = 0 \end{aligned}$$

where we used that $\eta(\eta(u', v'), w') + \text{c.p.} = 0$ since η satisfies the Jacobi identity. This now proves that $\text{GL}(\mathfrak{g})$ also acts on the set of Lie brackets on $\mathfrak{g}$.

Additionally, we can prove that the action of $\text{GL}(\mathfrak{g})$ on $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is in fact a Lie group action. As $\text{GL}(\mathfrak{g})$ is a Lie group, the map $A \mapsto A^{-1}$ is smooth, so that

$$\text{GL}(\mathfrak{g}) \times (\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}) \rightarrow \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} : (A, \eta) \mapsto A \cdot \eta$$

is smooth, as a composition of smooth maps.

Lastly, we note that $\text{GL}(\mathfrak{g})$ defines a Lie group action on any $\wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$ for $k \in \mathbb{N}$. This can be proven in the same way as for the case $k = 2$.

Definition 2.1.22. Let G be a Lie group acting on a manifold M , then for $x \in M$, we call $G \cdot x := \{g \cdot x | g \in G\}$ the G -**orbit** through x . These are immersed submanifolds of M .

For a Lie algebra $(\mathfrak{g}, \mu)$, the $\text{GL}(\mathfrak{g})$ -orbit through μ consists of all Lie algebra structures on $\mathfrak{g}$ that are isomorphic to $(\mathfrak{g}, \mu)$:

Proposition 2.1.23. Fix a Lie algebra $(\mathfrak{g}, \mu)$. A bracket η lies in $\text{GL}(\mathfrak{g}) \cdot \mu$ if and only if there exists a Lie algebra isomorphism $\varphi : (\mathfrak{g}, \mu) \rightarrow (\mathfrak{g}, \eta)$.

Proof. $\Rightarrow$

Suppose η lies in the orbit $\text{GL}(\mathfrak{g}) \cdot \mu$, then we can write $\eta = A \cdot \mu$ for some $A \in \text{GL}(\mathfrak{g})$. Let $\varphi : (\mathfrak{g}, \mu = [\cdot, \cdot]_1) \rightarrow (\mathfrak{g}, [\cdot, \cdot]_2 = (A \cdot \mu)) : x \mapsto Ax$, then

$$[\varphi(x), \varphi(y)]_2 = [Ax, Ay]_2 = A[A^{-1}Ax, A^{-1}Ay]_1 = A[x, y]_1 = \varphi([x, y]_1) \quad (2.9)$$

so that φ is a Lie algebra homomorphism. One can check that φ has inverse $(\mathfrak{g}, [\cdot, \cdot]_2 = (A \cdot \mu)) \rightarrow (\mathfrak{g}, \mu = [\cdot, \cdot]_1) : x \mapsto A^{-1}x$ which is also a Lie algebra homomorphism. From this we conclude that φ is a Lie algebra isomorphism.

$\Leftarrow$

Conversely, if $\varphi : (\mathfrak{g}, \mu = [\cdot, \cdot]_1) \rightarrow (\mathfrak{g}, [\cdot, \cdot]_2)$ is some Lie algebra isomorphism, then it's a linear isomorphism that preserves the bracket. The former states exactly that $\varphi(x) = Ax$ for some $A \in \text{GL}(\mathfrak{g})$. Take $u, v \in \mathfrak{g}$, and let $x = A^{-1}u, y = A^{-1}v$. Then clearly $u = \varphi(x)$ and $v = \varphi(y)$. As φ preserves the bracket, we have that

$$[\varphi(x), \varphi(y)]_2 = \varphi([x, y]_1)$$

or equivalently

$$[u, v]_2 = [Ax, Ay]_2 = A[x, y]_1$$

and thus

$$[u, v]_2 = A[A^{-1}u, A^{-1}v]_1 = (A \cdot \mu)(u, v)$$

so that the second bracket equals $[\cdot, \cdot]_2 = A \cdot \mu \in \text{GL}(\mathfrak{g}) \cdot \mu$. $\square$

Remark 2.1.24. For a Lie algebra $(\mathfrak{g}, \mu)$, denote by $\mathcal{J} : \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ the **Jacobiator** defined by

$$\mathcal{J}(\eta)(u, v, w) = \eta(\eta(u, v), w) + \text{c.p.} \quad (2.10)$$

for $u, v, w \in \mathfrak{g}$. Here, it's important to notice that an element $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ satisfies the Jacobi identity if and only if $\mathcal{J}(\eta) = 0$. Therefore, it's clear that the set of Lie brackets on $\mathfrak{g}$ is exactly $\mathcal{J}^{-1}(0)$.

Apart from analyzing this set of Lie brackets on $\mathfrak{g}$, we will also study when these Lie brackets are isomorphic. Hence, it makes sense to consider $\frac{\mathcal{J}^{-1}(0)}{\text{GL}(\mathfrak{g})}$, where isomorphic Lie brackets are identified:

Definition 2.1.25. For a Lie algebra $(\mathfrak{g}, \mu)$, we call $\frac{\mathcal{J}^{-1}(0)}{\text{GL}(\mathfrak{g})}$ the **moduli space** of $(\mathfrak{g}, \mu)$.

Example 2.1.26. Consider the case where $\mathfrak{g} = \mathbb{R}^2$. If we take a basis e_1, e_2 on $\mathbb{R}^2$, then we claim that the Lie brackets on $\mathbb{R}^2$ are exactly $[\cdot, \cdot]_v$, generated by the relation

$$[e_1, e_2]_v = v \text{ for } v \in \mathbb{R}^2.$$

Notice that elements in $\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ take three inputs, and give 0 if these inputs are linearly dependent. However, in $\mathbb{R}^2$, any three vectors are linearly dependent. It follows that $\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g} = \{0\}$, so that the Jacobiator

$$\mathcal{J} : \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g} \quad (2.11)$$

defined by

$$\mathcal{J}(\eta)(u, v, w) = \eta(\eta(u, v), w) + \text{c.p.} \quad (2.12)$$

is the zero map. This means exactly that every $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ trivially satisfies the Jacobi identity. As a consequence, any element in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is a Lie bracket on $\mathfrak{g} = \mathbb{R}^2$. Hence, the space of Lie brackets is isomorphic to $\mathbb{R}^2$:

$$\mathcal{J}^{-1}(0) \cong \mathbb{R}^2.$$

However, one can prove that there are exactly two isomorphism classes: the class of the zero Lie bracket $[\cdot, \cdot] = 0$, and the class of any other Lie bracket in $\mathcal{J}^{-1}(0)$,

$$[e_1, e_2]_v = v$$

for some $v \in \mathbb{R}^2 \setminus \{0\}$. Hence, the moduli space consists of two elements:

$$\frac{\mathcal{J}^{-1}(0)}{\text{GL}(\mathfrak{g})} \cong \{[0], [v \neq 0]\}.$$

2.2 Lie algebra homomorphisms

Just as for a vector space $\mathfrak{g}$ we could identify Lie brackets on $\mathfrak{g}$ whenever they were related by an element in the Lie group $\text{GL}(\mathfrak{g})$, we will now try to identify homomorphisms between two Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$ whenever they are related by an element in some suitable Lie group. To that end, observe the following remark:

Remark 2.2.1. Consider the Lie algebra $(\mathfrak{g}, \mu)$. By the Lie correspondence between Lie algebras and simply connected Lie groups, we can consider the unique simply connected Lie group G integrating $\mathfrak{g}$, meaning that

$$T_e G = \mathfrak{g}.$$

This Lie group G defines an action on itself by conjugation:

$$G \times G \rightarrow G : (g, h) \mapsto ghg^{-1}.$$

Then for any fixed $g \in G$, we can write down the Lie group homomorphism

$$C_g : G \rightarrow G : h \mapsto ghg^{-1}.$$

for which the derivative at e ,

$$d_e C_g : \mathfrak{g} \rightarrow \mathfrak{g}$$

is an element of $\mathrm{GL}(\mathfrak{g})$. This implies that we can define

$$\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g}) : g \mapsto \mathrm{Ad}_g := d_e C_g$$

which is called the adjoint representation of G on $\mathfrak{g}$, and is by the Lie correspondence a Lie algebra homomorphism. Hence, we have that

$$\mathrm{Ad} : G \rightarrow \mathrm{Aut}(\mathfrak{g}, \mu)$$

where $\mathrm{Aut}(\mathfrak{g}, \mu) := \{A \in \mathrm{GL}(\mathfrak{g}) \mid A \cdot \mu = \mu\} \subseteq \mathrm{GL}(\mathfrak{g})$ is the group of **automorphisms** of $(\mathfrak{g}, \mu)$, so that for all $g \in G$ and $u, v \in \mathfrak{g}$ holds that

$$\mathrm{Ad}_g \mu(u, v) = \mu(\mathrm{Ad}_g u, \mathrm{Ad}_g v).$$

Automorphisms that are in the image of Ad are called **inner automorphisms**. Moreover, we note that

$$d_e \mathrm{Ad} = \mathrm{ad}_{\mathfrak{g}} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}) : v \mapsto \mu(v, \cdot),$$

where $\mathrm{ad}_{\mathfrak{g}}$ is the adjoint representation as in example 2.1.16.

We now know that for Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, the simply connected Lie group G integrating $\mathfrak{g}$ defines an action on $\mathfrak{g}$ by adjoint action

$$G \times \mathfrak{g} \rightarrow \mathfrak{g} : (g, u) \mapsto \mathrm{Ad}_g u$$

We can extend this action to an action by G on all of $\wedge^{\bullet} \mathfrak{h}^* \otimes \mathfrak{g}$ by postcomposition:

$$\begin{aligned} G \times (\wedge^k \mathfrak{h}^* \otimes \mathfrak{g}) &\rightarrow \wedge^k \mathfrak{h}^* \otimes \mathfrak{g} : \\ (g, \eta) &\mapsto \mathrm{Ad}_g \circ \eta \end{aligned}$$

where

$$\mathrm{Ad}_g \circ \eta(u_1, \dots, u_k) = \mathrm{Ad}_g(\eta(u_1, \dots, u_k))$$

for all $g \in G$, $\eta \in \wedge^k \mathfrak{h}^* \otimes \mathfrak{g}$ and $u_1, \dots, u_k \in \mathfrak{h}$. Moreover, if we have a Lie algebra homomorphism $\eta : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, then $\mathrm{Ad}_g \circ \eta$ is again a Lie algebra homomorphism as a composition of Lie algebra homomorphisms

$$(\mathfrak{h}, \nu) \xrightarrow{\eta} (\mathfrak{g}, \mu) \xrightarrow{\mathrm{Ad}_g} (\mathfrak{g}, \mu)$$

so that the action restricts to the set of Lie algebra homomorphisms.

Remark 2.2.2. For a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, define the map $\mathcal{K} : \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ for $\eta \in \mathfrak{h}^* \otimes \mathfrak{g}$ and $u, v \in \mathfrak{h}$ as

$$\mathcal{K}(\eta)(u, v) = \mu(\eta(u), \eta(v)) - \eta(\nu(u, v)). \quad (2.13)$$

Then an element $\eta \in \mathfrak{h}^* \otimes \mathfrak{g}$ preserves the bracket if and only if $\mathcal{K}(\eta) = 0$. Therefore, one can see that the set of Lie algebra homomorphisms $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ is exactly $\mathcal{K}^{-1}(0)$.

Definition 2.2.3. For a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, and G the unique Lie group integrating $\mathfrak{g}$ acting on $\mathfrak{h}^* \otimes \mathfrak{g}$ by adjoint action, we call $\frac{\mathcal{K}^{-1}(0)}{G}$ the **moduli space** of ρ .

2.3 Lie subalgebras

This section diverges from our main source [CSS14], as we are interested in Lie subalgebras of a Lie algebra that are transverse to a fixed Lie subalgebra. Hence, this section is our own contribution and some notations may be new.

Definition 2.3.1. For a Lie algebra $(\mathfrak{g}, \mu)$, a vector subspace $\mathfrak{h} \subseteq \mathfrak{g}$ is called a **Lie subalgebra** of $(\mathfrak{g}, \mu)$ if $\mu(\mathfrak{h}, \mathfrak{h}) \subseteq \mathfrak{h}$. In particular, $(\mathfrak{h}, \mu|_{\mathfrak{h}})$ is a Lie algebra.

Example 2.3.2. For any $n \in \mathbb{N}$, the Lie algebra $\mathfrak{so}(n)$ from example 2.1.8 is a Lie subalgebra of the Lie algebra $\mathfrak{gl}(n, \mathbb{R})$ from example 2.1.5.

Example 2.3.3. The Heisenberg algebra $\mathfrak{h}_3(\mathbb{R})$ from example 2.1.9 is a Lie subalgebra of $\mathfrak{gl}(3, \mathbb{R})$ from example 2.1.5.

Definition 2.3.4. For a vector space V , denote by $\text{Gr}_k(V)$ the Grassmannian of k -dimensional subspaces of V :

$$\text{Gr}_k(V) := \{W \subseteq V \mid W \text{ is a } k\text{-dimensional vector subspace of } V\}$$

Remark 2.3.5. Let V be a vector space, W a k -dimensional vector subspace and let W^c be some complement of W , so that then

$$W \oplus W^c = V.$$

Then for any vector subspace $U \in \text{Gr}_k(V)$ which is transversal to W^c , meaning $W^c \oplus U = V$ and denoted $U \pitchfork W^c$, we can write U as the graph

$$U = \Gamma(\eta)$$

of some linear map $\eta : W \rightarrow W^c$. Conversely, the graph of any linear map $\eta : W \rightarrow W^c$ is a k -dimensional vector subspace of V that is transversal to W^c . Denote by $\text{Tr}_k(V, W^c) := \{U \in \text{Gr}_k(V) \mid U \pitchfork W^c\}$, then we have a bijection

$$\{\text{Linear maps } \eta : W \rightarrow W^c\} \longleftrightarrow \text{Tr}_k(V, W^c) \quad (2.14)$$

given by taking the graph $\eta \mapsto \Gamma(\eta)$. This notation $\text{Tr}_k(V, W^c)$ is our own, and thus is not in other literature.

Let $(\mathfrak{g}, \mu)$ be a Lie algebra and let G be the unique simply connected Lie group integrating $\mathfrak{g}$. The adjoint action of G on $\mathfrak{g}$ as in remark 2.2.1, induces for any $0 \leq k \leq \dim(\mathfrak{g})$, a Lie group action on the Grassmannian $\text{Gr}_k(\mathfrak{g})$: since $\text{Ad}_g \in \text{GL}(\mathfrak{g})$ for all $g \in G$, the map

$$G \times \text{Gr}_k(\mathfrak{g}) \rightarrow \text{Gr}_k(\mathfrak{g}) : (g, W) \mapsto \text{Ad}_g(W) := \{\text{Ad}_g w \mid w \in W\}$$

is a well-defined Lie group action. Let $\mathfrak{h} \subset \mathfrak{g}$ be a k -dimensional Lie subalgebra and suppose that $\mathfrak{h}$ has a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. As $\mathfrak{h}^c$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$ it forms a Lie algebra $(\mathfrak{h}^c, \mu|_{\mathfrak{h}^c})$. Therefore, by the Lie correspondence, we can take the unique connected Lie group K integrating $\mathfrak{h}^c$ which is a subgroup of G . By restricting the G -action on $\text{Gr}_k(\mathfrak{g})$ to K , the Lie group K defines a Lie group action on $\text{Gr}_k(\mathfrak{g})$ by adjoint action:

$$K \times \text{Gr}_k(\mathfrak{g}) \rightarrow \text{Gr}_k(\mathfrak{g}) : (l, W) \mapsto \text{Ad}_l(W).$$

Lemma 2.3.6. *The set $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c) := \{W \in \text{Gr}_k(\mathfrak{g}) \mid W \pitchfork \mathfrak{h}^c\}$ of k -dimensional subspaces of $\mathfrak{g}$ transversal to $\mathfrak{h}^c$ is preserved by the K -action. Hence K acts on $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$.*

Proof. Let $W \pitchfork \mathfrak{h}^c$ and $l \in K$. Since $\text{Ad}_l \in \text{Aut}(\mathfrak{h}^c, \mu|_{\mathfrak{h}^c})$, the adjoint action preserves $\mathfrak{h}^c$ as a set. Therefore

$$\text{Ad}_l(W) + \mathfrak{h}^c = \text{Ad}_l(W) + \text{Ad}_l(\mathfrak{h}^c) = \text{Ad}_l(W + \mathfrak{h}^c) = \text{Ad}_l(\mathfrak{g}) = \mathfrak{g},$$

so $\text{Ad}_l(W) \pitchfork \mathfrak{h}^c$. □

This defined action is a different one than the one defined in [CSS14], and thus is our own contribution. Similarly, the remainder of this section is our own.

By remark 2.3.5, the graph correspondence (2.14) gives a bijection

$$\mathfrak{h}^* \otimes \mathfrak{h}^c \longleftrightarrow \text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c) : \eta \mapsto \Gamma(\eta)$$

We use this bijection to transfer the K -action on $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$ to an action on $\mathfrak{h}^* \otimes \mathfrak{h}^c$: We define for all $l \in K$ and $\eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c$, the element $l \cdot \eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c$ is the unique linear map satisfying

$$\Gamma(l \cdot \eta) = \text{Ad}_l(\Gamma(\eta)).$$

From this follows that for all $h \in \mathfrak{h}$ holds that

$$h' + (l \cdot \eta)(h') = \text{Ad}_l(h + \eta(h)) \text{ for some } h' \in \mathfrak{h}$$

By taking the $\mathfrak{h}$ - and $\mathfrak{h}^c$ -component of this equation, this is equivalent to

$$\begin{cases} h' = p_{\mathfrak{h}}(\text{Ad}_l(h + \eta(h))) \\ (l \cdot \eta)(h') = p_{\mathfrak{h}^c}(\text{Ad}_l(h + \eta(h))) \end{cases}$$

where $p_{\mathfrak{h}} : \mathfrak{g} \rightarrow \mathfrak{h}$ and $p_{\mathfrak{h}^c} : \mathfrak{g} \rightarrow \mathfrak{h}^c$ denote the projections associated to the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^c$. After substituting the expression for h' into the second equation, we get the expression for $(l \cdot \eta)$

$$(l \cdot \eta)(p_{\mathfrak{h}}(\text{Ad}_l(h))) = p_{\mathfrak{h}^c}(\text{Ad}_l(h + \eta(h))). \quad (2.15)$$

Moreover, if we have that $W \in \text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$, then for any $l \in K$, it follows that $\text{Ad}_l(W)$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$ too, as $\text{Ad}_l \in \text{Aut}(\mathfrak{g}, \mu)$.

Definition 2.3.7. For a k -dimensional Lie subalgebra $\mathfrak{h}$ of a Lie algebra $(\mathfrak{g}, \mu)$ which admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra, define the map $\mathcal{L} : \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c$ for $\eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c$ and $u, v \in \mathfrak{h}$ as

$$\begin{aligned} \mathcal{L}(\eta)(u, v) = & p_{\mathfrak{h}^c}(\mu(u, \eta(v))) + p_{\mathfrak{h}^c}(\mu(\eta(u), v)) + p_{\mathfrak{h}^c}(\mu(\eta(u), \eta(v))) \\ & - \eta(\mu(u, v)) - \eta(p_{\mathfrak{h}}(\mu(u, \eta(v)))) - \eta(p_{\mathfrak{h}}(\mu(\eta(u), v))) \end{aligned} \quad (2.16)$$

The following lemma proves that the set of Lie subalgebras of $(\mathfrak{g}, \mu)$ transverse to $\mathfrak{h}^c$ is exactly $\Gamma(\mathcal{L}^{-1}(0))$.

Lemma 2.3.8. *If we have some $\eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c$, the graph $\Gamma(\eta)$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$ if and only if $\mathcal{L}(\eta) = 0$.*

Proof. We know that the graph $\Gamma(\eta)$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$ if and only if for all $u, v \in \mathfrak{h}$

$$\mu(u + \eta(u), v + \eta(v)) = w + \eta(w) \text{ for some } w \in \mathfrak{h}.$$

By taking the $\mathfrak{h}$ - and $\mathfrak{h}^c$ -component of

$$\mu(u + \eta(u), v + \eta(v)) = \mu(u, v) + \mu(\eta(u), v) + \mu(u, \eta(v)) + \mu(\eta(u), \eta(v)) = w + \eta(w)$$

we get the equations

$$\begin{aligned} \mu(u, v) + p_{\mathfrak{h}}(\mu(\eta(u), v)) + \mu(u, \eta(v)) &= w \\ p_{\mathfrak{h}^c}(\mu(\eta(u), v) + \mu(u, \eta(v))) + \mu(\eta(u), \eta(v)) &= \eta(w). \end{aligned}$$

After substituting the first equation into the second one, we get that

$$p_{\mathfrak{h}^c}(\mu(\eta(u), v) + \mu(u, \eta(v))) + \mu(\eta(u), \eta(v)) = \eta(\mu(u, v) + p_{\mathfrak{h}}(\mu(\eta(u), v) + \mu(u, \eta(v))))$$

and after rearranging the terms, and using linearity of the projections and η , we get

$$\begin{aligned} & p_{\mathfrak{h}^c}(\mu(u, \eta(v))) + p_{\mathfrak{h}^c}(\mu(\eta(u), v)) + p_{\mathfrak{h}^c}(\mu(\eta(u), \eta(v))) \\ & - \eta(\mu(u, v)) - \eta(p_{\mathfrak{h}}(\mu(u, \eta(v)))) - \eta(p_{\mathfrak{h}}(\mu(\eta(u), v))) = 0, \end{aligned} \quad (2.17)$$

but the expression on the left-hand side is exactly the definition for $\mathcal{L}(\eta)(u, v)$. Hence, we conclude that $\Gamma(\eta)$ is a Lie subalgebra if and only if $\mathcal{L}(\eta) = 0$. $\square$

Definition 2.3.9. Let $\mathfrak{h} \subset (\mathfrak{g}, \mu)$ be a k -dimensional Lie subalgebra with a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. Let K be the unique connected Lie group integrating $\mathfrak{h}^c$ that is a subgroup of G , the unique simply connected Lie group integrating $\mathfrak{g}$. Then the **moduli space** of $\mathfrak{h}$ is $\frac{\Gamma(\mathcal{L}^{-1}(0))}{K}$, where K acts on $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$ as in lemma 2.3.6.

2.4 Lie algebra Cohomology

Definition 2.4.1. A **differential complex of vector spaces** $(C^\bullet, d)$ consists of a sequence

$$\dots \rightarrow C^{m-1} \xrightarrow{d_{m-1}} C^m \xrightarrow{d_m} C^{m+1} \rightarrow \dots \quad (2.18)$$

where every C^i is a vector space and each linear map $d_n : C^m \rightarrow C^{m+1}$ satisfies $d_n \circ d_{n-1} = 0$.

The maps d_n are called the differentials of the complex. The elements of the group $Z^n(C^\bullet) := \ker(d_n)$ are called the **n -cocycles**, and the elements of the group $B^n(C^\bullet) := \operatorname{Im}(d_{n-1})$ are called the **n -coboundaries**. As all d_n are linear maps, every $Z^n(C^\bullet)$ and $B^n(C^\bullet)$ is a vector space. From the condition $d_n \circ d_{n-1} = 0$ follows that $B^n(C^\bullet) \subseteq Z^n(C^\bullet)$ for all $n \in \mathbb{Z}$, meaning that every n -coboundary is an n -cocycle. From this follows that the **n -th cohomology group**, defined by $H^n(C^\bullet) := Z^n(C^\bullet)/B^n(C^\bullet)$, is well defined. Notice that, as both $Z^n(C^\bullet)$ and $B^n(C^\bullet)$ are vector spaces, so is $H^n(C^\bullet)$.

Definition 2.4.2. Let $(\mathfrak{g}, \mu)$ be a Lie algebra. For any vector space V , we can define the **Chevalley-Eilenberg k -cochain** by $C^k(\mathfrak{g}, V) := \wedge^k \mathfrak{g}^* \otimes V$ for each $k \in \mathbb{N}$. If additionally, we have a representation $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, we can define the **Chevalley-Eilenberg differentials** $\delta_r : C^k(\mathfrak{g}, V) \rightarrow C^{k+1}(\mathfrak{g}, V)$ as

$$\begin{aligned} \delta_r \omega(u_0, \dots, u_k) &= \sum_{i=0}^k (-1)^i r(u_i) (\omega(u_0, \dots, \hat{u}_i, \dots, u_k)) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\mu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k) \end{aligned} \quad (2.19)$$

where the variables with hats are left out. We have to prove that δ_r is a differential, i.e., that it squares to 0. We postpone this until the end of chapter 3, where we only prove this in the case of $V = \mathfrak{g}$ with δ_r as in example 2.4.5. The differential in (2.19) leads to the differential complex

$$\dots \rightarrow C^{n-1}(\mathfrak{g}, V) \xrightarrow{\delta_r} C^n(\mathfrak{g}, V) \xrightarrow{\delta_r} C^{n+1}(\mathfrak{g}, V) \rightarrow \dots \quad (2.20)$$

which is called the **Chevalley-Eilenberg complex of $\mathfrak{g}$ with coefficients in V** , and is denoted by $(C^\bullet(\mathfrak{g}, V), \delta_r)$. For $n \in \mathbb{N}$, its cohomology group

$$H^n(\mathfrak{g}, V) = \frac{\ker(\delta_r : C^n(\mathfrak{g}, V) \rightarrow C^{n+1}(\mathfrak{g}, V))}{\operatorname{Im}(\delta_r : C^{n-1}(\mathfrak{g}, V) \rightarrow C^n(\mathfrak{g}, V))} \quad (2.21)$$

is said to be the **Lie algebra cohomology of $\mathfrak{g}$ with values in V** . Here, for $n = 0$, we take $\delta_r : C^{-1}(\mathfrak{g}, V) \rightarrow C^0(\mathfrak{g}, V)$ equal to the zero map.

Remark 2.4.3. The expression for δ_r makes sense even if $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is just a linear map, not necessarily a representation, and if μ is not a Lie bracket but merely an element in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$. In that case, δ_r is not a differential in general as the property $\delta_r^2 = 0$ will not necessarily be fulfilled.

Example 2.4.4. Let $(\mathfrak{g}, \mu)$ be a Lie algebra. If we take $V = \mathbb{R}$ with the trivial representation $r : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathbb{R}) : u \mapsto 0$, we recover the Chevalley-Eilenberg differential complex of $\mathfrak{g}$ with values in the trivial representation on $\mathbb{R}$. Its differential is denoted by $\delta_{\mathfrak{g}}$ and is defined for $\omega \in \wedge^k \mathfrak{g}^*$ and $u_0, \dots, u_k \in \mathfrak{g}$ as

$$\delta_{\mathfrak{g}} \omega(u_0, \dots, u_k) = \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\mu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k).$$

Example 2.4.5. Let $(\mathfrak{g}, \mu)$ be a Lie algebra. If we take $V = \mathfrak{g}$ with the adjoint representation $r = \operatorname{ad}_{\mathfrak{g}}$ from example 2.1.16, we obtain k -cochains $C^k(\mathfrak{g}, \mathfrak{g}) = \wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$ and

denote its Chevalley-Eilenberg differential by $\delta_\mu := \delta_{\text{ad}_\mathfrak{g}}$. Written down explicitly, this gives

$$\begin{aligned} \delta_\mu \omega(u_0, \dots, u_k) &= \sum_{i=0}^k (-1)^i \mu(u_i, \omega(u_0, \dots, \hat{u}_i, \dots, u_k)) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\mu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k) \end{aligned} \quad (2.22)$$

for $u_0, \dots, u_k \in \mathfrak{g}$ and $\omega \in \wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$. This yields the differential complex $(C^\bullet(\mathfrak{g}, \mathfrak{g}), \delta_\mu)$ and the cohomology groups $H^n(\mathfrak{g}, \mathfrak{g})$ which are the **Lie algebra cohomologies of $\mathfrak{g}$ with values in $\mathfrak{g}$** . These cohomologies will play a central role in the study of rigidity and local smoothness of Lie algebras, as we will explain in chapters 4 and 6.

Example 2.4.6. Let $(\mathfrak{g}, \mu)$ and $(\mathfrak{h}, \nu)$ be Lie algebras, and $\rho : \mathfrak{h} \rightarrow \mathfrak{g}$ a Lie algebra homomorphism. Then we can take the representation $r = \text{ad}_\mathfrak{g} \circ \rho : \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{g})$ from example 2.1.17 to obtain a differential on the k -cochains $\wedge^k \mathfrak{h}^* \otimes \mathfrak{g}$, and we denote it by $\delta_\rho := \delta_{\text{ad}_\mathfrak{g} \circ \rho}$. To get an expression for δ_ρ , we plug in r into the formula for the Chevalley-Eilenberg differential and we take $u_0, \dots, u_k \in \mathfrak{h}$ and $\omega \in \wedge^k \mathfrak{h}^* \otimes \mathfrak{g}$

$$\begin{aligned} \delta_\rho \omega(u_0, \dots, u_k) &= \sum_{i=0}^k (-1)^i \mu(\rho(u_i), \omega(u_0, \dots, \hat{u}_i, \dots, u_k)) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\nu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k) \end{aligned} \quad (2.23)$$

We obtain the differential complex $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_\rho)$ and cohomology groups $H^n(\mathfrak{h}, \mathfrak{g})$ called the **Lie algebra cohomologies of $\mathfrak{h}$ with values in $\mathfrak{g}$** , which will appear in the deformation theory of Lie algebra homomorphisms.

Example 2.4.7. Let $(\mathfrak{g}, \mu)$ be a Lie algebra. Suppose that there exists a Lie subalgebra $\mathfrak{h}$ that has a complement $\mathfrak{h}^c$ which is also a Lie subalgebra. Let $p_{\mathfrak{h}^c} : \mathfrak{g} \rightarrow \mathfrak{h}^c$ be the projection associated to the vector space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^c$. Then

$$r = p_{\mathfrak{h}^c} \circ \text{ad}_\mathfrak{g} : \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{h}^c)$$

given for $u \in \mathfrak{h}$ and $w \in \mathfrak{h}^c$

$$r(u)(w) = p_{\mathfrak{h}^c}(\mu(u, w))$$

defines a representation: For $u, v \in \mathfrak{h}$ and $w \in \mathfrak{h}^c$, we have that

$$\begin{aligned} [r(u), r(v)]_{\mathfrak{gl}(\mathfrak{h}^c)} &= r(u)(r(v)(w)) - r(v)(r(u)(w)) \\ &= p_{\mathfrak{h}^c}(\mu(u, p_{\mathfrak{h}^c}(\mu(v, w)))) - p_{\mathfrak{h}^c}(\mu(v, p_{\mathfrak{h}^c}(\mu(u, w)))) \end{aligned}$$

We can see that $p_{\mathfrak{h}^c}(\mu(u, p_{\mathfrak{h}^c}(\mu(v, w)))) = p_{\mathfrak{h}^c}(\mu(u, \mu(v, w)))$ because $p_{\mathfrak{h}^c}(\mu(u, p_{\mathfrak{h}}(\mu(v, w)))) = 0$ as $\mathfrak{h}$ is a Lie subalgebra. Analogously, we know that $p_{\mathfrak{h}^c}(\mu(v, p_{\mathfrak{h}^c}(\mu(u, w)))) = p_{\mathfrak{h}^c}(\mu(v, \mu(u, w)))$. Hence, by using the Jacobi identity and skew symmetry of μ , we get that

$$\begin{aligned} [r(u), r(v)]_{\mathfrak{gl}(\mathfrak{h}^c)} &= p_{\mathfrak{h}^c}(\mu(u, \mu(v, w))) - p_{\mathfrak{h}^c}(\mu(v, \mu(u, w))) \\ &= p_{\mathfrak{h}^c}(\mu(w, \mu(v, u))) \end{aligned}$$

$$\begin{aligned}
&= p_{\mathfrak{h}^c}(\mu(\mu(u, v), w)) \\
&= r(\mu(u, v))(w)
\end{aligned}$$

showing that r is a representation of $\mathfrak{h}$ on $\mathfrak{h}^c$. Hence, we can take δ_r with the representation $r = p_{\mathfrak{h}^c} \circ \text{ad}_{\mathfrak{g}} : \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{h}^c)$ on the k -cochains $\wedge^k \mathfrak{h}^* \otimes \mathfrak{h}^c$, and we denote it by $\delta_{\mathfrak{h}, \mathfrak{h}^c} := \delta_{p_{\mathfrak{h}^c} \circ \text{ad}_{\mathfrak{g}}}$. Explicitly, it's defined on $\omega \in \wedge^k \mathfrak{h}^* \otimes \mathfrak{h}^c$ and $u_0, \dots, u_k \in \mathfrak{h}$ as

$$\begin{aligned}
\delta_{\mathfrak{h}, \mathfrak{h}^c} \omega(u_0, \dots, u_k) &= \sum_{i=0}^k (-1)^i p_{\mathfrak{h}^c}(\mu(u_i, \omega(u_0, \dots, \hat{u}_i, \dots, u_k))) \\
&\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\mu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k)
\end{aligned} \tag{2.24}$$

which defines a differential complex that leads to **Lie algebra cohomologies of $\mathfrak{h}$ with values in $\mathfrak{h}^c$** .

Lemma 2.4.8 (Whitehead's lemma). *Let $(\mathfrak{g}, \mu)$ be a semisimple Lie algebra. Then holds that*

$$H^1(\mathfrak{g}, V) = 0 \text{ and } H^2(\mathfrak{g}, V) = 0$$

for any representation $r : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$.

Chapter 3

Differential Graded Lie Algebras

In this chapter, we develop the framework of differential graded Lie algebras (DGLAs), which provides a unified language to tackle deformation problems. We first develop the deformation theory of Lie algebras and Lie algebra homomorphisms according to the framework of Nijenhuis and Richardson in [NR67b] and [NR67a], where in section 3.1 we construct a suitable graded Lie algebra for these two deformation problems. In the deformation theory of Lie algebras, we start with the Chevalley-Eilenberg cochains $C^\bullet(V, V)$ for a vector space V , and for the deformation theory of Lie algebra homomorphisms with $C^\bullet(\mathfrak{h}, \mathfrak{g})$ for Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$.

After we have found the proper graded Lie algebras in section 3.1, we will further endow this with a compatible differential, turning this structure into a differential graded Lie algebra (DGLA). We follow [Tor22] for the case of Lie algebras and develop this in more detail, and try to work analogously for the case of Lie algebra homomorphisms. For the deformation theory of Lie subalgebras, we rely on [EJ25] to find a proper DGLA. Within this framework of DGLAs, the objects of interest will turn out to be the solutions of the Maurer-Cartan equation. Not only does this approach of using DGLAs streamline many computations, it is also the most universal perspective.

3.1 Graded Lie algebras

In this section, for a vector space V we will endow the Chevalley-Eilenberg cochains $C^\bullet(V, V)$ from example 2.4.5 with some extra structure. Notice that we do not need the Lie algebra structure that is present in example 2.4.5, only the vector space structure. The vector space $C^\bullet(V, V)$ is the direct sum of smaller vector spaces $C^k(V, V)$, which we all assign a degree k , naturally giving us a graded vector space. We will then equip it with a natural graded Lie bracket $\llbracket \cdot, \cdot \rrbracket$, the Gerstenhaber bracket, delivering us a graded Lie algebra. Similarly, for Lie algebras $(\mathfrak{g}, \mu)$ and $(\mathfrak{h}, \nu)$, the cochains $C^\bullet(\mathfrak{h}, \mathfrak{g})$ from example 2.4.6 will give rise to graded Lie algebra.

Definition 3.1.1. A **graded vector space** $V^\bullet$ is a vector space that is a direct sum of vector subspaces, meaning that $V^\bullet = \bigoplus_{i \in \mathbb{Z}} V^i$. The elements in V^i are said to have degree i .

Example 3.1.2. For any two vector spaces V and W , the Chevalley-Eilenberg complex $C^\bullet(V, W)$ of V with values in W , where $C^n(V, W) = \wedge^n V^* \otimes W$, forms naturally a graded vector space $\bigoplus_{n \in \mathbb{N}} C^n(V, W)$. The Chevalley-Eilenberg differential δ_r is of degree 1, in the sense that an element of degree n gets mapped to an element of degree $n + 1$.

Definition 3.1.3. A **graded Lie algebra** $(C^\bullet, \llbracket \cdot, \cdot \rrbracket)$ consists of a graded vector space $C^\bullet$, and a **graded Lie bracket** $\llbracket \cdot, \cdot \rrbracket : C^\bullet \times C^\bullet \rightarrow C^\bullet$, which is an $\mathbb{R}$ -bilinear map that is

- of degree 0, in the sense that $\llbracket C^h, C^k \rrbracket \subseteq C^{h+k}$,
- graded skew-symmetric, meaning that

$$\llbracket u, v \rrbracket = -(-1)^{hk} \llbracket v, u \rrbracket \quad (3.1)$$

for all $u \in C^h, v \in C^k$,

- and that satisfies the graded Jacobi identity:

$$\llbracket \llbracket u, v \rrbracket, w \rrbracket + (-1)^{(l+k)h} \llbracket \llbracket v, w \rrbracket, u \rrbracket + (-1)^{(h+k)l} \llbracket \llbracket w, u \rrbracket, v \rrbracket = 0 \quad (3.2)$$

for all $u \in C^h, v \in C^k$ and $w \in C^l$.

Denote for a graded vector space $V^\bullet = \bigoplus_{n \in \mathbb{Z}} V^n$, the shifted graded vector space $V^\bullet[k] = \bigoplus_{n \in \mathbb{Z}} W^n$ where $W^n = V^{n+k}$. Hence, in this graded vector space $V^\bullet[k]$, the elements of V^{n+k} have degree n . Furthermore, let $\mathcal{S}_n$ be the symmetric group of order n , consisting of all possible permutations on a set of n elements. We denote the k, l -unshuffle permutations by $\mathcal{S}_{k,l} \subset \mathcal{S}_{k+l}$ which is the set of all $\sigma \in \mathcal{S}_{k+l}$ for which $\sigma(1) < \sigma(2) < \dots < \sigma(k)$ and $\sigma(k+1) < \sigma(k+2) < \dots < \sigma(k+l)$, i.e., that preserve the order of the first k elements and the order of the last l elements. Consider for any vector space V the graded vector space $C^\bullet(V, V)$ from example 3.1.2 with $V = W$. We can now construct the Gerstenhaber product on the shifted graded vector space $C^\bullet(V, V)[1]$, as we defined above:

3.1.1 Lie algebras

This section follows [NR67b] and [Tor22] but is worked out in greater detail. We provide proofs where the source [NR67b] often omits them.

Definition 3.1.4. The **Gerstenhaber product** $\bullet : C^\bullet(V, V)[1] \times C^\bullet(V, V)[1] \rightarrow C^\bullet(V, V)[1]$ is the bilinear map, defined as

$$\begin{aligned} (\varphi \bullet \psi)(u_1, \dots, u_{h+k+1}) \\ = \sum_{\sigma \in \mathcal{S}_{k+1, h}} \text{sgn}(\sigma) \varphi(\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)}) \end{aligned} \quad (3.3)$$

for all $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$, $h, k \in \mathbb{N}$ and $u_1, \dots, u_{h+k+1} \in V$.

Remark 3.1.5. Equation (3.3) can be equivalently written as

$$\begin{aligned} (\varphi \bullet \psi)(u_1, \dots, u_{h+k+1}) \\ = \frac{1}{(k+1)!h!} \sum_{\sigma \in \mathcal{S}_{k+h+1}} \text{sgn}(\sigma) \varphi(\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)}) \end{aligned} \quad (3.4)$$

where the permutations now range over the full symmetric group of order $k + h + 1$. This is a consequence of the inputs φ and ψ being skew symmetric.

For this product to be well defined, we need to check three properties:

- The Gerstenhaber product $\bullet(\cdot, \cdot)$ is bilinear
- The map $\varphi \bullet \psi$ is multilinear
- The map $\varphi \bullet \psi$ is skew-symmetric

Linearity in the first component is immediate. Take any $c \in \mathbb{R}$, from multilinearity of φ it follows that

$$\begin{aligned} (\varphi \bullet c\psi)(u_1, \dots, u_{h+k+1}) &= \sum_{\sigma \in \mathcal{S}_{k+1, h}} \text{sgn}(\sigma) \varphi(c\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)}) \\ &= c \sum_{\sigma \in \mathcal{S}_{k+1, h}} \text{sgn}(\sigma) \varphi(\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)}) \\ &= c(\varphi \bullet \psi)(u_1, \dots, u_{h+k+1}) \end{aligned}$$

so that the product is also linear in the second component.

For $\sigma \in \mathcal{S}_{k+1, h}$, if we denote

$$T_\sigma(u_1, \dots, u_{h+k+1}) := \varphi(\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)})$$

then for any $c \in \mathbb{R}$ and $1 \leq i \leq h + k + 1$, we get that if $\sigma(i) \leq k + 1$, then

$$\begin{aligned} T_\sigma(u_1, \dots, cu_i, \dots, u_{h+k+1}) &= \varphi(\psi(u_{\sigma(1)}, \dots, cu_{\sigma(i)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, u_{\sigma(k+h+1)}) \\ &= cT_\sigma(u_1, \dots, u_i, \dots, u_{h+k+1}) \end{aligned}$$

which follows from multilinearity of φ and ψ . Similarly, if $\sigma(i) \geq k + 2$, it follows from multilinearity of φ that

$$\begin{aligned} T_\sigma(u_1, \dots, cu_i, \dots, u_{h+k+1}) &= \varphi(\psi(u_{\sigma(1)}, \dots, u_{\sigma(k+1)}), u_{\sigma(k+2)}, \dots, cu_{\sigma(i)}, \dots, u_{\sigma(k+h+1)}) \\ &= cT_\sigma(u_1, \dots, u_i, \dots, u_{h+k+1}) \end{aligned}$$

which proves that the sum

$$(\varphi \bullet \psi)(u_1, \dots, u_{h+k+1}) = \sum_{\sigma \in \mathcal{S}_{k+1, h}} \text{sgn}(\sigma) T_\sigma(u_1, \dots, u_{h+k+1})$$

is multilinear.

To prove skew symmetry, we use the equivalent formula from remark 3.1.5. Let $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$ and $u_1, \dots, u_{h+k+1} \in V$. Fix some index $m \in \{1, \dots, h + k + 1\}$ and let $\tau \in \mathcal{S}_{h+k+1}$ be the transposition swapping m and $m + 1$. We want to show that

$$(\varphi \bullet \psi)(u_1, \dots, u_m, u_{m+1}, \dots, u_{h+k+1}) = -(\varphi \bullet \psi)(u_1, \dots, u_{m+1}, u_m, \dots, u_{h+k+1}).$$

Using the formula from Remark 3.1.5, we compute

$$\begin{aligned} & (\varphi \bullet \psi)(u_1, \dots, u_{m+1}, u_m, \dots, u_{h+k+1}) \\ &= \frac{1}{(k+1)!h!} \sum_{\sigma \in \mathcal{S}_{h+k+1}} \text{sgn}(\sigma) \varphi(\psi(u_{\sigma(\tau(1))}, \dots, u_{\sigma(\tau(k+1))}), u_{\sigma(\tau(k+2))}, \dots, u_{\sigma(\tau(h+k+1))}). \end{aligned} \quad (3.5)$$

Since τ is a transposition, $\text{sgn}(\sigma \circ \tau) = -\text{sgn}(\sigma)$. As σ ranges over all of $\mathcal{S}_{h+k+1}$, so does $\sigma \circ \tau$, so substituting $\sigma' = \sigma \circ \tau$ gives that equation (3.5) is equal to

$$\begin{aligned} &= \frac{1}{(k+1)!h!} \sum_{\sigma' \in \mathcal{S}_{h+k+1}} (-\text{sgn}(\sigma')) \varphi(\psi(u_{\sigma'(1)}, \dots, u_{\sigma'(k+1)}), u_{\sigma'(k+2)}, \dots, u_{\sigma'(h+k+1)}) \\ &= -(\varphi \bullet \psi)(u_1, \dots, u_m, u_{m+1}, \dots, u_{h+k+1}). \end{aligned}$$

As any transposition of the inputs gives a relative minus sign, it follows that $(\varphi \bullet \psi)$ is skew-symmetric.

Lemma 3.1.6. *The Gerstenhaber product satisfies*

$$(\varphi \bullet \psi) \bullet \chi - \varphi \bullet (\psi \bullet \chi) = (-1)^{kl}((\varphi \bullet \chi) \bullet \psi - \varphi \bullet (\chi \bullet \psi)) \quad (3.6)$$

for all $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$ and $\chi \in C^{l+1}(V, V)$

Proof. This can be proven by writing it out. This is done in the appendix of [NR67b] by Nijenhuis and Richardson on pages 104 and 105. $\square$

Remark 3.1.7. This is the property of $(C^\bullet(V, V)[1], \bullet)$ to be a graded pre-Lie algebra, for example in [Man22] in section 5.8 on page 152.

Definition 3.1.8. The **Gerstenhaber bracket** $\llbracket \cdot, \cdot \rrbracket : C^\bullet(V, V)[1] \times C^\bullet(V, V)[1] \rightarrow C^\bullet(V, V)[1]$ is defined as

$$\llbracket \varphi, \psi \rrbracket := \psi \bullet \varphi - (-1)^{hk} \varphi \bullet \psi \quad (3.7)$$

for all $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$ and $h, k \in \mathbb{N}$.

Proposition 3.1.9. *The Gerstenhaber bracket satisfies the following three properties:*

- *The Gerstenhaber bracket is $\mathbb{R}$ -bilinear.*
- *The Gerstenhaber bracket is graded skew-symmetric, meaning that for all $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$ and $h, k \in \mathbb{N}$ holds that*

$$\llbracket \varphi, \psi \rrbracket = -(-1)^{hk} \llbracket \psi, \varphi \rrbracket \quad (3.8)$$

- *The Gerstenhaber bracket satisfies the graded Jacobi identity, meaning that for all $\varphi \in C^{h+1}(V, V)$, $\psi \in C^{k+1}(V, V)$, $\chi \in C^{l+1}(V, V)$ and $h, k, l \in \mathbb{N}$, we have the equation*

$$\llbracket \llbracket \varphi, \psi \rrbracket, \chi \rrbracket + (-1)^{(l+k)h} \llbracket \llbracket \psi, \chi \rrbracket, \varphi \rrbracket + (-1)^{(h+k)l} \llbracket \llbracket \chi, \varphi \rrbracket, \psi \rrbracket = 0 \quad (3.9)$$

Proof. The first property follows from $\mathbb{R}$ -bilinearity of the Gerstenhaber product, and graded skew-symmetry follows immediately from the definition of the Gerstenhaber bracket.

To prove the third property, we can compute

$$\begin{aligned} [[[\varphi, \psi], \chi]] &= [\psi \bullet \varphi - (-1)^{hk} \varphi \bullet \psi, \chi] \\ &= \chi \bullet (\psi \bullet \varphi - (-1)^{hk} \varphi \bullet \psi) - (-1)^{(h+k)l} (\psi \bullet \varphi - (-1)^{hk} \varphi \bullet \psi) \bullet \chi \\ &= \chi \bullet (\psi \bullet \varphi) - (-1)^{hk} \chi \bullet (\varphi \bullet \psi) - (-1)^{(h+k)l} (\psi \bullet \varphi) \bullet \chi + (-1)^{hl+kl+hk} (\varphi \bullet \psi) \bullet \chi. \end{aligned}$$

Analogously, we can compute the other two double brackets. It follows that the sum

$$\begin{aligned} &[[[\varphi, \psi], \chi]] + (-1)^{(l+k)h} [[[\psi, \chi], \varphi]] + (-1)^{(h+k)l} [[[\chi, \varphi], \psi]] \\ &= \chi \bullet (\psi \bullet \varphi) - (-1)^{hk} \chi \bullet (\varphi \bullet \psi) - (-1)^{(h+k)l} (\psi \bullet \varphi) \bullet \chi + (-1)^{hl+kl+hk} (\varphi \bullet \psi) \bullet \chi \\ &\quad + (-1)^{(l+k)h} \varphi \bullet (\chi \bullet \psi) - (-1)^{hl+kl+hk} \varphi \bullet (\psi \bullet \chi) - (\chi \bullet \psi) \bullet \varphi + (-1)^{kl} (\psi \bullet \chi) \bullet \varphi \\ &\quad + (-1)^{(h+k)l} \psi \bullet (\varphi \bullet \chi) - (-1)^{kl} \psi \bullet (\chi \bullet \varphi) - (-1)^{(k+l)h} (\varphi \bullet \chi) \bullet \psi + (-1)^{hk} (\chi \bullet \varphi) \bullet \psi \\ &= \chi \bullet (\psi \bullet \varphi) - (\chi \bullet \psi) \bullet \varphi - (-1)^{hk} (\chi \bullet (\varphi \bullet \psi) - (\chi \bullet \varphi) \bullet \psi) \\ &\quad + (-1)^{(l+k)h} (\varphi \bullet (\chi \bullet \psi) - (\varphi \bullet \chi) \bullet \psi) - (-1)^{hk+kl+hl} (\varphi \bullet (\chi \bullet \psi) - (\varphi \bullet \psi) \bullet \chi) \\ &\quad + (-1)^{(h+k)l} (\psi \bullet (\varphi \bullet \chi) - (\psi \bullet \varphi) \bullet \chi) - (-1)^{kl} (\psi \bullet (\chi \bullet \varphi) - (\psi \bullet \chi) \bullet \varphi) \\ &= 0 \end{aligned}$$

because in every line we can use the identity in Lemma 3.1.6, which gives that every line is zero, and thus their sum is too. $\square$

Corollary 3.1.10. *For any vector space V , the pair $(C^\bullet(V, V)[1], [[\cdot, \cdot]])$, where $[[\cdot, \cdot]]$ is the Gerstenhaber bracket on $C^\bullet(V, V)[1]$, forms a graded Lie algebra.*

Lemma 3.1.11. *For any graded Lie algebra $(C^\bullet, [[\cdot, \cdot]])$, we can write the graded Jacobi identity (3.2) equivalently as*

$$[\varphi, [[\psi, \chi]]] = [[[\varphi, \psi], \chi]] + (-1)^{hk} [\psi, [[\varphi, \chi]]] \quad (3.10)$$

for all $\varphi \in C^h$, $\psi \in C^k$ and $\chi \in C^l$.

Proof. Rewrite the the graded Jacobi identity (3.2) as

$$-(-1)^{(l+k)h} [[[\psi, \chi], \varphi]] = [[[\varphi, \psi], \chi]] + (-1)^{(h+k)l} [[[\chi, \varphi], \psi]].$$

If we use graded skew symmetry of the outer graded Lie bracket on the term on the left-hand side, this becomes

$$[[\varphi, [[\psi, \chi]]] = [[[\varphi, \psi], \chi]] + (-1)^{(h+k)l} [[[\chi, \varphi], \psi]].$$

Using graded skew symmetry of the inner and outer bracket on the last term gives the desired equation

$$[[\varphi, [[\psi, \chi]]] = [[[\varphi, \psi], \chi]] + (-1)^{hk} [\psi, [[\varphi, \chi]]].$$

$\square$

Furthermore, the Gerstenhaber bracket proves to be a convenient tool in finding Lie algebra structures on V :

Theorem 3.1.12. ([Tor22], Lemma 6.2) *An element $\eta \in C^2(V, V)$ satisfies $[\![\eta, \eta]\!] = 0$ if and only if it satisfies the Jacobi identity, meaning that it defines a Lie algebra structure.*

One can also compare theorem 3.1.12 with equation (3) in [NR67b].

Proof. Recall the Jacobiator $\mathcal{J} : \wedge^2 V^* \otimes V \rightarrow \wedge^3 V^* \otimes V$ defined by

$$\mathcal{J}(\eta)(u, v, w) = \eta(\eta(u, v), w) + \text{c.p.} \quad (3.11)$$

and remember that an element $\eta \in \wedge^2 V^* \otimes V$ satisfies the Jacobi identity if and only if $\mathcal{J}(\eta) = 0$. We claim that $\frac{1}{2}[\![\eta, \eta]\!] = \mathcal{J}(\eta)$, then the assertion follows. Indeed, if we compute

$$\frac{1}{2}[\![\eta, \eta]\!] = \eta \bullet \eta$$

and apply this on $u_1, u_2, u_3 \in V$, we find that

$$(\eta \bullet \eta)(u_1, u_2, u_3) = \sum_{\sigma \in \mathcal{S}_{2,1}} \text{sgn}(\sigma) \eta(\eta(u_{\sigma(1)}, u_{\sigma(2)}), u_{\sigma(3)})$$

where the possible permutations in cyclic notation are Id , $(2\ 3)$ and $(1\ 2\ 3)$, where only $(2\ 3)$ has negative sign. If we expand this sum, we find that this is equal to

$$\begin{aligned} &= \eta(\eta(u_1, u_2), u_3) - \eta(\eta(u_1, u_3), u_2) + \eta(\eta(u_2, u_3), u_1) \\ &= \eta(\eta(u_1, u_2), u_3) + \eta(\eta(u_3, u_1), u_2) + \eta(\eta(u_2, u_3), u_1) \\ &= \mathcal{J}(\eta)(u_1, u_2, u_3). \end{aligned}$$

Now for any $\eta \in C^2(V, V)$, the vanishing of $[\![\eta, \eta]\!]$ is equivalent to the vanishing of $\mathcal{J}(\eta)$, which means exactly that η satisfies the Jacobi identity. $\square$

3.1.2 Lie algebra homomorphisms

As in the case of Lie algebras, we will construct a graded Lie algebra for any Lie algebra. We will try to follow the same structure as in the previous subsection. While the main results of this subsection also appear in [NR67a], we try to be more complete.

Definition 3.1.13. Let $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$ be Lie algebras. On $C^\bullet(\mathfrak{h}, \mathfrak{g})$ from example 2.4.6, we define the bracket $[\![\cdot, \cdot]\!] : C^\bullet(\mathfrak{h}, \mathfrak{g}) \times C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ as

$$[\![\varphi, \psi]\!](u_1, \dots, u_{h+k}) := \sum_{\sigma \in \mathcal{S}_{h,k}} \text{sgn}(\sigma) \mu(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h)}), \psi(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})) \quad (3.12)$$

for $u_1, \dots, u_{h+k} \in \mathfrak{h}$, $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$.

Remark 3.1.14. Equation (3.12) can be equivalently written as

$$[\![\varphi, \psi]\!](u_1, \dots, u_{h+k}) := \frac{1}{k!h!} \sum_{\sigma \in \mathcal{S}_{k+h}} \text{sgn}(\sigma) \mu(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h)}), \psi(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})) \quad (3.13)$$

where the permutations now range over the full symmetric group of order $k + h$.

The following lemma says that the bracket $\llbracket \cdot, \cdot \rrbracket$ from definition 3.1.13 is well defined and of degree 0, meaning $\llbracket C^h(\mathfrak{h}, \mathfrak{g}), C^k(\mathfrak{h}, \mathfrak{g}) \rrbracket \subseteq C^{h+k}(\mathfrak{h}, \mathfrak{g})$.

Lemma 3.1.15. *Let $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$ be Lie algebras. Then for any $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$, we have that $\llbracket \varphi, \psi \rrbracket \in C^{h+k}(\mathfrak{h}, \mathfrak{g})$, meaning that:*

- The map $\llbracket \varphi, \psi \rrbracket$ is $(h+k)$ -linear.
- The map $\llbracket \varphi, \psi \rrbracket$ is skew-symmetric

Proof. Linearity in the $h+k$ components of $\llbracket \varphi, \psi \rrbracket$ follows from h -linearity from φ , k -linearity of ψ , and bilinearity of μ . To prove skew-symmetry, We use the equivalent formula from Remark 3.1.14. Let $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$ and $u_1, \dots, u_{h+k} \in \mathfrak{h}$. Fix some index $m \in \{1, \dots, h+k\}$ and let $\tau \in \mathcal{S}_{h+k}$ be the transposition swapping m and $m+1$. We want to show that

$$\llbracket \varphi, \psi \rrbracket(u_1, \dots, u_m, u_{m+1}, \dots, u_{h+k}) = -\llbracket \varphi, \psi \rrbracket(u_1, \dots, u_{m+1}, u_m, \dots, u_{h+k}).$$

Using the formula from Remark 3.1.14, we compute

$$\begin{aligned} & \llbracket \varphi, \psi \rrbracket(u_1, \dots, u_{m+1}, u_m, \dots, u_{h+k}) \\ &= \frac{1}{h!k!} \sum_{\sigma \in \mathcal{S}_{h+k}} \text{sgn}(\sigma) \mu(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h)}), \psi(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})) \\ &= \frac{1}{h!k!} \sum_{\sigma \in \mathcal{S}_{h+k}} \text{sgn}(\sigma) \mu(\varphi(u_{(\sigma \circ \tau)(1)}, \dots, u_{(\sigma \circ \tau)(h)}), \psi(u_{(\sigma \circ \tau)(h+1)}, \dots, u_{(\sigma \circ \tau)(h+k)})). \end{aligned}$$

Since τ is a transposition, $\text{sgn}(\sigma \circ \tau) = -\text{sgn}(\sigma)$. As σ ranges over all of $\mathcal{S}_{h+k}$, so does $\sigma \circ \tau$, so substituting $\sigma' = \sigma \circ \tau$ gives

$$\begin{aligned} &= \frac{1}{h!k!} \sum_{\sigma' \in \mathcal{S}_{h+k}} (-\text{sgn}(\sigma')) \mu(\varphi(u_{\sigma'(1)}, \dots, u_{\sigma'(h)}), \psi(u_{\sigma'(h+1)}, \dots, u_{\sigma'(h+k)})) \\ &= -\llbracket \varphi, \psi \rrbracket(u_1, \dots, u_m, u_{m+1}, \dots, u_{h+k}). \end{aligned}$$

As any transposition of the inputs gives a relative minus sign, it follows that $\llbracket \varphi, \psi \rrbracket$ is skew-symmetric. $\square$

To prove that this bracket defines a graded Lie bracket, we first prove the following lemma.

Lemma 3.1.16. *For decomposable elements $\varphi := \alpha \otimes u \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi := \beta \otimes v \in C^k(\mathfrak{h}, \mathfrak{g})$, we have that*

$$\llbracket \varphi, \psi \rrbracket = (\alpha \wedge \beta) \otimes \mu(u, v) \quad (3.14)$$

where $\wedge$ denotes the wedge product on $\wedge^\bullet \mathfrak{h}^*$.

Proof. Let $\varphi = \alpha \otimes u$ and $\psi = \beta \otimes v$, where $\alpha \in \wedge^h \mathfrak{h}^*$, $\beta \in \wedge^k \mathfrak{h}^*$, and $u, v \in \mathfrak{h}$. By the definition, we have:

$$\begin{aligned} \varphi(u_1, \dots, u_h) &= \alpha(u_1, \dots, u_h)u \\ \psi(u_1, \dots, u_k) &= \beta(u_1, \dots, u_k)v \end{aligned}$$

If we evaluate the bracket $[\![\varphi, \psi]\!]$ on arbitrary $u_1, \dots, u_{h+k} \in \mathfrak{h}$, then

$$\begin{aligned} [\![\varphi, \psi]\!](u_1, \dots, u_{h+k}) &= \sum_{\sigma \in S_{h,k}} \operatorname{sgn}(\sigma) \mu(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h)}), \psi(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})) \\ &= \sum_{\sigma \in S_{h,k}} \operatorname{sgn}(\sigma) \mu(\alpha(u_{\sigma(1)}, \dots, u_{\sigma(h)})u, \beta(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})v) \\ &= \sum_{\sigma \in S_{h,k}} \operatorname{sgn}(\sigma) \alpha(u_{\sigma(1)}, \dots, u_{\sigma(h)}) \beta(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)}) \mu(u, v) \end{aligned}$$

where we used that $\alpha(u_{\sigma(1)}, \dots, u_{\sigma(h)})$ and $\beta(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)})$ are just elements in $\mathbb{R}$, and thus can be factored out of μ . As $\mu(u, v)$ does not depend on σ , it can be factored out of the entire summation:

$$= \left(\sum_{\sigma \in S_{h,k}} \operatorname{sgn}(\sigma) \alpha(u_{\sigma(1)}, \dots, u_{\sigma(h)}) \beta(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)}) \right) \mu(u, v)$$

The term inside the parentheses is exactly $(\alpha \wedge \beta)(u_1, \dots, u_{h+k})$. Therefore it follows that

$$[\![\varphi, \psi]\!](u_1, \dots, u_{h+k}) = (\alpha \wedge \beta)(u_1, \dots, u_{h+k}) \cdot \mu(u, v)$$

and thus

$$[\![\varphi, \psi]\!] = (\alpha \wedge \beta) \otimes \mu(u, v)$$

which proves the statement. $\square$

Proposition 3.1.17. *For Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, the bracket $[\![\cdot, \cdot]\!] : C^\bullet(\mathfrak{h}, \mathfrak{g}) \times C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ satisfies the following three properties:*

- *It is $\mathbb{R}$ -bilinear.*
- *It is graded skew-symmetric, meaning that for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$ and $h, k \in \mathbb{N}$ holds that*

$$[\![\varphi, \psi]\!] = -(-1)^{hk} [\![\psi, \varphi]\!]. \quad (3.15)$$

- *It satisfies the graded Jacobi identity, meaning that for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$, $\chi \in C^l(\mathfrak{h}, \mathfrak{g})$ and $h, k, l \in \mathbb{N}$, we have the equation*

$$[[[\varphi, \psi], \chi]] + (-1)^{(l+k)h} [[[\psi, \chi], \varphi]] + (-1)^{(h+k)l} [[[\chi, \varphi], \psi]] = 0. \quad (3.16)$$

proving that the pair $(C^\bullet(\mathfrak{h}, \mathfrak{g}), [\![\cdot, \cdot]\!])$ forms a graded Lie algebra.

Proof. From $\mathbb{R}$ -bilinearity of μ follows directly the $\mathbb{R}$ -bilinearity of $[\![\cdot, \cdot]\!]$.

If we take decomposable $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$, then we can write

$$\begin{aligned} \varphi &= \alpha \otimes u \text{ for some } \alpha \in \wedge^h \mathfrak{h}^* \text{ and } u \in \mathfrak{g} \\ \psi &= \beta \otimes v \text{ for some } \beta \in \wedge^k \mathfrak{h}^* \text{ and } v \in \mathfrak{g}. \end{aligned}$$

We find that

$$\begin{aligned} \llbracket \psi, \varphi \rrbracket &= \llbracket \beta \otimes v, \alpha \otimes u \rrbracket = (\beta \wedge \alpha) \otimes \mu(v, u) \\ &= -(\beta \wedge \alpha) \otimes \mu(u, v) = -(-1)^{hk}(\alpha \wedge \beta) \otimes \mu(u, v) \\ &= -(-1)^{hk} \llbracket \varphi, \psi \rrbracket \end{aligned}$$

where we first used skew symmetry of μ and then graded commutativity of the wedge product. This proves that the bracket is graded skew symmetric.

If we additionally take decomposable $\chi \in C^l(\mathfrak{h}, \mathfrak{g})$, then we can write

$$\chi = \gamma \otimes w \text{ for some } \gamma \in \wedge^l \mathfrak{h}^* \text{ and } w \in \mathfrak{g}.$$

Then we can compute that

$$\llbracket \llbracket \varphi, \psi \rrbracket, \chi \rrbracket = \llbracket \llbracket \alpha \otimes u, \beta \otimes v \rrbracket, \gamma \otimes w \rrbracket = (\alpha \wedge \beta \wedge \gamma) \otimes \mu(\mu(u, v), w).$$

Analogously, by using graded commutativity of the wedge product, the other terms are

$$\llbracket \llbracket \psi, \chi \rrbracket, \varphi \rrbracket = (\beta \wedge \gamma \wedge \alpha) \otimes \mu(\mu(v, w), u) = (-1)^{h(k+l)}(\alpha \wedge \beta \wedge \gamma) \otimes \mu(\mu(v, w), u)$$

and

$$\llbracket \llbracket \chi, \varphi \rrbracket, \psi \rrbracket = (\gamma \wedge \alpha \wedge \beta) \otimes \mu(\mu(w, u), v) = (-1)^{l(h+k)}(\alpha \wedge \beta \wedge \gamma) \otimes \mu(\mu(w, u), v).$$

Taking the graded sum of these three terms gives

$$\begin{aligned} &\llbracket \llbracket \varphi, \psi \rrbracket, \chi \rrbracket + (-1)^{(l+k)h} \llbracket \llbracket \psi, \chi \rrbracket, \varphi \rrbracket + (-1)^{(h+k)l} \llbracket \llbracket \chi, \varphi \rrbracket, \psi \rrbracket \\ &= (\alpha \wedge \beta \wedge \gamma) \otimes (\mu(\mu(u, v), w) + \mu(\mu(v, w), u) + \mu(\mu(w, u), v)) \\ &= 0 \end{aligned}$$

where we used that μ satisfies the Jacobi identity. As the decomposable elements span each $C^i(\mathfrak{h}, \mathfrak{g})$, by bilinearity of the bracket $\llbracket \cdot, \cdot \rrbracket$, this proves that the bracket $\llbracket \cdot, \cdot \rrbracket : C^\bullet(\mathfrak{h}, \mathfrak{g}) \times C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ satisfies the graded Jacobi identity. $\square$

Remark 3.1.18. Notice that the construction of $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \llbracket \cdot, \cdot \rrbracket)$ does not use the Lie algebra structure of $(\mathfrak{h}, \nu)$, only its vector space structure.

The constructed bracket can be used to find Lie algebra homomorphisms:

Theorem 3.1.19. Suppose we have a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, then for an element $\eta \in C^1(\mathfrak{h}, \mathfrak{g})$ holds that

$$\delta_0 \eta + \frac{1}{2} \llbracket \eta, \eta \rrbracket = 0 \iff \eta \text{ is a Lie algebra homomorphism.}$$

Proof. A linear map $\eta : \mathfrak{h} \rightarrow \mathfrak{g}$ is a Lie algebra homomorphism if and only if it satisfies for all $u, v \in \mathfrak{h}$

$$\mu(\eta(u), \eta(v)) - \eta(\nu(u, v)) = 0.$$

Furthermore, we calculate for

$$\delta_0 : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g})$$

that

$$\delta_0 \eta(u, v) = -\eta(\nu(u, v))$$

and

$$[[\eta, \eta]](u, v) = \mu(\eta(u), \eta(v)) - \mu(\eta(v), \eta(u)) = 2\mu(\eta(u), \eta(v))$$

by skew symmetry of μ . Now it's clear that

$$\delta_0 \eta(u, v) + \frac{1}{2} [[\eta, \eta]](u, v) = \mu(\eta(u), \eta(v)) - \eta(\nu(u, v)).$$

Therefore, the condition for $\eta \in C^1(\mathfrak{h}, \mathfrak{g})$ to be a Lie algebra homomorphism is exactly the condition for η to satisfy

$$\delta_0 \eta + \frac{1}{2} [[\eta, \eta]] = 0.$$

□

This is discussed in chapter 1 of [NR67b] on page 176.

3.2 Differential graded Lie algebras

From the graded Lie algebras defined in the previous section, relevant for the deformation theory of Lie algebras and Lie algebra homomorphisms, we will be able to naturally build DGLAs, by equipping the graded Lie algebras with a compatible differential. Here, the objects of interest will be given by the Maurer-Cartan elements of the corresponding DGLA. We will see that twisting the differential by a Maurer-Cartan element yields yet another DGLA structure.

For the deformation theory of Lie subalgebras, [EJ25] will help us get to a suitable DGLA in section 3.2.3.

Definition 3.2.1. A differential graded Lie algebra $(L^\bullet, d, [[\cdot, \cdot]])$ consists of

- a graded vector space $L^\bullet = \bigoplus_{k \in \mathbb{Z}} L^k$
- a **graded Lie bracket** $[[\cdot, \cdot]] : L^\bullet \times L^\bullet \rightarrow L^\bullet$ as in definition 3.1.3, which is a degree 0 bilinear map that is graded skew-symmetric, meaning that

$$[[u, v]] = -(-1)^{hk} [[v, u]] \quad (3.17)$$

for all $u \in L^h, v \in L^k$ and that satisfies the graded Jacobi identity:

$$[[[u, v], w]] + (-1)^{(l+k)h} [[v, w], u] + (-1)^{(h+k)l} [[w, u], v] = 0 \quad (3.18)$$

for all $u \in L^h, v \in L^k$ and $w \in L^l$.

- a differential d , which is a degree 1 linear map $d : L^\bullet \rightarrow L^\bullet$ for which $d^2 = 0$ and is a degree 1 derivation with respect to $[\![\cdot, \cdot]\!]$, meaning that

$$d[\![u, v]\!] = [\![du, v]\!] + (-1)^h [\![u, dv]\!] \quad (3.19)$$

for all $u \in L^h$ and $v \in L^k$.

Here, $[\![\cdot, \cdot]\!]$ being of degree 0 means that $[\![C^h, C^k]\!] \subseteq C^{h+k}$. Similarly d being of degree 1 means that $d(C^h) \subseteq C^{h+1}$.

3.2.1 Lie algebras

Thus far, for a vector space V we have constructed the graded Lie algebra $(C^\bullet(V, V)[1], [\![\cdot, \cdot]\!])$. This forms trivially a DGLA by taking the zero differential. If we have a Lie algebra $(\mathfrak{g}, \mu)$, we will be able to obtain another DGLA. For that, we do make use of the Lie algebra structure of $(\mathfrak{g}, \mu)$. Indeed, if we define the map $d : C^\bullet(\mathfrak{g}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{g}, \mathfrak{g})$ by

$$d\varphi := [\![\mu, \varphi]\!] \quad (3.20)$$

for $\varphi \in C^\bullet(\mathfrak{g}, \mathfrak{g})$, we can prove that this is a differential that converts the triple

$$(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], d = [\![\mu, \cdot]\!], [\![\cdot, \cdot]\!])$$

into a differential graded Lie algebra.

Proposition 3.2.2. *The map $d : C^\bullet(\mathfrak{g}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{g}, \mathfrak{g})$, defined by*

$$d\varphi := [\![\mu, \varphi]\!]$$

for $\varphi \in C^\bullet(\mathfrak{g}, \mathfrak{g})$ is a differential and degree 1 derivation of the graded Lie algebra $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], [\![\cdot, \cdot]\!])$, meaning that

- d is a linear map
- d is of degree 1, in the sense that for all $k \in \mathbb{N}$,

$$d(C^{k+1}(\mathfrak{g}, \mathfrak{g})) \subseteq C^{k+2}(\mathfrak{g}, \mathfrak{g}). \quad (3.21)$$

- $d^2 = 0$
- d is compatible with $[\![\cdot, \cdot]\!]$, in the sense that

$$d[\![\varphi, \psi]\!] = [\![d\varphi, \psi]\!] + (-1)^k [\![\varphi, d\psi]\!] \quad (3.22)$$

for all $\varphi \in C^{h+1}(\mathfrak{g}, \mathfrak{g})$, $\psi \in C^{k+1}(\mathfrak{g}, \mathfrak{g})$ and $h, k \in \mathbb{N}$.

Proof. From bilinearity of $[\![\cdot, \cdot]\!]$ immediately follows that d is linear.

For $\varphi \in C^{h+1}(\mathfrak{g}, \mathfrak{g})$, $\psi \in C^{k+1}(\mathfrak{g}, \mathfrak{g})$ and $h, k \in \mathbb{N}$, we know that $[\![\varphi, \psi]\!] \in C^{h+k+1}(\mathfrak{g}, \mathfrak{g})$. Because now $\mu \in C^2(\mathfrak{g}, \mathfrak{g})$, it follows that $d\psi = [\![\mu, \psi]\!] \in C^{k+2}(\mathfrak{g}, \mathfrak{g})$, which means that d is of degree 1.

Take any $\varphi \in C^{k+1}(\mathfrak{g}, \mathfrak{g})$ for some $k \in \mathbb{N}$, then by the graded Jacobi identity in lemma 3.1.11

$$d^2\varphi = d\llbracket\mu, \varphi\rrbracket = \llbracket\mu, \llbracket\mu, \varphi\rrbracket\rrbracket = \llbracket\llbracket\mu, \mu\rrbracket, \varphi\rrbracket - \llbracket\mu, \llbracket\mu, \varphi\rrbracket\rrbracket$$

which implies

$$d^2\varphi = \frac{1}{2}\llbracket\llbracket\mu, \mu\rrbracket, \varphi\rrbracket = 0$$

as μ is a Lie bracket and thus $\llbracket\mu, \mu\rrbracket = 0$ by theorem 3.1.12.

Lastly, if we let $\varphi \in C^{h+1}(\mathfrak{g}, \mathfrak{g}), \psi \in C^{k+1}(\mathfrak{g}, \mathfrak{g})$ for $h, k \in \mathbb{N}$, then the graded Jacobi identity in lemma 3.1.11 gives

$$\begin{aligned} d\llbracket\varphi, \psi\rrbracket &= \llbracket\mu, \llbracket\varphi, \psi\rrbracket\rrbracket = \llbracket\llbracket\mu, \varphi\rrbracket, \psi\rrbracket + (-1)^k \llbracket\varphi, \llbracket\mu, \psi\rrbracket\rrbracket \\ &= \llbracket d\varphi, \psi\rrbracket + (-1)^k \llbracket\varphi, d\psi\rrbracket \end{aligned}$$

which is exactly the last assertion. $\square$

As mentioned before, this now turns

$$(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], d = \llbracket\mu, \cdot\rrbracket, \llbracket\cdot, \cdot\rrbracket)$$

into a differential graded Lie algebra. Additionally, we have a relation between the differential $\llbracket\mu, \cdot\rrbracket$ and the Chevalley-Eilenberg differential.

Proposition 3.2.3. *For any $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ and $\varphi \in C^{h+1}(\mathfrak{g}, \mathfrak{g})$, we have the relation*

$$\delta_\eta \varphi = \llbracket\eta, \varphi\rrbracket.$$

Proof. This is a straight forward calculation: For $u_1, \dots, u_{h+2} \in \mathfrak{g}$, we have that

$$\begin{aligned} \llbracket\eta, \varphi\rrbracket(u_1, \dots, u_{h+2}) &= (\varphi \bullet \eta - (-1)^h \eta \bullet \varphi)(u_1, \dots, u_{h+2}) \\ &= \sum_{\sigma \in \mathcal{S}_{2,h}} \text{sgn}(\sigma) \varphi(\eta(u_{\sigma(1)}, u_{\sigma(2)}), u_{\sigma(3)}, \dots, u_{\sigma(h+2)}) \\ &\quad + (-1)^{h+1} \sum_{\sigma \in \mathcal{S}_{h+1,1}} \text{sgn}(\sigma) \eta(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h+1)}), u_{\sigma(h+2)}). \end{aligned}$$

We can convert these sums over permutations into sums over indices, so that:

$$\begin{aligned} \varphi \bullet \eta(u_1, \dots, u_{h+2}) &= \sum_{i < j} (-1)^{i+j+1} \varphi(\eta(u_i, u_j), u_1, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_{h+2}) \\ \eta \bullet \varphi(u_1, \dots, u_{h+2}) &= \sum_{i=1}^{h+2} (-1)^{h+2-i} \eta(\varphi(u_1, \dots, \hat{u}_i, \dots, u_{h+2}), u_i). \end{aligned}$$

If we substitute this back into the expression for the Gerstenhaber bracket, we get

$$\llbracket\eta, \varphi\rrbracket(u_1, \dots, u_{h+2}) = \sum_{i < j} (-1)^{i+j+1} \varphi(\eta(u_i, u_j), u_1, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_{h+2})$$

$$\begin{aligned}
& + (-1)^{h+1} \sum_{i=1}^{h+2} (-1)^{h+2-i} \eta(\varphi(u_1, \dots, \hat{u}_i, \dots, u_{h+2}), u_i) \\
& = \sum_{i < j} (-1)^{i+j+1} \varphi(\eta(u_i, u_j), u_1, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_{h+2}) \\
& + \sum_{i=1}^{h+2} (-1)^i \eta(\varphi(u_1, \dots, \hat{u}_i, \dots, u_{h+2}), u_i) \\
& = \delta_\eta \varphi(u_1, \dots, u_{h+2}),
\end{aligned}$$

proving the assertion. $\square$

Corollary 3.2.4. *The Chevalley-Eilenberg differential δ_μ and the differential $\llbracket \mu, \cdot \rrbracket$ are equal:*

$$\delta_\mu = \llbracket \mu, \cdot \rrbracket. \quad (3.23)$$

Corollary 3.2.5. *Let $(\mathfrak{g}, \mu)$ be a Lie algebra. Then, the map δ_μ defined in example 2.4.5 is a differential: $\delta_\mu \circ \delta_\mu = 0$.*

Definition 3.2.6. For a DGLA $(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$, the elements in $\xi \in L^1$ that satisfy the **Maurer-Cartan equation**

$$d\xi + \frac{1}{2} \llbracket \xi, \xi \rrbracket = 0 \quad (3.24)$$

are called the **Maurer-Cartan elements** of the DGLA. We denote this set by $\mathcal{MC}(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$.

Remark 3.2.7. In the Maurer-Cartan equation (3.24), ξ having degree 1 ensures that both terms are of degree 2. Therefore, the vanishing of the sum does not imply the vanishing of both terms separately.

Remark 3.2.8. Any graded Lie algebra $(C^\bullet, \llbracket \cdot, \cdot \rrbracket)$ can naturally be turned into a DGLA, by taking the zero differential $d = 0$. Therefore, one can also define Maurer-Cartan elements of any graded Lie bracket as elements $\xi \in C^1$ for which

$$\llbracket \xi, \xi \rrbracket = 0.$$

Let $(\mathfrak{g}, \mu)$ be a Lie algebra and $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], \llbracket \cdot, \cdot \rrbracket)$ from corollary 3.1.10. Then theorem 3.1.12 can be stated alternatively as follows:

Corollary 3.2.9. *The Lie brackets on $\mathfrak{g}$ are exactly the Maurer-Cartan elements of the graded Lie algebra $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], \llbracket \cdot, \cdot \rrbracket)$ or equivalently, of the DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], 0, \llbracket \cdot, \cdot \rrbracket)$.*

Definition 3.2.10. For a DGLA $(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$, and an element $x \in L^1$, one can define

$$d^x := d + \llbracket x, \cdot \rrbracket$$

as the **differential twisted by x** .

Proposition 3.2.11. *Let $(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$ be a DGLA and x a Maurer-Cartan element. Then the twisted differential*

$$d^x := d + \llbracket x, \cdot \rrbracket \quad (3.25)$$

is

1. of degree 1,
2. a degree 1 derivation with respect to $[\![\cdot, \cdot]\!]$,
3. a differential,

making $(L^\bullet, d^x, [\![\cdot, \cdot]\!])$ into a DGLA.

Proof. As d is of degree 1, $x \in L^1$ and $[\![\cdot, \cdot]\!]$ of degree zero, it follows immediately that d^x is of degree 1.

We now show statement 2. It follows from the fact that d is a derivation with respect to $[\![\cdot, \cdot]\!]$ and by graded Jacobi identity of $[\![\cdot, \cdot]\!]$ that for all $a \in L^h$ and $b \in L^k$

$$\begin{aligned} d^x([\![a, b]\!]) &= d([\![a, b]\!]) + [x, [\![a, b]\!]] \\ &= [da, b] + (-1)^h [a, db] + [[x, a], b] + (-1)^h [a, [x, b]] \\ &= [d^x a, b] + (-1)^h [a, d^x b] \end{aligned}$$

proving the second statement.

Lastly, by the computation

$$\begin{aligned} (d^x \circ d^x)(a) &= d(da + [x, a]) + [x, da + [x, a]] \\ &= d^2 a + d[x, a] + [x, da] + [x, [x, a]] \end{aligned}$$

which is, by using that d is a differential and a degree 1 derivation of $[\![\cdot, \cdot]\!]$, and by using the graded Jacobi identity of $[\![\cdot, \cdot]\!]$, equal to

$$\begin{aligned} &= [dx, a] - [x, da] + [x, da] + \frac{1}{2} [[x, x], a] \\ &= [dx + \frac{1}{2} [x, x], a]. \end{aligned}$$

By using that x is a Maurer-Cartan element, it follows that $dx + \frac{1}{2} [x, x] = 0$ and thus $(d^x \circ d^x)(a) = 0$. $\square$

Example 3.2.12. For the DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], 0, [\![\cdot, \cdot]\!])$ with Maurer-Cartan element μ we have that

$$d^\mu = 0 + [\![\mu, \cdot]\!] = \delta_\mu$$

by corollary 3.2.4.

Theorem 3.2.13. For a DGLA $(L^\bullet, d, [\![\cdot, \cdot]\!])$ and a Maurer-Cartan element x , we have that

$$x + y \in \mathcal{MC}(L^\bullet, d, [\![\cdot, \cdot]\!]) \iff y \in \mathcal{MC}(L^\bullet, d^x, [\![\cdot, \cdot]\!]). \quad (3.26)$$

Proof. We can see that $x + y \in \mathcal{MC}(L^\bullet, d, [\![\cdot, \cdot]\!])$ if and only if

$$\begin{aligned} 0 &= d(x + y) + \frac{1}{2} [x + y, x + y] = dx + dy + \frac{1}{2} [x, x] + \frac{1}{2} [x, y] + \frac{1}{2} [y, x] + \frac{1}{2} [y, y] \\ &= dy + \frac{1}{2} [x, y] + \frac{1}{2} [y, x] + \frac{1}{2} [y, y] \end{aligned}$$

$$\begin{aligned}
&= dy + \llbracket x, y \rrbracket + \frac{1}{2} \llbracket y, y \rrbracket \\
&= d^x y + \frac{1}{2} \llbracket y, y \rrbracket
\end{aligned}$$

if and only if $y \in \mathcal{MC}(L^\bullet, d^x, \llbracket \cdot, \cdot \rrbracket)$. Here, in the first equation we used linearity of d , in the second equation the fact that $x \in \mathcal{MC}(L^\bullet, d, \llbracket \cdot, \cdot \rrbracket)$ and in the third equation graded skew symmetry of $\llbracket \cdot, \cdot \rrbracket$. $\square$

Corollary 3.2.14. ([Tor22], Lemma 6.4) *For a Lie algebra $(\mathfrak{g}, \mu)$, the Lie brackets on $\mathfrak{g}$ are exactly the elements $\mu + \xi$ for Maurer-Cartan elements ξ of the DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], \delta_\mu, \llbracket \cdot, \cdot \rrbracket)$.*

This is also stated in equation 4 of [NR67b].

Proof. By corollary 3.1.10, we know that the Lie brackets on $\mathfrak{g}$ are the Maurer-Cartan elements of the DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], 0, \llbracket \cdot, \cdot \rrbracket)$. Then by using theorem 3.2.13, we know that the Lie brackets on $\mathfrak{g}$ are the elements $\mu + \xi$ for Maurer-Cartan elements ξ of the DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], \delta_\mu, \llbracket \cdot, \cdot \rrbracket)$.

$$\eta \text{ is a Lie bracket} \iff \llbracket \eta, \eta \rrbracket = 0. \quad (3.27)$$

If we write $\eta = \mu + \xi$ for some $\xi \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, then

$$\mu + \xi \text{ is a Lie bracket} \iff \delta_\mu \xi + \frac{1}{2} \llbracket \xi, \xi \rrbracket = 0 \quad (3.28)$$

which concludes the proof. $\square$

3.2.2 Lie algebra homomorphisms

Similar to how we constructed a DGLA for any vector space V , we will now create a suitable DGLA for any Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$. In section 3.1, we constructed a graded Lie algebra $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \llbracket \cdot, \cdot \rrbracket)$ given Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$. We will see that δ_0 on $C^\bullet(\mathfrak{h}, \mathfrak{g})$ will be a compatible differential, making $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$ into a DGLA. If moreover we have a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, we will obtain another DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_\rho, \llbracket \cdot, \cdot \rrbracket)$.

The constructions in this section are described in chapter 1 in [NR67b], and we will try to rigorously develop this theory as we did in subsection 3.2.1.

Before we show that δ_0 is a differential compatible with $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \llbracket \cdot, \cdot \rrbracket)$ in proposition 3.2.21, we first prove some intermediate results that we will use.

Lemma 3.2.15. *For any $\eta \in C^1(\mathfrak{h}, \mathfrak{g})$ and $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, we have the relation*

$$\delta_\eta \varphi = \delta_0 \varphi + \llbracket \eta, \varphi \rrbracket$$

where δ_0 is the Chevalley-Eilenberg differential defined on $C^\bullet(\mathfrak{h}, \mathfrak{g})$ as in example 2.4.6 with the zero homomorphism.

Proof. For $\eta \in C^1(\mathfrak{h}, \mathfrak{g})$ and $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, we have for $u_1, \dots, u_{h+1}$ that

$$\begin{aligned} \llbracket \eta, \varphi \rrbracket(u_1, \dots, u_{h+1}) &= \sum_{\sigma \in S_{1,h}} \text{sgn}(\sigma) \mu(\eta(u_{\sigma(1)}), \varphi(u_{\sigma(2)}, \dots, u_{\sigma(h+1)})) \\ &= \sum_{i=1}^{h+1} (-1)^{i-1} \mu(\eta(u_i), \varphi(u_1, \dots, \hat{u}_i, \dots, u_{h+1})) \end{aligned}$$

which corresponds to the first sum of $\delta_\eta \varphi(u_1, \dots, u_{h+1})$. The second sum corresponds to

$$\delta_0 \varphi(u_1, \dots, u_{h+1}) = \sum_{i < j} \varphi(\nu(u_i, u_j), u_1, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_{h+1}).$$

From this we conclude that

$$\delta_\eta \varphi = \delta_0 \varphi + \llbracket \eta, \varphi \rrbracket.$$

□

Corollary 3.2.16. *For a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, the Chevalley-Eilenberg differential δ_ρ satisfies*

$$\delta_\rho = \delta_0 + \llbracket \rho, \cdot \rrbracket.$$

Lemma 3.2.17. *For any Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$ the Chevalley-Eilenberg differential $\delta_0 : C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ can be written as*

$$\delta_0 = \delta_{\mathfrak{h}} \otimes \text{Id}_{\mathfrak{g}}$$

where $\delta_{\mathfrak{h}}$ is the Chevalley-Eilenberg differential of $\mathfrak{h}$ with values in the trivial representation on $\mathbb{R}$.

Proof. Take a decomposable element $\omega = \alpha \otimes u \in C^k(\mathfrak{h}, \mathfrak{g})$. We know that $\delta_{\mathfrak{h}} : \wedge^k \mathfrak{h}^* \rightarrow \wedge^{k+1} \mathfrak{h}^*$ is defined by

$$(\delta_{\mathfrak{h}} \alpha)(u_0, \dots, u_k) = \sum_{0 \leq i < j \leq k} (-1)^{i+j} \alpha(\nu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k).$$

As $\rho = 0$, the representation $r = \text{ad}_{\mathfrak{g}} \circ \rho = 0$, so the first sum in the Chevalley-Eilenberg formula for δ_0 vanishes, leaving

$$\delta_0 \omega(u_0, \dots, u_k) = \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\nu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_k),$$

which on a decomposable element $\omega = \alpha \otimes u$ gives exactly $(\delta_{\mathfrak{h}} \alpha) \otimes u$. As both δ_0 and $\delta_{\mathfrak{h}} \otimes \text{Id}_{\mathfrak{g}}$ are linear, they coincide on all of $C^k(\mathfrak{h}, \mathfrak{g})$. □

Lemma 3.2.18. *For Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, the differential $\delta_0 : C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ is a degree 1 derivation with respect to $\llbracket \cdot, \cdot \rrbracket$ on $C^\bullet(\mathfrak{h}, \mathfrak{g})$, meaning that*

$$\delta_0 \llbracket \varphi, \psi \rrbracket = \llbracket \delta_0 \varphi, \psi \rrbracket + (-1)^k \llbracket \varphi, \delta_0 \psi \rrbracket \quad (3.29)$$

for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$ and $h, k \in \mathbb{N}$.

Proof. For decomposable elements $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$, we can write

$$\begin{aligned}\varphi &= \alpha \otimes u \text{ for } \alpha \in \wedge^h \mathfrak{h}^* \text{ and } u \in \mathfrak{g} \\ \psi &= \beta \otimes v \text{ for } \beta \in \wedge^k \mathfrak{h}^* \text{ and } v \in \mathfrak{g}.\end{aligned}$$

Then, by using the that $\delta_0 = \delta_{\mathfrak{h}} \otimes \text{Id}_{\mathfrak{g}}$ for the differential $\delta_{\mathfrak{h}}$ on $\mathfrak{h}$, we get that

$$\begin{aligned}\delta_0 \llbracket \varphi, \psi \rrbracket &= \delta_0 \llbracket \alpha \otimes u, \beta \otimes v \rrbracket \\ &= \delta_0(\alpha \wedge \beta) \otimes \mu(u, v) \\ &= ((\delta_{\mathfrak{h}} \alpha) \wedge \beta + (-1)^k \alpha \wedge (\delta_{\mathfrak{h}} \beta)) \otimes \mu(u, v) \\ &= ((\delta_{\mathfrak{h}} \alpha) \wedge \beta) \otimes \mu(u, v) + (-1)^k (\alpha \wedge (\delta_{\mathfrak{h}} \beta)) \otimes \mu(u, v) \\ &= \llbracket \delta_{\mathfrak{h}} \alpha \otimes u, \beta \otimes v \rrbracket + (-1)^k \llbracket \alpha \otimes u, \delta_{\mathfrak{h}} \beta \otimes v \rrbracket \\ &= \llbracket \delta_0 \varphi, \psi \rrbracket + (-1)^k \llbracket \varphi, \delta_0 \psi \rrbracket\end{aligned}$$

where we used in the third equality the fact that the Chevalley-Eilenberg differential of $\mathfrak{h}$ with values in the trivial representation on $\mathbb{R}$ is a degree 1 derivation with respect to the wedge product, as stated in [FO07] in equation (2) of section 1.2. As the decomposable elements generate the vector spaces $C^i(\mathfrak{h}, \mathfrak{g})$, by linearity of δ_0 and bilinearity of $\llbracket \cdot, \cdot \rrbracket$ holds that the equation

$$\delta_0 \llbracket \varphi, \psi \rrbracket = \llbracket \delta_0 \varphi, \psi \rrbracket + (-1)^k \llbracket \varphi, \delta_0 \psi \rrbracket$$

holds for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$. □

Corollary 3.2.19. *For any Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, the triple $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$ forms a DGLA.*

Theorem 3.1.19 can now be stated alternatively as the following.

Corollary 3.2.20. *For Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, the Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$ are exactly the Maurer-Cartan elements of the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$.*

Proposition 3.2.21. *The differential $\delta_\rho : C^\bullet(\mathfrak{h}, \mathfrak{g}) \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{g})$ is a derivation with respect to $\llbracket \cdot, \cdot \rrbracket$, meaning that*

$$\delta_\rho \llbracket \varphi, \psi \rrbracket = \llbracket \delta_\rho \varphi, \psi \rrbracket + (-1)^k \llbracket \varphi, \delta_\rho \psi \rrbracket \quad (3.30)$$

for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$, $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$ and $h, k \in \mathbb{N}$.

Proof. By corollary 3.2.16, we can write

$$\delta_\rho = \delta_0 + \llbracket \rho, \cdot \rrbracket$$

so that it is sufficient to prove that both terms, δ_0 and $\llbracket \rho, \cdot \rrbracket$, are a derivation with respect to $\llbracket \cdot, \cdot \rrbracket$. We already proved that for the first term in lemma 3.2.18, so we only need to do so for the second term: We need to prove that for all $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$ holds that

$$\llbracket \rho, \llbracket \varphi, \psi \rrbracket \rrbracket = \llbracket \llbracket \rho, \varphi \rrbracket, \psi \rrbracket + (-1)^k \llbracket \varphi, \llbracket \rho, \psi \rrbracket \rrbracket.$$

This now follows from 3.1.11 as $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \llbracket \cdot, \cdot \rrbracket)$ is a graded Lie algebra. Therefore, we proved that δ_0 and $\llbracket \rho, \cdot \rrbracket$ are derivations with respect to $\llbracket \cdot, \cdot \rrbracket$, and thus δ_ρ too. □

Corollary 3.2.22. *For Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, for the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$ and Maurer-Cartan element ρ , the statement says that δ_ρ is the differential δ_0 twisted by ρ .*

Corollary 3.2.23. *For any Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, the triple $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_\rho, \llbracket \cdot, \cdot \rrbracket)$ forms a DGLA.*

Theorem 3.2.24. *Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism, then $\rho + \eta$ is a Lie algebra homomorphism if and only if η is a Maurer-Cartan element of the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_\rho, \llbracket \cdot, \cdot \rrbracket)$, i.e.*

$$\delta_\rho \eta + \frac{1}{2} \llbracket \eta, \eta \rrbracket = 0.$$

Proof. The Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$ are exactly the Maurer-Cartan elements of the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$. By theorem 3.2.13, we then know that $\rho + \eta$ is a Lie algebra homomorphism from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$ if and only if η is a Maurer-Cartan element of the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_\rho, \llbracket \cdot, \cdot \rrbracket)$. $\square$

The former theorem is also described in [NR67a] in chapter 2 on page 178.

3.2.3 Lie subalgebras

In principle for a general Lie subalgebra $\mathfrak{h}$ of a Lie algebra $(\mathfrak{g}, \mu)$, we do not obtain a DGLA as we did in the previous two cases. Instead, one can get an L_∞ -algebra, which is a generalization of a DGLA. This is shown in Lemma 3.2.4 in [EJ25]. As the theory about L_∞ -algebra is beyond the scope of this thesis, we restrict ourselves to the special case where this Lie subalgebra $\mathfrak{h}$ admits a complement, $\mathfrak{h}^c$, that is also a Lie subalgebra of $(\mathfrak{g}, \mu)$. In that case, we can prove that the L_∞ -algebra from Lemma 3.2.4 in [EJ25], reduces to an honest DGLA. Because of the extra assumption we make, this subsection is not explicitly in the literature.

Proposition 3.2.25. *In the setting of Lemma 3.2.4 in [EJ25], if $\mathfrak{h}^c$ is also a Lie subalgebra of $(\mathfrak{g}, \mu)$, then $m_3 = 0$, so that the $L_\infty[1]$ -algebra $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c)[1], m_1, m_2, m_3)$ reduces to a differential graded Lie algebra $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), m_1, m_2)$.*

Proof. By Lemma 3.2.4 in [EJ25], the multibrackets of the $L_\infty[1]$ -algebra are defined by

$$m_l(a_1, \dots, a_l) = P[\dots [\llbracket \mu, I(a_1) \rrbracket_G, I(a_2) \rrbracket_G, \dots, I(a_l) \rrbracket_G$$

for $l \geq 1$, where $\llbracket \cdot, \cdot \rrbracket_G$ is the Gerstenhaber bracket from definition 3.1.8 on $C^\bullet(\mathfrak{g}, \mathfrak{g})$. Furthermore, if we denote by $p_{\mathfrak{h}} : \mathfrak{g} \rightarrow \mathfrak{h}$ and $p_{\mathfrak{h}^c} : \mathfrak{g} \rightarrow \mathfrak{h}^c$ denotes the projections associated to the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^c$ and $\iota : \mathfrak{h} \rightarrow \mathfrak{g}$ the inclusion, then

$$P : C^\bullet(\mathfrak{g}, \mathfrak{g})[1] \rightarrow C^\bullet(\mathfrak{h}, \mathfrak{h}^c)[1] : \xi \mapsto p_{\mathfrak{h}^c} \circ \xi \circ \iota$$

and

$$I : C^\bullet(\mathfrak{h}, \mathfrak{h}^c)[1] \rightarrow C^\bullet(\mathfrak{g}, \mathfrak{g})[1] : \xi \mapsto \xi \circ p_{\mathfrak{h}}$$

The computations in the proof of Lemma 3.2.4 in [EJ25] show that for any $\xi \in \wedge^k \mathfrak{h}^* \otimes \mathfrak{h}^c$, we have that

$$m_1(\xi) = P(\delta_\mu(I(\xi))) = p_{\mathfrak{h}^c}(\delta_\mu(\xi \circ p_{\mathfrak{h}})).$$

By taking $h_1, \dots, h_{k+1} \in \mathfrak{h}$, we see that

$$\begin{aligned} m_1(\xi)(h_1, \dots, h_{k+1}) &= p_{\mathfrak{h}^c}(\delta_\mu(\xi \circ p_{\mathfrak{h}})) \\ &= \sum_{i=1}^{k+1} (-1)^{i+1} p_{\mathfrak{h}^c} \mu(h_i, \xi(h_1, \dots, \hat{h}_i, \dots, h_{k+1})) \\ &\quad + \sum_{i < j} (-1)^{i+j} \xi(\mu(h_i, h_j), h_1, \dots, \hat{h}_i, \dots, \hat{h}_j, \dots, h_{k+1}) \\ &= \delta_{\mathfrak{h}, \mathfrak{h}^c}(\xi)(h_1, \dots, h_{k+1}) \end{aligned}$$

so that $m_1 = \delta_{\mathfrak{h}, \mathfrak{h}^c}$. Furthermore, denote $\llbracket \cdot, \cdot \rrbracket := m_2$. Then we see that this graded Lie bracket $\llbracket \cdot, \cdot \rrbracket = m_2$ is defined as

$$\llbracket \xi, \eta \rrbracket = m_2(\xi, \eta) = P(\llbracket \delta_\mu(I(\xi)), I(\eta) \rrbracket_G) \quad (3.31)$$

Since images of elements of degree 1 determine m_3 and $m_k = 0$ for $k \geq 4$, it suffices to prove $m_3(\xi, \eta, \nu) = 0$ for all $\xi, \eta, \nu \in C^1(\mathfrak{h}, \mathfrak{h}^c) = \mathfrak{h}^* \otimes \mathfrak{h}^c$. From the computation in the proof of lemma 3.2.4 in [EJ25], for all $h_1, h_2 \in \mathfrak{h}$:

$$\begin{aligned} m_3(\xi, \eta, \nu)(h_1, h_2) &= -\xi(p_{\mathfrak{h}}[\nu(h_1), \eta(h_2)]) - \eta(p_{\mathfrak{h}}[\nu(h_1), \xi(h_2)]) \\ &\quad - \xi(p_{\mathfrak{h}}[\eta(h_1), \nu(h_2)]) - \eta(p_{\mathfrak{h}}[\xi(h_1), \nu(h_2)]) \\ &\quad - \nu(p_{\mathfrak{h}}([\xi(h_1), \eta(h_2)]) + p_{\mathfrak{h}}([\eta(h_1), \xi(h_2)])), \end{aligned}$$

Now suppose $\mathfrak{h}^c$ is a Lie subalgebra of $(\mathfrak{g}, \mu)$. Then $\mu(\mathfrak{h}^c, \mathfrak{h}^c) \subseteq \mathfrak{h}^c$, which means that for the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^c$ we have

$$p_{\mathfrak{h}}([\mathfrak{h}^c, \mathfrak{h}^c]) = 0,$$

since any bracket of two elements in $\mathfrak{h}^c$ lies again in $\mathfrak{h}^c$, and thus has zero $\mathfrak{h}$ -component. Therefore $m_3(\xi, \eta, \nu)(h_1, h_2) = 0$ for all $h_1, h_2 \in \mathfrak{h}$ and all $\xi, \eta, \nu \in \mathfrak{h}^* \otimes \mathfrak{h}^c$, which gives $m_3 = 0$. Consequently, the L_∞ -algebra structure reduces to the differential graded Lie algebra $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, \llbracket \cdot, \cdot \rrbracket)$. $\square$

The following theorem follows from proposition 3.5 in [EJ25], given the extra assumption that $\mathfrak{h}^c$ is also a Lie subalgebra.

Theorem 3.2.26. *Given a k -dimensional Lie subalgebra $\mathfrak{h}$ with a complement $\mathfrak{h}^c$ that is a Lie subalgebra, we have a bijection*

$$\left\{ \begin{array}{l} \text{Maurer-Cartan elements of} \\ \text{the DGLA } (C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, \llbracket \cdot, \cdot \rrbracket) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Lie subalgebras } \mathfrak{h}' \subset \mathfrak{h} \text{ such} \\ \text{that } \mathfrak{h}' \in \text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c) \end{array} \right\}$$

given by taking the graph $\eta \mapsto \Gamma(\eta)$. Here $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c) = \{\mathfrak{h}' \in \text{Gr}_k(\mathfrak{g}) \mid \mathfrak{h}' \cap \mathfrak{h}^c\}$ as defined in remark 2.3.5.

Proof. Take some $\eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c$, and consider $\mathcal{L}$ from definition 2.3.7. Then $\mathcal{L}(\eta)$ is defined for all $u, v \in \mathfrak{h}$

$$\begin{aligned} \mathcal{L}(\eta)(u, v) &= p_{\mathfrak{h}^c}(\mu(u, \eta(v))) + p_{\mathfrak{h}^c}(\mu(\eta(u), v)) + \mu(\eta(u), \eta(v)) \\ &\quad - \eta(\mu(u, v)) - \eta(p_{\mathfrak{h}}(\mu(u, \eta(v)))) - \eta(p_{\mathfrak{h}}(\mu(\eta(u), v))). \end{aligned}$$

On the other hand, we compute that

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} \eta(u, v) = p_{\mathfrak{h}^c}(\mu(u, \eta(v))) - p_{\mathfrak{h}^c}(\mu(v, \eta(u))) - \eta(\mu(u, v))$$

and by equation (3.31)

$$\begin{aligned} \llbracket \eta, \eta \rrbracket(u, v) &= p_{\mathfrak{h}^c}(\delta_{\mu}(\eta \circ p_{\mathfrak{h}})(\eta \circ p_{\mathfrak{h}}(u), v)) - \delta_{\mu}(\eta \circ p_{\mathfrak{h}})(\eta(v), u) - \eta(\delta_{\mu}(\eta \circ p_{\mathfrak{h}})(u, v)) \\ &= p_{\mathfrak{h}^c}(\mu(\eta(u), \eta(v)) - \mu(v, \eta(p_{\mathfrak{h}}(\eta(u)))) - \eta(p_{\mathfrak{h}}(\mu(\eta(u), v))) \\ &\quad - \mu(\eta(v), \eta(u)) + \mu(u, \eta(p_{\mathfrak{h}}(\eta(v)))) + \eta(p_{\mathfrak{h}}(\mu(\eta(v), u))) \\ &\quad - \eta(\mu(\eta(v), \eta(u))) - \eta(p_{\mathfrak{h}}(\mu(v, \eta(u)))) + \eta(p_{\mathfrak{h}}(\eta(\mu(u, v)))) \\ &= 2p_{\mathfrak{h}^c}(\mu(\eta(u), \eta(v))) - 2\eta(p_{\mathfrak{h}}(\mu(\eta(u), v))) + 2\eta(p_{\mathfrak{h}}(\mu(\eta(v), u))). \end{aligned}$$

Hence, we can see that

$$\mathcal{L}(\eta) = \delta_{\mathfrak{h}, \mathfrak{h}^c} \eta + \frac{1}{2} \llbracket \eta, \eta \rrbracket \quad (3.32)$$

We know by lemma 2.3.8 that its graph $\Gamma(\eta)$ is a Lie subalgebra if and only if $\mathcal{L}(\eta) = 0$ and thus by the former equation, if and only if

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} \eta + \frac{1}{2} \llbracket \eta, \eta \rrbracket = 0$$

which proves the assertion. □

Chapter 4

Deformations

This chapter rigorously develops the deformation theory of Lie algebras, Lie algebra homomorphisms, and Lie subalgebras, following the framework of Crainic, Schätz and Struchiner in chapter 3 of [CSS14] closely for these first two theories. For the latter, by making an extra assumption, the statements formed in this thesis differ from the statements in [CSS14] but still follow the same pattern. Hence, section 4.3 is our own contribution. The central question throughout is the same in all three deformation theories: given a fixed structure, how does it behave under small perturbations, and when should two such perturbations be considered essentially the same?

For each of the three objects, the answer follows a common pattern. A *deformation* is a smooth family of structures starting at the given one, and two deformations are *equivalent* if they are related by an appropriate Lie group action. This leads naturally to description of infinitesimal deformations and their equivalence classes in terms of Lie algebra cohomology. This cohomological interpretation sets the stage for solving the main problems stated in chapter 1.

In section 4.4, we recognize the DGLA theory behind the three deformation theories in sections 4.1, 4.2 and 4.3. The results in section 4.4 are classical, but the presentation, showing the connection between the DGLA theory and the deformation theories in 4.1, 4.2 and 4.3, is our own.

4.1 Deformations of Lie algebras

This section is based on section 3.1 in [CSS14] and was also done in [NR67b].

Definition 4.1.1. Fix a Lie algebra $(\mathfrak{g}, \mu)$. A **deformation** of $(\mathfrak{g}, \mu)$ is a smooth family $\{\mu_t \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \mid t \in [0, 1]\}$ of Lie brackets on $\mathfrak{g}$ so that $\mu_0 = \mu$. We say that two deformations μ_t and μ'_t are **equivalent** if there exists a smooth family of Lie algebra isomorphisms $\{\varphi_t : (\mathfrak{g}, \mu_t) \rightarrow (\mathfrak{g}, \mu'_t) \mid t \in [0, 1]\}$ so that $\varphi_0 = \text{Id}_{\mathfrak{g}}$.

In the previous definition, the condition that $\{\mu_t\}_{t \in [0, 1]}$ and $\{\varphi_t\}_{t \in [0, 1]}$ are smooth families, allows that the derivatives $\dot{\mu}_t := \frac{d}{dt} \mu_t$ and $\dot{\varphi}_t := \frac{d}{dt} \varphi_t$ are well defined.

Proposition 4.1.2. ([CSS14], Proposition 3.2) *If μ_t is a deformation of $(\mathfrak{g}, \mu)$, then $\delta_\mu \dot{\mu}_0 = 0$, for the Chevalley-Eilenberg differential δ_μ from example 2.4.5. In particular, this means that $\dot{\mu}_0 \in Z^2(\mathfrak{g}, \mathfrak{g})$.*

Proof. Clearly, if μ_t is a deformation, $\mu_t \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$. Then, it's clear that $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is a Lie bracket if and only if $\mathcal{J}(\eta) = 0$, where $\mathcal{J}$ is the Jacobiator as in equation (2.10). Hence, if we take a deformation μ_t of $(\mathfrak{g}, \mu)$, then for all $t \in [0, 1]$, the Jacobi identity on μ_t is equivalent to

$$\mathcal{J}(\mu_t) = 0$$

meaning that for all $u, v, w \in \mathfrak{g}$

$$\mathcal{J}(\mu_t)(u, v, w) = \mu_t(\mu_t(u, v), w) + \text{c.p.} = 0.$$

Taking the derivative with respect to t gives the equation

$$\dot{\mu}_t(\mu_t(u, v), w) + \mu_t(\dot{\mu}_t(u, v), w) + \text{c.p.} = 0$$

which, evaluated at $t = 0$, is

$$\dot{\mu}_0(\mu_0(u, v), w) + \mu_0(\dot{\mu}_0(u, v), w) + \text{c.p.} = 0.$$

Written in full and using $\mu_0 = \mu$, this equation becomes

$$\begin{aligned} & \dot{\mu}_0(\mu(u, v), w) + \dot{\mu}_0(\mu(v, w), u) + \dot{\mu}_0(\mu(w, u), v) \\ & + \mu(\dot{\mu}_0(u, v), w) + \mu(\dot{\mu}_0(v, w), u) + \mu(\dot{\mu}_0(w, u), v) = 0. \end{aligned} \tag{4.1}$$

On the other hand, if we apply δ_μ to $\dot{\mu}_0 \in C^2(\mathfrak{g}, \mathfrak{g})$, we get for all $u, v, w \in \mathfrak{g}$

$$\begin{aligned} \delta_\mu \dot{\mu}_0(u, v, w) &= \sum_{i=0}^2 (-1)^i \mu(u_i, \dot{\mu}_0(u_0, \dots, \hat{u}_i, \dots, u_2)) \\ &+ \sum_{0 \leq i < j \leq 2} (-1)^{i+j} \dot{\mu}_0(\mu(u_i, u_j), u_0, \dots, \hat{u}_i, \dots, \hat{u}_j, \dots, u_2) \\ &= \mu(u, \dot{\mu}_0(v, w)) - \mu(v, \dot{\mu}_0(u, w)) + \mu(w, \dot{\mu}_0(u, v)) \\ &- \dot{\mu}_0(\mu(u, v), w) + \dot{\mu}_0(\mu(u, w), v) - \dot{\mu}_0(\mu(v, w), u) \end{aligned}$$

If we compare this with the left-hand side of equation (4.1), and use skew symmetry and bilinearity of μ and $\dot{\mu}_0$, we see that

$$\delta_\mu \dot{\mu}_0 = - \frac{d}{dt} \Big|_{t=0} \mathcal{J}(\mu_t) = -d_\mu \mathcal{J}(\dot{\mu}_0) = 0 \tag{4.2}$$

which means exactly that $\dot{\mu}_0 \in Z^2(\mathfrak{g}, \mathfrak{g})$. □

Remark 4.1.3. ([Tor22], Proof of Proposition 6.8) Notice that we can find the same result a lot quicker if we use the DGLA formalism: The condition that μ_t is a deformation implies by theorem 3.1.12 that for all t , we have that

$$[\mu_t, \mu_t] = 0,$$

where $\llbracket \cdot, \cdot \rrbracket$ is the Gerstenhaber bracket from definition 3.1.8. As a result, if we differentiate with respect to t , we get by graded skew symmetry of the Gerstenhaber bracket that

$$0 = \frac{d}{dt} \llbracket \mu_t, \mu_t \rrbracket = \llbracket \dot{\mu}_t, \mu_t \rrbracket + \llbracket \mu_t, \dot{\mu}_t \rrbracket = 2 \llbracket \mu_t, \dot{\mu}_t \rrbracket$$

so that, evaluated at $t = 0$, this implies

$$0 = 2 \llbracket \mu, \dot{\mu}_0 \rrbracket = -2 \delta_\mu \dot{\mu}_0.$$

by corollary 3.2.4.

Proposition 4.1.4. ([CSS14], Proposition 3.2) *If μ_t and ν_t are two deformations of $(\mathfrak{g}, \mu)$ that are equivalent, then $[\dot{\mu}_0] = [\dot{\nu}_0]$ in $H^2(\mathfrak{g}, \mathfrak{g})$.*

Proof. As μ_t and ν_t are equivalent, there exists a smooth family of Lie algebra isomorphisms φ_t so that for all $u, v \in \mathfrak{g}$

$$\mu_t(u, v) = (\varphi_t \cdot \nu_t)(u, v) = \varphi_t^{-1}(\nu_t(\varphi_t(u), \varphi_t(v)))$$

which is equivalent to

$$\varphi_t(\mu_t(u, v)) = \nu_t(\varphi_t(u), \varphi_t(v)).$$

Differentiating this expression with respect to t gives

$$\dot{\varphi}_t(\mu_t(u, v)) + \varphi_t(\dot{\mu}_t(u, v)) = \dot{\nu}_t(\varphi_t(u), \varphi_t(v)) + \nu_t(\dot{\varphi}_t(u), \varphi_t(v)) + \nu_t(\varphi_t(u), \dot{\varphi}_t(v)).$$

If we evaluate this in $t = 0$, and use that $\mu_0 = \nu_0 = \mu$ and $\varphi_0 = \text{Id}_{\mathfrak{g}}$, we get

$$\dot{\varphi}_0(\mu(u, v)) + \dot{\mu}_0(u, v) = \dot{\nu}_0(u, v) + \mu(\dot{\varphi}_0(u), v) + \mu(u, \dot{\varphi}_0(v))$$

or equivalently

$$\dot{\mu}_0(u, v) - \dot{\nu}_0(u, v) = \mu(u, \dot{\varphi}_0(v)) + \mu(\dot{\varphi}_0(u), v) - \dot{\varphi}_0(\mu(u, v)). \quad (4.3)$$

On the other hand, if we apply δ_μ to $\dot{\varphi}_0 \in C^1(\mathfrak{g}, \mathfrak{g})$ and compare

$$\delta_\mu \dot{\varphi}_0(u, v) = \mu(u, \dot{\varphi}_0(v)) - \mu(v, \dot{\varphi}_0(u)) - \dot{\varphi}_0(\mu(u, v))$$

to equation (4.3), we see that the right-hand sides of both equations are the same. It follows that

$$\dot{\mu}_0(u, v) - \dot{\nu}_0(u, v) = \delta_\mu \dot{\varphi}_0(u, v)$$

so that

$$\dot{\mu}_0 - \dot{\nu}_0 = \delta_\mu \dot{\varphi}_0 \in B^2(\mathfrak{g}, \mathfrak{g}).$$

This means that in $H^2(\mathfrak{g}, \mathfrak{g})$,

$$[\dot{\mu}_0 - \dot{\nu}_0] = 0$$

or

$$[\dot{\mu}_0] = [\dot{\nu}_0].$$

□

Remark 4.1.5. ([CSS14], Remark 3.3) We can interpret propositions 4.1.2 and 4.1.4 in the following way. Deformations of μ are essentially smooth curves in $\{\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \mid \mathcal{J}(\eta) = 0\}$ starting at μ . The formal tangent space of the algebraic variety $\{\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \mid \mathcal{J}(\eta) = 0\}$ at the point μ is defined as

$$T_\mu(\mathcal{J}^{-1}(0)) := \{\xi \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \mid d_\mu \mathcal{J}(\xi) = 0\}$$

but as $d_\mu \mathcal{J} = -\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g})$ by equation (4.2) in proposition 4.1.2, we know that

$$T_\mu(\mathcal{J}^{-1}(0)) = Z^2(\mathfrak{g}, \mathfrak{g}),$$

meaning that the velocities of deformations of μ at $t = 0$ are the 2-cocycles. If we divide this formal tangent space by the honest tangent space to the $\mathrm{GL}(\mathfrak{g})$ -orbit in the point μ , we can see by proposition 4.1.4 that

$$T_{[\mu]} \frac{\mathcal{J}^{-1}(0)}{\mathrm{GL}(\mathfrak{g})} = H^2(\mathfrak{g}, \mathfrak{g}).$$

In proposition 4.1.2 we proved that every deformation gives rise to a 2-cocycle. The converse is not true. As $H^2(\mathfrak{g}, \mathfrak{g})$ is interpreted as *formal* tangent space, and not as honest tangent space; not every element comes from the velocity of a deformation at $t = 0$. In particular, not every element in $Z^2(\mathfrak{g}, \mathfrak{g})$ integrates to a deformation. We then say such an element is ‘obstructed’. To detect such obstructions, we define the Kuranishi map:

Definition 4.1.6. For any Lie algebra $(\mathfrak{g}, \mu)$ we define the Kuranishi map as

$$\Phi : Z^2(\mathfrak{g}, \mathfrak{g}) \rightarrow H^3(\mathfrak{g}, \mathfrak{g}) : \eta \mapsto \mathcal{J}(\eta) + B^3(\mathfrak{g}, \mathfrak{g}) = \llbracket \eta, \eta \rrbracket + B^3(\mathfrak{g}, \mathfrak{g}). \quad (4.4)$$

We claim that it is well defined. We know the Jacobiator is a map $\mathcal{J} : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g})$. If we can prove that the Jacobiator maps every element in $Z^2(\mathfrak{g}, \mathfrak{g})$ to an element in $Z^3(\mathfrak{g}, \mathfrak{g})$, we’re done. Then the Kuranishi map Φ is just a composition of the restricted Jacobiator $\mathcal{J}|_{Z^2(\mathfrak{g}, \mathfrak{g})} : Z^2(\mathfrak{g}, \mathfrak{g}) \rightarrow Z^3(\mathfrak{g}, \mathfrak{g})$ and the projection $\pi : Z^3(\mathfrak{g}, \mathfrak{g}) \rightarrow H^3(\mathfrak{g}, \mathfrak{g})$. Therefore, take some $\eta \in Z^2(\mathfrak{g}, \mathfrak{g})$, and $u, v, w \in \mathfrak{g}$. Taking the Jacobiator of η is just the Gerstenhaber bracket of η with itself: $\mathcal{J}(\eta) = \llbracket \eta, \eta \rrbracket$. From this follows that

$$\delta_\mu(\mathcal{J}(\eta)) = \delta_\mu \llbracket \eta, \eta \rrbracket = \llbracket \delta_\mu \eta, \eta \rrbracket - \llbracket \eta, \delta_\mu \eta \rrbracket$$

as δ_μ is a derivation that is compatible with the Gerstenhaber bracket. Now because $\eta \in Z^2(\mathfrak{g}, \mathfrak{g})$, we know that $\delta_\mu \eta = 0$, so that

$$\delta_\mu(\mathcal{J}(\eta)) = 2 \llbracket \delta_\mu \eta, \eta \rrbracket = 0$$

and thus $\mathcal{J}(\eta) \in Z^3(\mathfrak{g}, \mathfrak{g})$, which is what we had to prove.

Proposition 4.1.7. ([CSS14], Proposition 3.4) Fix some $\xi \in Z^2(\mathfrak{g}, \mathfrak{g})$. If there exists some deformation μ_t of $\mathfrak{g}$ with $\dot{\mu}_0 = \xi$, then $\Phi(\xi) = 0$.

Proof. As μ_t is a deformation, it suffices the Jacobi identity, meaning that for all $u, v, w \in \mathfrak{g}$

$$0 = \mathcal{J}(\mu_t)(u, v, w) = \mu_t(\mu_t(u, v), w) + \text{c.p.} \quad (4.5)$$

Since $C^2(\mathfrak{g}, \mathfrak{g})$ is a vector space and μ_t depends smoothly on t , we can write μ_t as Taylor expansion around $t = 0$:

$$\mu_t = \mu_0 + t\dot{\mu}_0 + \frac{t^2}{2}\ddot{\mu}_0 + O(t^3) = \mu + t\xi + \frac{t^2}{2}\ddot{\mu}_0 + O(t^3),$$

where we used big O notation. Substituting this into equation (4.5), we get

$$0 = (\mu + t\xi + \frac{t^2}{2}\ddot{\mu}_0 + O(t^3))(\mu + t\xi + \frac{t^2}{2}\ddot{\mu}_0 + O(t^3))(u, v), w) + \text{c.p.}$$

By using that $\mu, \xi, \ddot{\mu}_0, \dots$ are bilinear we get that

$$0 = \mu(\mu(u, v), w) + \text{c.p.} + O(t).$$

As the coefficients of t^0 on the left- and right-hand side should be equal, we recover that

$$0 = \mu(\mu(u, v), w) + \text{c.p.}$$

which just means that μ satisfies the Jacobi identity, but this was already true. Similarly, we find

$$\begin{aligned} 0 &= \mu(\mu(u, v), w) + \mu(t\xi(u, v), w) + t\xi(\mu(u, v), w) + \text{c.p.} + O(t^2) \\ &= t(\mu(\xi(u, v), w) + \xi(\mu(u, v), w)) + \text{c.p.} + O(t^2) \\ &= -t\delta_\mu\xi(u, v, w) + O(t^2) \end{aligned}$$

so that also $\delta_\mu\xi = 0$. Additionally, in the same way we get that

$$\begin{aligned} 0 &= \mu\left(\frac{t^2}{2}\ddot{\mu}_0(u, v), w\right) + t\xi(t\xi(u, v), w) + \frac{t^2}{2}\ddot{\mu}_0(\mu(u, v), w) + \text{c.p.} + O(t^3) \\ &= t^2\left(\frac{1}{2}\ddot{\mu}_0(u, v), w + \xi(\xi(u, v), w) + \frac{1}{2}\ddot{\mu}_0(\mu(u, v), w)\right) + \text{c.p.} + O(t^3) \\ &= t^2\left(\mathcal{J}(\xi) - \frac{1}{2}\delta_\mu\ddot{\mu}_0\right) + O(t^3) \end{aligned}$$

meaning that also

$$0 = \mathcal{J}(\xi) - \frac{1}{2}\delta_\mu\ddot{\mu}_0$$

or equivalently

$$\mathcal{J}(\xi) = \frac{1}{2}\delta_\mu\ddot{\mu}_0 \in \text{Im}(\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g})) = B^3(\mathfrak{g}, \mathfrak{g}).$$

This proves that the Kuranishi map annihilates ξ :

$$\Phi(\xi) = \mathcal{J}(\xi) + B^3(\mathfrak{g}, \mathfrak{g}) = 0 + B^3(\mathfrak{g}, \mathfrak{g}).$$

□

Corollary 4.1.8. *Let $(\mathfrak{g}, \mu)$ be a Lie algebra and Φ the Kuranishi map from definition 4.1.6. If*

$$\Phi(\xi) \neq 0 \text{ for some } \xi \in Z^2(\mathfrak{g}, \mathfrak{g}),$$

then ξ is obstructed, i.e., there does not exist a deformation μ_t of μ such that $\dot{\mu}_0 = \xi$.

4.2 Deformations of Lie algebra homomorphisms

This section follows section 3.2 in [CSS14].

Definition 4.2.1. Let $(\mathfrak{g}, \mu)$ and $(\mathfrak{h}, \nu)$ be Lie algebras and $\rho : \mathfrak{h} \rightarrow \mathfrak{g}$ a Lie algebra homomorphism. A **deformation** of ρ is a smooth family $\{\rho_t \in \mathfrak{h}^* \otimes \mathfrak{g} \mid t \in [0, 1]\}$ of Lie algebra homomorphisms such that $\rho_0 = \rho$. Let G be the unique simply connected Lie group integrating $\mathfrak{g}$. Two deformations ρ_t and ρ'_t are called **equivalent** if there exists a smooth curve g_t in G such that $\rho'_t = \text{Ad}_{g_t} \circ \rho_t$ for all $t \in [0, 1]$ and $g_0 = e$.

Proposition 4.2.2. ([CSS14], Proposition 3.7) *If $\{\rho_t\}$ is a deformation of a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, then $\delta_\rho \dot{\rho}_0 = 0$, where δ_ρ is the Chevalley-Eilenberg differential from example 2.4.6. In particular, this means that $\dot{\rho}_0 \in Z^1(\mathfrak{h}, \mathfrak{g})$.*

Proof. As $\{\rho_t\}$ is a deformation of ρ , for all $t \in [0, 1]$ and $u, v \in \mathfrak{h}$ holds that

$$\rho_t(\nu(u, v)) = \mu(\rho_t(u), \rho_t(v)).$$

If we define the map $\mathcal{K} : \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ as in (2.13), then the former condition is equivalent to saying for all $t \in [0, 1]$ and $u, v \in \mathfrak{h}$ that

$$\mathcal{K}(\rho_t)(u, v) = \mu(\rho_t(u), \rho_t(v)) - \rho_t(\nu(u, v)) = 0$$

Differentiating this with respect to t , evaluated at $t = 0$ gives the equation

$$d_\rho \mathcal{K}(\dot{\rho}_0)(u, v) = \mu(\dot{\rho}_0(u), \rho_0(v)) + \mu(\rho_0(u), \dot{\rho}_0(v)) - \dot{\rho}_0(\nu(u, v)) = 0$$

Together with the computation that

$$\delta_\rho \dot{\rho}_0(u, v) = \mu(\rho_0(u), \dot{\rho}_0(v)) - \mu(\rho_0(v), \dot{\rho}_0(u)) - \dot{\rho}_0(\nu(u, v)),$$

this shows that

$$\delta_\rho \dot{\rho}_0(u, v) = d_\rho \mathcal{K}(\dot{\rho}_0)(u, v) = 0 \tag{4.6}$$

As $u, v \in \mathfrak{h}$ were arbitrary, we have that $\delta_\rho \dot{\rho}_0 = 0$. $\square$

Proposition 4.2.3. ([CSS14], Proposition 3.7) *If $\{\rho_t\}$ and $\{\rho'_t\}$ are two equivalent deformations of $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, then $[\dot{\rho}_0] = [\dot{\rho}'_0] \in H^1(\mathfrak{h}, \mathfrak{g})$.*

Proof. As $\{\rho_t\}$ and $\{\rho'_t\}$ are two equivalent deformations of ρ , for all $t \in [0, 1]$ and $u \in \mathfrak{h}$ holds that

$$\rho'_t(u) = \text{Ad}_{g_t} \circ \rho_t(u)$$

Differentiating this expression with respect to t and evaluating in $t = 0$ gives

$$\dot{\rho}'_0(u) = \text{Ad}_{g_0} \circ \dot{\rho}_0(u) + \text{ad}_{\dot{g}_0} \circ \rho_0(u)$$

as $\text{ad}_{\dot{g}}(g) = d_e \text{Ad}_g : T_e G \rightarrow T_e G$. Therefore,

$$\begin{aligned} \dot{\rho}'_0(u) &= \text{Ad}_e \circ \dot{\rho}_0(u) + \mu(\dot{g}_0, \rho(u)) \\ &= \dot{\rho}_0(u) + \mu(\dot{g}_0, \rho(u)) \\ &= \dot{\rho}_0(u) - \delta_\rho(\dot{g}_0), \end{aligned}$$

from which we conclude that $[\dot{\rho}_0] = [\dot{\rho}'_0] \in H^1(\mathfrak{h}, \mathfrak{g})$. $\square$

Remark 4.2.4. ([CSS14], Remark 3.8) We can interpret propositions 4.2.2 and 4.2.3 in the following way. Deformations of ρ are essentially smooth curves in $\{\eta \in \mathfrak{h}^* \otimes \mathfrak{g} \mid \mathcal{K}(\eta) = 0\}$. The formal tangent space of the algebraic variety $\{\eta \in \mathfrak{h}^* \otimes \mathfrak{g} \mid \mathcal{K}(\eta) = 0\}$ at the point ρ is defined as

$$T_\rho(\mathcal{K}^{-1}(0)) := \{\xi \in \mathfrak{h}^* \otimes \mathfrak{g} \mid d_\rho \mathcal{K}(\xi) = 0\}$$

but as $d_\rho \mathcal{K} = -\delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g})$ by equation (4.6) in proposition 4.2.2, we know that

$$T_\rho(\mathcal{K}^{-1}(0)) = Z^1(\mathfrak{h}, \mathfrak{g}),$$

meaning that the velocities of deformations of ρ at $t = 0$ are the 1-cocycles. If we divide this formal tangent space by the honest tangent space to the G -orbit in the point ρ , we can see by proposition 4.2.3 that

$$T_{[\rho]} \frac{\mathcal{K}^{-1}(0)}{G} = H^1(\mathfrak{h}, \mathfrak{g}).$$

We proved in proposition 4.2.2 that every deformation of ρ gives rise to a 1-cocycle. Conversely, it is not true that every 1-cocycle can be integrated to a genuine deformation. This is because $H^1(\mathfrak{h}, \mathfrak{g})$ is interpreted as *formal* tangent space, and not as honest tangent space. In particular, not every element in $Z^1(\mathfrak{h}, \mathfrak{g})$ integrates to a deformation. This gives rise to a Kuranishi map.

Definition 4.2.5. For any Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, we define the Kuranishi map as

$$\Phi : Z^1(\mathfrak{h}, \mathfrak{g}) \rightarrow H^2(\mathfrak{h}, \mathfrak{g}) : \xi \mapsto \llbracket \xi, \xi \rrbracket + B^2(\mathfrak{h}, \mathfrak{g})$$

for the bracket $\llbracket \cdot, \cdot \rrbracket$ from definition 3.1.13 defined on $C^\bullet(\mathfrak{h}, \mathfrak{g})$.

Proposition 4.2.6. ([CSS14], Proposition 3.9) If $\{\rho_t\}$ is a deformation of a Lie algebra homomorphism $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$, then $\Phi(\dot{\rho}_0) = 0$.

Proof. We can write $\rho_t = \rho + t\dot{\rho}_0 + \frac{t^2}{2}\ddot{\rho}_0 + O(t^3)$ in the vector space $\mathfrak{h}^* \otimes \mathfrak{g}$. Then, because ρ_t is a deformation of ρ , we have for all $t \in [0, 1]$ and $u, v \in \mathfrak{h}$ that

$$\mu(\rho_t(u), \rho_t(v)) - \rho_t(\nu(u, v)) = 0.$$

Substituting the Taylor expansion into this condition gives the equality

$$\begin{aligned} 0 &= \mu(\rho(u), \rho(v)) - \rho(\nu(u, v)) + t(\mu(\dot{\rho}_0(u), \rho(v)) + \mu(\rho(u), \dot{\rho}_0(v)) - \dot{\rho}_0(\nu(u, v))) \\ &\quad + t^2 \left(\frac{1}{2}\mu(\ddot{\rho}_0(u), \rho(v)) + \mu(\dot{\rho}_0(u), \dot{\rho}_0(v)) + \frac{1}{2}\mu(\rho(u), \ddot{\rho}_0(v)) - \frac{1}{2}\ddot{\rho}_0(\nu(u, v)) \right) + O(t^3) \end{aligned}$$

as Taylor series. In particular, the coefficient of t^2 on the right-hand side should be zero, which gives the condition that

$$\mu(\dot{\rho}_0(u), \dot{\rho}_0(v)) = \frac{1}{2}(-\mu(\rho(u), \ddot{\rho}_0(v)) + \mu(\rho(v), \ddot{\rho}_0(u)) + \ddot{\rho}_0(\nu(u, v))).$$

Now we can recognize the left-hand side as $-\frac{1}{2}[\dot{\rho}_0, \dot{\rho}_0](u, v)$, and the right-hand side as $\delta_\rho \ddot{\rho}_0(u, v)$. Therefore, it follows that

$$[\dot{\rho}_0, \dot{\rho}_0] = -\delta_\rho \ddot{\rho}_0$$

and thus

$$\Phi(\dot{\rho}_0) = 0 \in H^2(\mathfrak{h}, \mathfrak{g}).$$

□

Corollary 4.2.7. *Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism and Φ the Kuranishi map from definition 4.2.5. If*

$$\Phi(\xi) \neq 0 \text{ for some } \xi \in Z^1(\mathfrak{h}, \mathfrak{g}),$$

then ξ is obstructed, i.e., there does not exist a deformation ρ_t of ρ such that $\dot{\rho}_0 = \xi$.

4.3 Deformations of Lie subalgebras

This section is inspired by section 3.3 in [CSS14]. However, as we make the extra assumption for the Lie subalgebra $\mathfrak{h}$ having a complement $\mathfrak{h}^c$ that is a Lie subalgebra too, the contents really differ from the ones in [CSS14]. In that sense, the results in this section are our own contributions, but are implied by the more general results made in [CSS14], after making the identification $\mathfrak{g}/\mathfrak{h} \cong \mathfrak{h}^c$.

Definition 4.3.1. Let $(\mathfrak{g}, \mu)$ be a Lie algebra and $\mathfrak{h}$ a k -dimensional Lie subalgebra with a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. Recall the set $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$ from remark 2.3.5. A **deformation** of $\mathfrak{h}$ is a smooth family $\{\mathfrak{h}_t \in \text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c) | t \in [0, 1]\}$ of Lie subalgebras of $(\mathfrak{g}, \mu)$ such that $\mathfrak{h}_0 = \mathfrak{h}$. Let G be the unique simply connected Lie group integrating $\mathfrak{g}$, and let K be the unique connected Lie group integrating $\mathfrak{h}^c$ that is a subgroup of G . Then two deformations $\mathfrak{h}_t$ and $\mathfrak{h}'_t$ are called **equivalent** if there exists a smooth curve $l_t \in K$ such that

$$\mathfrak{h}'_t = \text{Ad}_{l_t}(\mathfrak{h}_t)$$

for all $t \in [0, 1]$ and $l_0 = e$.

Remark 4.3.2. In the light of the correspondence in remark 2.3.5, we see that these deformations $\mathfrak{h}_t$ can be identified with smooth curves $\eta_t \in \mathfrak{h}^* \otimes \mathfrak{h}^c$ with $\eta_0 = 0$, for which we then can write $\mathfrak{h}_t$ as the graph $\mathfrak{h}_t = \Gamma(\eta_t)$ for all t . Under this correspondence, two such curves η_t and η'_t are equivalent if there exists a smooth curve $l_t \in K$ such that for all $h \in \mathfrak{h}$ holds that

$$\eta'_t(p_{\mathfrak{h}}(\text{Ad}_{l_t}(h))) = p_{\mathfrak{h}^c}(\text{Ad}_{l_t}(h)) + \text{Ad}_{l_t}(\eta_t(h))$$

as in equation (2.15), where we used linearity of $p_{\mathfrak{h}^c}$ and Ad_{l_t} , and the fact that $\text{Ad}_{l_t}(\eta_t(h)) \in \mathfrak{h}^c$.

Proposition 4.3.3. *Let $\mathfrak{h}$ be a Lie subalgebra of a Lie algebra $(\mathfrak{g}, \mu)$, which admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. If $\mathfrak{h}_t = \Gamma(\eta_t)$ is a deformation of $\mathfrak{h}$, then $\dot{\eta}_0 \in Z^1(\mathfrak{h}, \mathfrak{h}^c)$.*

Proof. Let $\mathfrak{h}_t$ be a deformation of $\mathfrak{h}$. Then we know that $\mathfrak{h}_t = \Gamma(\eta_t)$ for some smooth curve $\eta_t \in \mathfrak{h}^* \otimes \mathfrak{h}^c$ with $\eta_0 = 0$. If we take the map $\mathcal{L} : \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ from definition 2.3.7, then by lemma 2.3.8 we know for all $t \in [0, 1]$ and $u, v \in \mathfrak{h}$ that

$$0 = \mathcal{L}(\eta_t)(u, v) = p_{\mathfrak{h}^c}(\mu(u, \eta_t(v))) + p_{\mathfrak{h}^c}(\mu(\eta_t(u), v)) + p_{\mathfrak{h}^c}(\mu(\eta_t(u), \eta_t(v))) \\ - \eta_t(\mu(u, v)) - \eta_t(p_{\mathfrak{h}}(\mu(u, \eta_t(v)))) - \eta_t(p_{\mathfrak{h}}(\mu(\eta_t(u), v))).$$

Differentiating the former equation with respect to t , evaluated at $t = 0$ gives the equality

$$0 = \frac{d}{dt} \Big|_{t=0} \mathcal{L}(\eta_t)(u, v) = d_0 \mathcal{L}(\dot{\eta}_0)(u, v) \\ = p_{\mathfrak{h}^c}(\mu(u, \dot{\eta}_0(v))) + p_{\mathfrak{h}^c}(\mu(\dot{\eta}_0(u), v)) + p_{\mathfrak{h}^c}(\mu(\eta_0(u), \dot{\eta}_0(v))) + p_{\mathfrak{h}^c}(\mu(\dot{\eta}_0(u), \eta_0(v))) - \dot{\eta}_0(\mu(u, v)) \\ - \dot{\eta}_0(p_{\mathfrak{h}}(\mu(u, \eta_0(v)))) - \eta_0(p_{\mathfrak{h}}(\mu(u, \dot{\eta}_0(v)))) - \dot{\eta}_0(p_{\mathfrak{h}}(\mu(\eta_0(u), v))) - \eta_0(p_{\mathfrak{h}}(\mu(\dot{\eta}_0(u), v))) \\ = p_{\mathfrak{h}^c}(\mu(u, \dot{\eta}_0(v))) + p_{\mathfrak{h}^c}(\mu(\dot{\eta}_0(u), v)) - \dot{\eta}_0(\mu(u, v)) \\ = \delta_{\mathfrak{h}, \mathfrak{h}^c} \dot{\eta}_0(u, v) \tag{4.7}$$

where we used that $\eta_0 = 0$ in the third equation. This proves that $\delta_{\mathfrak{h}, \mathfrak{h}^c} \dot{\eta}_0 = 0$, from which the assertion follows. $\square$

Proposition 4.3.4. *If $\mathfrak{h}_t = \Gamma(\eta_t)$ and $\mathfrak{h}'_t = \Gamma(\eta'_t)$ are two equivalent deformations of a Lie subalgebra $\mathfrak{h}$ with as a complement a Lie subalgebra $\mathfrak{h}^c$, then $[\dot{\eta}_0] = [\dot{\eta}'_0] \in H^1(\mathfrak{h}, \mathfrak{h}^c)$.*

Proof. We know by 4.3.2 that $\mathfrak{h}_t$ and $\mathfrak{h}'_t$ are equivalent deformations of $\mathfrak{h}$ if and only if for all $h \in \mathfrak{h}$ holds that

$$\eta'_t(p_{\mathfrak{h}}(\text{Ad}_{l_t}(h))) = p_{\mathfrak{h}^c}(\text{Ad}_{l_t}(h)) + \text{Ad}_{l_t}(\eta_t(h))$$

for some curve $l_t \in K$ with $l_0 = e$. Differentiating this equation with respect to t , and evaluating at $t = 0$ gives

$$\dot{\eta}'_0(p_{\mathfrak{h}}(\text{Ad}_{l_0}(h))) + \eta'_0(p_{\mathfrak{h}}(\text{ad}_{\mathfrak{g}}(\dot{l}_0)(h))) = p_{\mathfrak{h}^c}(\text{ad}_{\mathfrak{g}}(\dot{l}_0)(h)) + \text{ad}_{\mathfrak{g}}(\dot{l}_0)(\eta_0(h)) + \text{Ad}_{l_0}(\dot{\eta}_0(h))$$

and after using that $l_0 = e$ and $\eta_0 = 0$, this becomes

$$\dot{\eta}'_0(h) = p_{\mathfrak{h}^c}(\mu(\dot{l}_0, h)) + \dot{\eta}_0(h) = \delta_{\mathfrak{h}, \mathfrak{h}^c} \dot{l}_0(h) + \dot{\eta}_0(h).$$

This implies that

$$\dot{\eta}'_0 = \dot{\eta}_0 + \delta_{\mathfrak{h}, \mathfrak{h}^c} \dot{l}_0 \tag{4.8}$$

which proves the statement. $\square$

Remark 4.3.5. Propositions 4.3.3 and 4.3.4 can be interpreted in the following way. Deformations of $\mathfrak{h}$ can be identified with smooth curves in $\{\eta \in \mathfrak{h}^* \otimes \mathfrak{h}^c | \mathcal{L}(\eta) = 0\}$. The formal tangent space of this algebraic variety at the point 0 is defined as

$$T_0(\mathcal{L}^{-1}(0)) := \{\xi \in \mathfrak{h}^* \otimes \mathfrak{h}^c | d_0 \mathcal{L}(\xi) = 0\}.$$

We know from (4.7) that $d_0 \mathcal{L} = \delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c)$ and thus

$$T_0(\mathcal{L}^{-1}(0)) = Z^1(\mathfrak{h}, \mathfrak{h}^c)$$

meaning that the velocities of these η_t at $t = 0$ are the 1-cocycles. If we divide this formal tangent space by the honest tangent space to the K -orbit in the point 0, we can see by proposition 4.3.4 that

$$T_{[0]} \frac{\mathcal{L}^{-1}(0)}{K} = H^1(\mathfrak{h}, \mathfrak{h}^c)$$

as formal tangent space.

Again, we can construct a Kuranishi map to detect elements in $Z^1(\mathfrak{h}, \mathfrak{h}^c)$ that fail to be integrated to a genuine deformation.

Definition 4.3.6. For a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ which admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra, we define the Kuranishi map as

$$\Phi : Z^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow H^2(\mathfrak{h}, \mathfrak{h}^c) : \eta \mapsto \llbracket \eta, \eta \rrbracket + B^2(\mathfrak{h}, \mathfrak{h}^c)$$

for the bracket $\llbracket \cdot, \cdot \rrbracket$ on $C^\bullet(\mathfrak{h}, \mathfrak{h}^c)$ from equation (3.31).

Proposition 4.3.7. If $\mathfrak{h}_t = \Gamma(\eta_t)$ is a deformation of a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ that has a complement $\mathfrak{h}^c$ that is a Lie subalgebra too, then $\Phi(\dot{\eta}_0) = 0$.

Proof. As $\mathfrak{h}_t = \Gamma(\eta_t)$ is a deformation of $\mathfrak{h}$, we know by lemma 2.3.8 for all $t \in [0, 1]$ and $u, v \in \mathfrak{h}$ that

$$\begin{aligned} 0 = \mathcal{L}(\eta_t)(u, v) = & p_{\mathfrak{h}^c}(\mu(u, \eta_t(v))) + p_{\mathfrak{h}^c}(\mu(\eta_t(u), v)) + p_{\mathfrak{h}^c}(\mu(\eta_t(u), \eta_t(v))) \\ & - \eta_t(\mu(u, v)) - \eta_t(p_{\mathfrak{h}}(\mu(u, \eta_t(v)))) - \eta_t(p_{\mathfrak{h}}(\mu(\eta_t(u), v))). \end{aligned} \quad (4.9)$$

Since $\mathfrak{h}^* \otimes \mathfrak{h}^c$ is a vector space and η_t depends smoothly on t , we can write η_t as a Taylor expansion around $t = 0$:

$$\eta_t = t\dot{\eta}_0 + \frac{t^2}{2}\ddot{\eta}_0 + O(t^3).$$

Using this in equation (4.9), then we get from $0 = \mathcal{L}(t\dot{\eta}_0 + \frac{t^2}{2}\ddot{\eta}_0 + O(t^3))(u, v)$ a condition on every coefficient of the Taylor series in t . As the coefficient of t^0 on the left- and right-hand side of the equation should be equal, we get that

$$0 = \mathcal{L}(0)(u, v)$$

which corresponds to the fact that $\mathfrak{h} = \Gamma(0)$ is a Lie subalgebra of $\mathfrak{g}$ transverse to $\mathfrak{h}$. Similarly, the coefficient of t^1 on the right-hand side should be zero, so

$$0 = d_0 \mathcal{L}(\dot{\eta}_0)(u, v)$$

which is just the statement of proposition 4.3.3. Lastly, the coefficient of t^2 should be zero, which gives

$$\begin{aligned} 0 = & \frac{1}{2} p_{\mathfrak{h}^c}(\mu(u, \ddot{\eta}_0(v))) + \frac{1}{2} p_{\mathfrak{h}^c}(\mu(\ddot{\eta}_0(u), v)) + p_{\mathfrak{h}^c}(\mu(\dot{\eta}_0(u), \dot{\eta}_0(v))) \\ & - \frac{1}{2} \ddot{\eta}_0(\mu(u, v)) - \dot{\eta}_0(p_{\mathfrak{h}}(\mu(u, \dot{\eta}_0(v)))) - \dot{\eta}_0(p_{\mathfrak{h}}(\mu(\dot{\eta}_0(u), v))) \end{aligned}$$

$$= \frac{1}{2} \delta_{\mathfrak{h}, \mathfrak{h}^c} \ddot{\eta}_0(u, v) + \frac{1}{2} \llbracket \dot{\eta}_0, \dot{\eta}_0 \rrbracket(u, v)$$

As $u, v \in \mathfrak{h}$ were arbitrary, it follows that

$$\llbracket \dot{\eta}_0, \dot{\eta}_0 \rrbracket = -\delta_{\mathfrak{h}, \mathfrak{h}^c} \ddot{\eta}_0 \in B^2(\mathfrak{h}, \mathfrak{h}^c),$$

proving the statement. $\square$

Corollary 4.3.8. *Let $\mathfrak{h} \subset \mathfrak{g}$ be a Lie subalgebra which admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra and Φ the Kuranishi map from definition 4.3.6. If*

$$\Phi(\xi) \neq 0 \text{ for some } \xi \in Z^1(\mathfrak{h}, \mathfrak{h}^c),$$

then ξ is obstructed, i.e., there does not exist a deformation $\mathfrak{h}_t = \Gamma(\eta_t)$ of $\mathfrak{h}$ such that $\dot{\eta}_0 = \xi$.

4.4 DGLAs and Deformations

This section identifies the DGLA theory hidden in the previous three sections. The material in this section is classical and well known in the literature on deformation theory. For example, it can be found in [DM25]. The presentation here is our own, organized to make the connection with the three deformation theories of the preceding sections explicit.

One can see quite a lot of similarities in the deformation theories we worked out in sections 4.1, 4.2 and 4.3. This is not only at a superficial level. What's behind this, is, in each case, a DGLA governing the deformation problem. This DGLA is called the **controlling DGLA** of the deformation problem.

Example 4.4.1. We can identify some controlling DGLAs that we have come across:

1. In the deformation theory of Lie algebras, the controlling DGLA is $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], 0, \llbracket \cdot, \cdot \rrbracket)$, for a vector space $\mathfrak{g}$ and where $\llbracket \cdot, \cdot \rrbracket$ is the Gerstenhaber bracket on $C^\bullet(\mathfrak{g}, \mathfrak{g})[1]$ from definition 3.1.8.
2. In the deformation theory of Lie algebra homomorphisms, the controlling DGLA is $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket)$, for Lie algebras $(\mathfrak{h}, \nu)$ and $(\mathfrak{g}, \mu)$, where δ_0 is the Chevalley-Eilenberg differential on $C^\bullet(\mathfrak{h}, \mathfrak{g})$ and $\llbracket \cdot, \cdot \rrbracket$ is the graded bracket on $C^\bullet(\mathfrak{h}, \mathfrak{g})$ from definition 3.1.13.
3. For a Lie subalgebra $\mathfrak{h} \subseteq (\mathfrak{g}, \mu)$ that has a complements which is a Lie subalgebra too, the controlling DGLA is $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, \llbracket \cdot, \cdot \rrbracket)$, where $\delta_{\mathfrak{h}, \mathfrak{h}^c}$ is the Chevalley-Eilenberg differential on $C^\bullet(\mathfrak{h}, \mathfrak{h}^c)$ from example 2.4.7 and $\llbracket \cdot, \cdot \rrbracket$ is the bracket on $C^\bullet(\mathfrak{h}, \mathfrak{h}^c)$ defined as in 3.2.25.

Let $(L^\bullet, d, [\cdot, \cdot])$ be a DGLA, and denote by $\mathcal{MC}(L^\bullet) = \mathcal{MC}(L^\bullet, d, [\cdot, \cdot])$ its Maurer-Cartan set, i.e., the elements $x \in L^1$ for which $dx + \frac{1}{2}[x, x] = 0$.

Definition 4.4.2. For an element $x \in \mathcal{MC}(L^\bullet)$ of a DGLA $(L^\bullet, d, [\cdot, \cdot])$, a **deformation** of x is a smooth curve $x_t \in \mathcal{MC}(L^\bullet)$ such that $x_0 = x$.

Definition 4.4.3. If we have a DGLA $(L^\bullet, d, [\cdot, \cdot])$, then $(L^0, [\cdot, \cdot]_{L^0})$ is an honest Lie algebra. Hence, we can integrate it to a connected Lie group G . As $[\cdot, \cdot]$ is of degree 0, for all $i \in \mathbb{N}$, we have that L^0 defines a Lie algebra action on L^i

$$L^0 \times L^i \rightarrow L^i : (x, \varphi) \mapsto [x, \varphi], \quad (4.10)$$

which integrates to a Lie group action of G on L^i . For $i = 2$, this action, the **adjoint action**, is the Lie group action of G we take on L^i . On L^1 , we take a different action. From definition 6.3.3 in [Man22], we know that the (non-linear) action

$$L^0 \times L^1 \rightarrow L^1 : (x, \varphi) \mapsto [x, \varphi] - dx \quad (4.11)$$

integrates to a Lie group action $G \times L^1 \rightarrow L^1$. This action is called the **gauge action** of G on L^1 .

Remark 4.4.4. Manetti's construction of the gauge action via the exponential formula requires the DGLA $(L^\bullet, d, [\cdot, \cdot])$ to be nilpotent, in order for the exponent map to be convergent and thus the product in definition 6.3.3 to be well defined. In our setting, the controlling DGLAs are not nilpotent in general. However, the components of these DGLAs are finitedimensional $\mathbb{R}$ -vector spaces. In that case, the exponential is always convergent, so that the gauge action is still well defined.

Remark 4.4.5. One can check that on each of the three objects of interest, the action we defined on these objects in chapter 2 coincides with the action defined in definition 4.4.3 induced by the corresponding DGLA. One thing that does differ, is that for Lie algebras on a vector space $\mathfrak{g}$, we took the Lie group $\mathrm{GL}(\mathfrak{g})$, which is not connected, acting on $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$. However, one could have taken a connected Lie group acting on $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, which would have led to the same results. Hence, this is not really a problem.

Definition 4.4.6. Two deformations x_t and y_t of x are called **equivalent** if there exists a smooth family $g_t \in G$ such that $y_t = g_t \cdot x_t$ and $g_0 = e$.

One can see the former three deformation theories in sections 4.1, 4.2 and 4.3 as examples of the following general deformation theory for a given DGLA. Recall the twisted differential of definition 3.2.10.

Proposition 4.4.7. Let $(L^\bullet, d, [\cdot, \cdot])$ be a DGLA and $x \in \mathcal{MC}(L^\bullet)$. If x_t is a deformation of x , then $d^x \dot{x}_0 = 0$. Therefore, $\dot{x}_0 \in Z^1(L^\bullet, d^x)$.

Proof. As x_t is a deformation of $x \in \mathcal{MC}(L^\bullet)$, we know that

$$dx_t + \frac{1}{2}[x_t, x_t] = 0 \text{ for all } t \in [0, 1]$$

Differentiating this equation with respect to t , and evaluating in $t = 0$ gives

$$0 = \frac{d}{dt}|_{t=0}(dx_t + \frac{1}{2}[x_t, x_t]) = d\dot{x}_0 + \frac{1}{2}[x, \dot{x}_0] + \frac{1}{2}[\dot{x}_0, x] = d\dot{x}_0 + [x, \dot{x}_0] = d^x \dot{x}_0$$

proving the statement. □

Remark 4.4.8. One should compare this statement to

1. proposition 4.1.2, for DGLA $(C^\bullet(\mathfrak{g}, \mathfrak{g})[1], 0, [\cdot, \cdot])$ from the first example in 4.4.1 with Maurer-Cartan element μ .
2. proposition 4.2.2, for DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, [\cdot, \cdot])$ from the second example in 4.4.1 with Maurer-Cartan element ρ .
3. proposition 4.3.3, for DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, [\cdot, \cdot])$ from the third example in 4.4.1 with Maurer-Cartan element 0.

Then it becomes clear that propositions 4.1.2, 4.2.2 and 4.3.3 are examples of proposition 4.4.7.

Proposition 4.4.9. *If x_t and y_t are equivalent deformations of $x \in \mathcal{MC}(L^\bullet)$, then $\dot{y}_0 = \dot{x}_0 + dz$ for some $z \in L^0$. It follows that $[\dot{x}_0] = [\dot{y}_0] \in H^1(L^\bullet, d^x)$.*

Proof. If x_t and y_t are equivalent deformations, then

$$y_t = g_t \cdot x_t \text{ for some curve } g_t \in G \text{ with } g_0 = e.$$

Differentiating this equation with respect to t , and evaluating in $t = 0$ gives

$$\dot{y}_0 = \left. \frac{d}{dt} \right|_{t=0} y_t = \left. \frac{d}{dt} \right|_{t=0} (g_t \cdot x_t) = \dot{x}_0 - d\dot{g}_0 + [\dot{g}_0, x] = \dot{x}_0 - d\dot{g}_0 - [x, \dot{g}_0] = \dot{x}_0 - d^x \dot{g}_0$$

□

Remark 4.4.10. Analogously to remark 4.4.8, we can see that propositions 4.1.4, 4.2.3 and 4.3.4 are examples of proposition 4.4.9.

Remark 4.4.11. One can interpret the former two results in the following way. The formal tangent space to $\mathcal{MC}(L^\bullet)$ at the point x is equal to

$$T_x \mathcal{MC}(L^\bullet) = Z^1(L^\bullet, d^x).$$

By dividing by the genuine tangent space of the G orbit through x in x , we get that

$$T_{[x]} \frac{\mathcal{MC}(L^\bullet)}{G} = H^1(L^\bullet, d^x)$$

as formal tangent space.

We can define a Kuranishi map to detect elements in $Z^1(L^\bullet, d^x)$ that cannot be integrated to genuine deformations.

Definition 4.4.12. For any DGLA $(L^\bullet, d, [\cdot, \cdot])$ and $x \in \mathcal{MC}(L^\bullet)$, we define the Kuranishi map as

$$\Phi : Z^1(L^\bullet, d^x) \rightarrow H^2(L^\bullet, d^x) : y \mapsto [y, y] + B^2(L^\bullet, d^x) \quad (4.12)$$

We check definition 4.4.12 is well defined. For $y \in Z^1(L^\bullet, d^x)$ we must show that $[y, y] \in Z^2(L^\bullet, d^x)$, that is, $d^x[y, y] = 0$. Since x is a Maurer–Cartan element, the twisted differential $d^x = d + [x, \cdot]$ is a graded derivation of the bracket. Hence

$$d^x[y, y] = [d^x y, y] - [y, d^x y] = 0,$$

because $d^x y = 0$, so that $[y, y] \in Z^2(L^\bullet, d^x)$.

Proposition 4.4.13. *If x_t is a deformation of $x \in \mathcal{MC}(L^\bullet)$ then $\Phi(\dot{x}_0) = 0$.*

Proof. As x_t is a deformation, we know that

$$dx_t + \frac{1}{2}[x_t, x_t] = 0$$

for all $t \in [0, 1]$. In the vector space L^1 , we can write $x_t = x + t\dot{x}_0 + \frac{1}{2}t^2\ddot{x}_0 + O(t^3)$ as Taylor expansion around $t = 0$. If we substitute this Taylor expansion in the Maurer-Cartan equation, we get that

$$0 = d(x + t\dot{x}_0 + \frac{1}{2}t^2\ddot{x}_0 + O(t^3)) + \frac{1}{2}[x + t\dot{x}_0 + \frac{1}{2}t^2\ddot{x}_0 + O(t^3), x + t\dot{x}_0 + \frac{1}{2}t^2\ddot{x}_0 + O(t^3)].$$

The equation, to 0'th order in t , states that $x \in \mathcal{MC}(L^\bullet)$. To first order in t , we recover proposition 4.4.7. To second order in t , it gives the condition

$$0 = \frac{1}{2}d\ddot{x}_0 + \frac{1}{4}[\ddot{x}_0, x] + \frac{1}{2}[\dot{x}_0, \dot{x}_0] + \frac{1}{4}[x, \ddot{x}_0].$$

By using graded skew symmetry of the graded Lie bracket $[\cdot, \cdot]$ and by multiplying by 2, we get that

$$d\ddot{x}_0 + [x, \ddot{x}_0] + [\dot{x}_0, \dot{x}_0] = 0$$

or equivalently

$$[\dot{x}_0, \dot{x}_0] = -d^x\ddot{x}_0 \in B^2(L^\bullet, d^x)$$

and thus $\Phi(\dot{x}_0) = 0$. □

Corollary 4.4.14. *Let $(L^\bullet, d, [\cdot, \cdot])$ be a DGLA, $x \in \mathcal{MC}(L^\bullet)$, and Φ the Kuranishi map. If $\Phi(\xi) \neq 0$ for some $\xi \in Z^1(L^\bullet, d^x)$, then ξ is obstructed: there is no deformation x_t of x with $\dot{x}_0 = \xi$.*

Remark 4.4.15. Just as in remark 4.4.8, the Kuranishi maps of definitions 4.1.6, 4.2.5 and 4.3.6, together with the propositions stating that Φ annihilates infinitesimal deformations, are the instances of the abstract Kuranishi map and proposition 4.4.13 for the three controlling DGLAs of Example 4.4.1.

Taken together, the results of this section show that for a DGLA $(L^\bullet, d, [\cdot, \cdot])$ and Maurer-Cartan element x , $H^1(L^\bullet, d^x)$ is the space of infinitesimal deformations modulo equivalence and $H^2(L^\bullet, d^x)$ contains the obstructions to integrating them. Sections 4.1, 4.2 and 4.3 are the three instances of this principle.

Chapter 5

Geometric Tools

Now, we take a step back from the cohomological and DGLA-based deformation theory we developed so far. In this chapter, we will introduce some geometric definitions that for the reader may not appear to be related to the DGLAs or deformation theory. With that setup, we shall prove some technical results, which will become essential in the next chapter: It will become clear that the structures introduced previously can be naturally applied into the geometric setup developed here, allowing us to exploit these results in the deformation theory of Lie algebras, Lie algebra homomorphisms and Lie subalgebras. This chapter is based on chapter 4 in [CSS14], although we give more rigorous and extensive proofs. In the proofs, we will also make use of some theorems from [Lee12].

5.1 Openness of orbits

Definition 5.1.1. A **vector bundle** of rank $n \in \mathbb{N}$ consists of manifolds E and B , and a smooth surjection $\pi : E \rightarrow B$ such that

- $E_p := \pi^{-1}(p)$ is an n -dimensional vector space for all $p \in B$
- For all $p \in B$, there exists an open neighborhood U of p and a diffeomorphism $\psi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ so that the following diagram commutes

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\psi} & U \times \mathbb{R}^n \\ & \searrow \pi & \swarrow \pi_1 \\ & U & \end{array}$$

where for every $x \in U$ the map $v \mapsto \psi^{-1}(x, v)$ is a linear isomorphism. Here, $\pi_1 : U \times \mathbb{R}^n \rightarrow U$ denotes the projection onto B . Such a map ψ is called a local trivialization.

Example 5.1.2. For any manifold B and any $n \in \mathbb{N}$, $E := B \times \mathbb{R}^n$ is a vector bundle of rank n . Such a vector bundle is called a trivial vector bundle.

Example 5.1.3. For any manifold B of dimension n , the tangent bundle

$$TB := \bigsqcup_{p \in B} T_p B$$

is a vector bundle of rank n .

Definition 5.1.4. A **section** is a smooth map $\sigma : B \rightarrow E$ so that $\pi \circ \sigma = \text{Id}_B$.

Example 5.1.5. The section $\sigma : B \rightarrow E$ that sends every q to $0_q \in E_q$ is called the zero section

Remark 5.1.6. If we refer to a zero of σ , we mean an element $p \in B$ for which $\sigma(p) = 0_p$. Similarly, we define $\sigma^{-1}(0) = \{p \in B \mid \sigma(p) = 0_p\}$.

In what follows, we take E to be a vector bundle over a manifold M , and G a Lie group acting on E that fixes the zero section $Z : M \rightarrow E$, meaning that

$$G \cdot Z = Z. \quad (5.1)$$

From this follows that M inherits a G -action from E , namely $m|_M : G \times M \rightarrow M$.

Definition 5.1.7. A section $\sigma : M \rightarrow E$ is called **equivariant** if

$$\sigma(g \cdot x) = g \cdot \sigma(x) \quad (5.2)$$

for all $g \in G$ and $x \in M$.

Definition 5.1.8. Let $\sigma : M \rightarrow E$ be an equivariant section. A zero of σ , i.e. an $x \in M$ for which $\sigma(x) = 0_x$, is called **non-degenerate** if the sequence

$$T_e G \xrightarrow{d_e m_x} T_x M \xrightarrow{d_x^{\text{vert}} \sigma} E_x \quad (5.3)$$

is exact, where

- $T_e G$ is the Lie algebra of G .
- $m_x(g) = g \cdot x$ and $d_e m_x : T_e G \rightarrow T_x M$ is the derivative of m_x at e .
- $d_x^{\text{vert}} \sigma : T_x M \rightarrow E_x$ is the vertical derivative of σ at x , defined as the composition

$$T_x M \xrightarrow{d_x \sigma} T_{0_x} E \cong T_x M \oplus E_x \xrightarrow{\pi_{E_x}} E_x$$

where π_{E_x} is the projection onto E_x . We also used that $T_x E_x \cong E_x$ as E_x is a vector space.

An argument we will repeatedly use throughout the proofs in this chapter is that the rank map is lower semicontinuous. This essentially means that locally, the rank of a matrix can only stay the same or increase. We will prove this in the next lemma. This lemma and the proceeding corollary are not in [CSS14], we added it ourselves for clarity and completeness. However, they are well known.

Lemma 5.1.9. The rank map $\text{rank} : \mathbb{R}^{m \times n} \rightarrow \mathbb{N}$ is **lower semicontinuous**, meaning that for every $0 \leq r \leq m, n$, the set $\{A \in \mathbb{R}^{m \times n} \mid \text{rank}(A) \geq r\}$ is open in $\mathbb{R}^{m \times n}$.

Proof. We know for $A \in \mathbb{R}^{m \times n}$ that

$$\text{rank}(A) = \max_{M \text{ minor of } A} \text{rank}(M) \quad (5.4)$$

and thus for any $0 \leq r \leq m, n$ that $\text{rank}(A) \geq r$ if and only if there exists an r -by- r minor M of A that has nonzero determinant. Additionally, we know that every determinant $\det_r : \mathbb{R}^{r \times r} \rightarrow \mathbb{R}$ is continuous. For each subset $I \subseteq \{1, \dots, m\}$ and $J \subseteq \{1, \dots, n\}$ with $|I| = |J| = r$, define the map

$$\pi_{I,J}^r : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{r \times r} : A \mapsto (A_{i_k, j_l})_{k,l=1}^r \quad (5.5)$$

which is linear so continuous. Then we have that

$$\{A \in \mathbb{R}^{m \times n} \mid \text{rank}(A) \geq r\} = \bigcup_{\substack{I, J \\ |I|=|J|=r}} (\det_r \circ \pi_{I,J}^r)^{-1}(\mathbb{R} \setminus \{0\}). \quad (5.6)$$

By continuity, every inverse image $(\det_r \circ \pi_{I,J}^r)^{-1}(\mathbb{R} \setminus \{0\})$ is open, and thus the set $\{A \in \mathbb{R}^{m \times n} \mid \text{rank}(A) \geq r\}$ is open as finite union of open sets. $\square$

Corollary 5.1.10. *For a smooth map $F : M_1 \rightarrow M_2$, the map $p \mapsto \text{rank}(d_p F)$ is lower semicontinuous. For a specific $q \in M_1$, we can find an open neighborhood U around q in M_1 , so that*

$$\text{rank}(d_p F) \geq \text{rank}(d_q F) \quad (5.7)$$

for all $p \in U$.

Theorem 5.1.11. ([CSS14], Proposition 4.3) (Openness of orbits) *Let M be a manifold, E a vector bundle over M and σ be an equivariant section of the vector bundle E .*

$$\begin{array}{c} G \curvearrowright E \\ \sigma \nearrow \downarrow \\ G \curvearrowright M \end{array}$$

Let x be a non-degenerate zero of σ . Then there exist an open neighborhood $U \subseteq M$ of x and a map $h : U \rightarrow G$ so that for all zeros y of σ in U , we have that

$$h(y) \cdot x = y. \quad (5.8)$$

In particular,

$$\sigma^{-1}(0) \cap U = G \cdot x \cap U, \quad (5.9)$$

stating that the zero set of σ and the G -orbit of x coincide in a neighborhood around x in M .

Proof. We will start by proving equation (5.8). From this, equation (5.9) will easily follow.

Step 1:

We want to find an open neighborhood U_G of e in G where m_x has constant rank and $\sigma|_{\text{Im}(m_x)}$ has constant rank on $m_x(U_G)$, so that we can apply the constant rank theorem 4.12 in [Lee12]. As E is a vector bundle, we can locally trivialize E , meaning that locally

$E \cong U_M \times \mathbb{R}^n$ for some neighborhood $U_M \subseteq M$ of x and some $n \in \mathbb{N}$ (one should read this as $U_E \cong U_M \times \mathbb{R}^n$ for some open set $U_E \subseteq E$). This allows us to write $\sigma = (\text{Id}, \sigma_2)$ where now $\sigma_2 : U_M \rightarrow \mathbb{R}^n$. Notice that the zeros of σ_2 are exactly the zeros of σ . We shall consistently apply this strategy of using σ_2 instead of σ throughout this entire section. With this notation, we can easily see that in this neighborhood $d_x^{\text{vert}}\sigma = d_x\sigma_2$. We know from corollary 5.1.10 that the maps $g \mapsto \text{rank}(d_g m_x)$ and $y \mapsto \text{rank}(d_y \sigma_2)$ are lower semicontinuous. Consequently, there exists an open neighborhood $U_G \subseteq G$ around e so that

$$\begin{cases} \text{rank}(d_g m_x) \geq \text{rank}(d_e m_x) \text{ for all } g \in U_G \\ \text{rank}(d_y \sigma_2) \geq \text{rank}(d_x \sigma_2) \text{ for all } y \in U_M \end{cases} \quad (5.10)$$

after shrinking U_M if needed. Additionally, as σ is equivariant, for all $g \in G$, it holds that

$$\sigma(g \cdot x) = g \cdot \sigma(x) = g \cdot 0_x = 0_{g \cdot x} \quad (5.11)$$

and thus for all $g \in U_G$, this reduces to

$$\sigma_2 \circ m_x(g) = 0.$$

This then implies that

$$\text{Im}(d_g m_x) \subseteq \ker(d_{m_x(g)} \sigma_2). \quad (5.12)$$

Furthermore, the assumption that x is a non-degenerate zero of σ implies that

$$\ker(d_x \sigma_2) = \text{Im}(d_e m_x). \quad (5.13)$$

As U_M is a neighborhood of $x = m_x(e)$, and m_x is continuous, $m_x^{-1}(U_M)$ is an open neighborhood of e . Then we can shrink U_G so that it is contained in $m_x^{-1}(U_M)$, while still being a neighborhood of e in G . Take some $g \in U_G$, then we can use all the implications we just found. It follows that we get a sequence

$$\begin{aligned} \text{rank}(d_x \sigma_2) + \text{rank}(d_e m_x) &\leq \text{rank}(d_{m_x(g)} \sigma_2) + \text{rank}(d_g m_x) \\ &\leq \text{rank}(d_{m_x(g)} \sigma_2) + \dim(\ker(d_{m_x(g)} \sigma_2)) = \dim(M) \\ &= \text{rank}(d_x \sigma_2) + \dim(\ker(d_x \sigma_2)) \leq \text{rank}(d_x \sigma_2) + \text{rank}(d_e m_x) \end{aligned}$$

where we used equation (5.10) in the first inequality, (5.12) in the second inequality, and (5.13) in the last inequality. We also used the dimension theorem for linear maps in the two equalities. We can see that the left- and right-hand side of the inequality are the same, so that all inequalities are actually equalities, implying that

$$\text{rank}(d_x \sigma_2) + \text{rank}(d_e m_x) = \text{rank}(d_{m_x(g)} \sigma_2) + \text{rank}(d_g m_x).$$

If we use equation (5.10) again, we can see that

$$\text{rank}(d_x \sigma_2) - \text{rank}(d_{g \cdot x} \sigma_2) = \text{rank}(d_g m_x) - \text{rank}(d_e m_x) \geq 0$$

so that this forces that

$$\text{rank}(d_{m_x(g)} \sigma_2) = \text{rank}(d_x \sigma_2).$$

Similarly we find that

$$\text{rank}(d_g m_x) = \text{rank}(d_e m_x).$$

We have therefore found a neighborhood U_G of e for which m_x has constant rank on U_G , and $\sigma_2|_{\text{Im}(m_x)}$ has constant rank on $m_x(U_G)$.

Step 2:

Now, we can use the constant rank theorem in [Lee12] on m_x and $\sigma_2|_{\text{Im}(m_x)}$. Denote by $m = \dim(G)$, $r = \text{rank}(d_e m_x)$ and $s = \text{rank}(d_x \sigma_2)$. The theorem says that there exist charts $\phi_1 : U_G \rightarrow \mathbb{R}^m \cong \mathbb{R}^r \times \mathbb{R}^{m-r}$ around e and $\phi_2 : U_M \rightarrow \mathbb{R}^{\dim(M)} \cong \mathbb{R}^r \times \mathbb{R}^s$ around x so that locally

$$\begin{cases} m_x : \mathbb{R}^r \times \mathbb{R}^{m-r} \rightarrow \mathbb{R}^r \times \mathbb{R}^s : (y, z) \mapsto (y, 0) \\ \sigma_2 : \mathbb{R}^r \times \mathbb{R}^s \rightarrow \mathbb{R}^n : (y, w) \mapsto (0, w). \end{cases}$$

so that in particular

$$\sigma_2(y, 0) = 0. \quad (5.14)$$

Here, one should read the maps m_x and σ_2 in the charts ϕ_1 and ϕ_2 as $\phi_2 \circ m_x \circ \phi_1^{-1}$ and $\sigma_2 \circ \phi_2^{-1}$ respectively.

Step 3:

Now we will apply the implicit function theorem C.40 in [Lee12]. It's clear that the s by s matrix $\left(\frac{\partial(\sigma_2)_i}{\partial w_j} \right)$ for $r+1 \leq i \leq r+s$ and $1 \leq j \leq s$ is equal to $\text{Id}_{\mathbb{R}^s}$ in any point. Since this matrix is in particular non-singular in $(0, 0)$ and as $\sigma_2(0, 0) = 0$, the implicit function theorem C.40 in [Lee12] says that there exist neighborhoods $W_1 \subseteq \mathbb{R}^r$ of 0, $W_2 \subseteq \mathbb{R}^s$ of 0 and a smooth function $F : W_1 \rightarrow W_2$ so that

$$\sigma_2(y, w) = 0 \text{ if and only if } w = F(y). \quad (5.15)$$

As $x \in U_M$ corresponds to $(0, 0) \in \mathbb{R}^r \times \mathbb{R}^s$, and W_1 and W_2 are open neighborhoods of 0, it follows that we can shrink U_M if needed so that it is contained in $W_1 \times W_2$. Now we claim that for $p = (y, w) \in U_M$,

$$\sigma_2(y, w) = 0 \text{ if and only if } w = 0.$$

We know that for $p = (y, w) \in U_M$ holds that

$$\text{if } w = 0, \text{ then } \sigma_2(p) = \sigma_2(y, 0) = 0.$$

Conversely, we can fill in $w = 0$ in equation (5.15) and find that from $\sigma_2(y, 0) = 0$ follows that $F(y) = 0$ independently of y , so that F is the zero map. Hence,

$$\text{if } \sigma_2(y, w) = 0, \text{ then } w = F(y) = 0.$$

This proves that for $p = (y, w) \in U_M$,

$$\sigma_2(y, w) = 0 \text{ if and only if } w = 0.$$

Step 4:

Now we will define h in the charts, and then transfer it back to the manifolds. In the

charts, we can define $h : \mathbb{R}^r \times \mathbb{R}^s \rightarrow \mathbb{R}^r \times \mathbb{R}^{m-r} : (y, w) \mapsto (y, 0)$, which is clearly smooth. Whenever we have a $p = (y, w) \in \mathbb{R}^r \times \mathbb{R}^s$ with $\sigma_2(y, w) = 0$, it follows by the previous condition that $w = 0$, so that $p = (y, 0)$. Then it follows that

$$m_x \circ h(p) = m_x(h(y, 0)) = m_x(y, 0) = (y, 0) = p.$$

If we transfer this back into the initial manifolds through the charts, we get that for $p \in U_M$ with $\sigma_2(p) = 0$, and thus $\sigma(p) = 0_p$, it follows that

$$m_x(h(p)) = p$$

or equivalently

$$h(p) \cdot x = p$$

which proves equation (5.8).

Step 5:

This means in particular that $\sigma^{-1}(0) \cap U_M \subseteq G \cdot x \cap U_M$. From equation (5.11), the other inclusion follows, so that

$$\sigma^{-1}(0) \cap U_M = G \cdot x \cap U_M$$

meaning that the zero set of σ and the G -orbit of x coincide in a neighborhood of x . This shows equation (5.9). $\square$

5.2 Stability of zeros

Theorem 5.2.1. ([CSS14], Proposition 4.4)(Stability of zeros) *Let E and F be vector bundles over M . Let $\sigma : M \rightarrow E$ be a section and $\phi \in \Gamma(\text{Hom}(E, F))$ be a vector bundle map that satisfies $\phi \circ \sigma = 0$.*

$$\begin{array}{ccc} E & \xrightarrow{\phi} & F \\ \sigma \uparrow \downarrow & \swarrow & \\ M & & \end{array}$$

Additionally, suppose that $x \in M$ is a zero of σ for which the sequence

$$T_x M \xrightarrow{d_x^{\text{vert}} \sigma} E_x \xrightarrow{\phi_x} F_x \quad (5.16)$$

is exact. Then the following statements hold:

1. The set $\sigma^{-1}(0)$ is locally a manifold around x of dimension $\dim(\ker(d_x^{\text{vert}} \sigma))$.
2. If τ is another section of E which is $\mathcal{C}^0$ -close to σ , and $\psi \in \Gamma(\text{Hom}(E, F))$ is another vector bundle map which is $\mathcal{C}^0$ -close to ϕ such that $\psi \circ \tau = 0$, then there exists a $y \in M$ close to x for which $\tau(y) = 0$.
3. Moreover, if τ from statement 2 is a section of E that is $\mathcal{C}^1$ -close to σ , then $\tau^{-1}(0)$ is also locally a manifold around y of the same dimension as $\sigma^{-1}(0)$.

Proof. Step 1:

We start by proving the first statement. We want to apply the transversality theorem 6.30 in [Lee12]. To that end, we first look for a suitable manifold transverse to σ_2 . We first trivialize E and F , i.e., let locally $E \cong U_M \times \mathbb{R}^n$ and $F \cong U_M \times \mathbb{R}^m$ for some neighborhood $U_M \subseteq M$ of x . Like before, we then obtain a map $\sigma_2 : U_M \rightarrow \mathbb{R}^n$ for which $d_x \sigma_2 = d_x^{\text{vert}} \sigma$. We also now have $\phi : U_1 \rightarrow \text{Hom}(\mathbb{R}^n, \mathbb{R}^m)$. Let $A := \text{Im}(d_x \sigma_2)$ and choose a linear complement B to A in $\mathbb{R}^n$, so that now

$$\mathbb{R}^n = A \oplus B.$$

Step 2:

In principle, now we can apply the transversality theorem 6.30 in [Lee12] to the map σ_2 and the transversal submanifold B of $\mathbb{R}^n$. However, we first want to find a neighborhood U_M of x such that $\sigma_2^{-1}(0) = \sigma_2^{-1}(B)$ in U_M . From the assumption that $\text{Im}(d_x^{\text{vert}} \sigma) = \ker(\phi_x)$ follows that $\phi_x|_B$ is injective. Indeed, for any $v \in B$, if $\phi_x(v) = 0$, then $v \in \ker(\phi_x) = \text{Im}(d_x^{\text{vert}} \sigma) = A$, meaning that $v \in A \cap B = \{0\}$. Since $\phi_x|_B$ is a linear map, this means exactly that it is injective. In particular, this means that it has maximal rank $\dim(B)$. By lower semicontinuity of the map $y \mapsto \text{rank}(\phi_y|_B)$, after possibly shrinking U_M , we have that

$$\text{rank}(\phi_y|_B) \geq \text{rank}(\phi_x|_B) \text{ for all } y \in U_M.$$

However, as $\text{rank}(\phi_x|_B)$ was already maximal,

$$\text{rank}(\phi_y|_B) = \text{rank}(\phi_x|_B) = n \text{ for all } y \in U_M$$

proving that $\phi_y|_B$ is injective for $y \in U_M$. Furthermore, for $y \in U_M$, if $\sigma_2(y) \in B$, then from the assumption that $\phi \circ \sigma = 0$ follows that

$$\phi_y|_B(\sigma_2(y)) = 0.$$

Injectivity of $\phi_y|_B$ then implies that $\sigma_2(y) = 0$, and thus $y \in \sigma_2^{-1}(0)$. Hence, in U_M , we see that $\sigma_2^{-1}(0) = \sigma_2^{-1}(B)$ because the inclusion $\sigma_2^{-1}(0) \subseteq \sigma_2^{-1}(B)$ is trivial.

Step 3:

Now we will apply the transversality theorem in [Lee12], and use the fact that $\sigma_2^{-1}(0) = \sigma_2^{-1}(B)$ in U_M to prove statement 1. As σ_2 is transverse to B , by theorem 6.30a in [Lee12], $\sigma_2^{-1}(B)$ is locally a submanifold of M of codimension equal to $\dim(\text{Im}(d_x \sigma_2))$. This means that its dimension is $n - \dim(\text{Im}(d_x \sigma_2)) = \dim(\ker(d_x \sigma_2)) = \dim(\ker(d_x^{\text{vert}} \sigma))$. Because in U_M , $\sigma_2^{-1}(0) = \sigma_2^{-1}(B)$, it follows that here, $\sigma_2^{-1}(0)$ is a submanifold of dimension equal to $\dim(\ker(d_x^{\text{vert}} \sigma))$. This proves statement 1.

Step 4:

To prove statement 2, we keep the same setup, but have an additional $\tau_2 : U_M \rightarrow \mathbb{R}^n$, and a $\psi : U_M \rightarrow \text{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ that is $\mathcal{C}^0$ -close to ϕ . We first want to find a neighborhood U_M of x in M such that for all $y \in U_M$, $\psi_y|_B$ is injective. As $\phi_x|_B$ is injective and ψ is $\mathcal{C}^0$ -close to ϕ , it follows that $\psi_x|_B$ is also injective. To see this, we take the Euclidean norm on $\mathbb{R}^m$, and let

$$\varepsilon := \min_{v \in S^{n-1}} \|\phi_x(v)\|.$$

where the minimum is well defined as the unit $n - 1$ -sphere S^{n-1} is compact and nonzero by injectivity. As ψ is $\mathcal{C}^0$ -close to ϕ , we can let $\|\phi_x - \psi_x\| < \frac{\varepsilon}{2}$ where the norm is the supremum norm on $\text{Hom}(\mathbb{R}^n, \mathbb{R}^m)$, so that then

$$\sup_{v \in S^{n-1}} \|(\phi_x - \psi_x)(v)\| < \frac{\varepsilon}{2}.$$

It follows that for any element in the unit $n - 1$ -sphere $v \in S^{n-1}$

$$\|\psi_x(v)\| \geq \|\phi_x(v)\| - \|\phi(v) - \psi(v)\| > \varepsilon - \frac{\varepsilon}{2} = \frac{\varepsilon}{2}.$$

As a consequence, for any $v \in \mathbb{R}^n \setminus \{0\}$, we have by linearity of ψ_x that

$$\|\psi_x(v)\| = \|v\| \left\| \psi_x \left(\frac{v}{\|v\|} \right) \right\| > \|v\| \frac{\varepsilon}{2} \neq 0$$

which proves that ψ_x is injective too. Now as shown in the first part of the proof, we can further shrink U_M so that $\psi_y|_B$ is injective for all $y \in U_M$.

Step 5:

If $y \in \overline{U_M}$ satisfies $\tau_2(y) \in B$, then by assumption

$$\psi_y|_B(\tau(y)) = 0$$

so from injectivity follows that $\tau_2(y) = 0$. Therefore, it suffices to show that there exists a $y \in U_M$ close to x such that $\tau_2(y) \in B$. Let π_A and π_B be the projections from $\mathbb{R}^m$ onto A and B respectively. Now we define $\sigma_A := \pi_A \circ \sigma_2$, $\sigma_B := \pi_B \circ \sigma_2$. Analogously, we define τ_A and τ_B . We can pick charts

$$\varphi_1 : U_M \rightarrow \mathbb{R}^{\dim(A)} \times \mathbb{R}^{\dim(M) - \dim(A)}$$

and

$$\varphi_2 : \mathbb{R}^n \rightarrow \mathbb{R}^{\dim(A)} \times \mathbb{R}^{\dim(B)}$$

so that

$$\sigma_2 : \mathbb{R}^{\dim(A)} \times \mathbb{R}^{\dim(M) - \dim(A)} \rightarrow \mathbb{R}^{\dim(A)} \times \mathbb{R}^{\dim(B)} : (u, v) \mapsto (u, \sigma_B(u, v))$$

and in particular $\sigma_A(u, v) = u$. In these charts, we also have that

$$\tau_2 = (\tau_A, \tau_B)$$

As τ is $\mathcal{C}^0$ -close to σ , we can write $\tau_A(u, v) = \sigma_A(u, v) + f(u, v) = u + f(u, v)$ for some f and predetermined precision ϵ for which $\|f(u, v)\| < \epsilon$. The assertion we have to prove now boils down to finding a pair $(u, v) \in M \cong \mathbb{R}^{\dim(A)} \times \mathbb{R}^{\dim(M) - \dim(A)}$ for which $u + f(u, v)$ vanishes.

Step 6:

Now we will show by contradiction that we can find such a pair $(u, v) \in M$ for which $u + f(u, v) = 0$. For the sake of finding a contradiction, take any $r > \epsilon$ and suppose

that there does not exist such a solution of $u + f(u, v)$ for any possible v and any u in the ball $B_r = \{u \in A : \|u\| \leq r\}$. Then for any such v , we obtain a continuous map

$$g_v : rS^{\dim(A)-1} \rightarrow S^{\dim(A)-1} : u \mapsto \frac{u + f(u, v)}{\|u + f(u, v)\|}$$

where $rS^{\dim(A)-1} = \{u \in \mathbb{R}^{\dim(A)} : \|u\| = r\} \cong S^{\dim(A)-1}$ and $u + f(u, v) \neq 0$ by assumption, implying that g_v is well defined. Consider the homotopy

$$H_1(u, t) := \frac{u + tf(u, v)}{\|u + tf(u, v)\|} \text{ for } t \in [0, 1].$$

This is well defined too, as

$$\|u\| = r > \epsilon > \|f(u, v)\| \geq \|tf(u, v)\| \text{ for all } t \in [0, 1]$$

so that $u + tf(u, v)$ cannot vanish for $t \in [0, 1]$. Clearly $H_1(u, 0)$ is the identity on $S^{\dim(A)-1}$, and $H_1(u, 1) = g_v(u)$. Because homotopic maps have the same degree, it follows that $\deg(g_v) = \deg(\text{Id}_{S^{\dim(A)-1}}) = 1$. On the other hand, we can define the homotopy

$$H_2(u, t) := \frac{tu + f(tu, v)}{\|tu + f(tu, v)\|} \text{ for } t \in [0, 1].$$

For any $t \in [0, 1]$, $tu + f(tu, v) = u' + f(u', v)$ for $u' = tu \in B_r$, and we assumed that in that case $u' + f(u', v) \neq 0$. This means that H_2 is also well defined. Moreover, $H_1(u, 1) = g_v(u)$ and $H_2(u, 0) = \frac{f(0, v)}{\|f(0, v)\|}$ is constant in u , so that $\deg(g_v) = 0$. This now gives a contradiction because we obtained $1 = \deg(g_v) = 0$, proving the assumption that there doesn't exist such a solution in B_r was wrong. We thus conclude that for any possible v , we can find a u in a ball B_r of radius $r > \epsilon$ for which $u + f(u, v) = 0$. As mentioned before, from this follows that for that solution $p = (u, v) \in U_M$ holds that

$$\tau_A(p) = \tau_A(u, v) = 0$$

and thus $\tau(p) \in B$ for which we already proved that then $\tau(p) = 0$. This is exactly what proves statement 2.

Step 7:

Regarding statement 3, we again consider the splitting

$$\mathbb{R}^n = A \oplus B$$

as in the proof of the first statement. If we let $\pi : \mathbb{R}^n \rightarrow A$ be the projection onto A , then

$$\text{rank}(\pi \circ d_x \sigma_2)$$

is maximal. Hence, by lemma 5.1.9 and as we have the condition that τ is $\mathcal{C}^1$ -close to σ , we can choose τ such that the rank of $\pi \circ d_x \tau_2$ is still maximal. This means in particular that

$$\text{Im}(d_x \tau_2) + B = \mathbb{R}^n.$$

If we use that τ is a fortiori C^0 -close to σ , then by the second statement of this theorem and corollary 5.1.10 follows that we can find a y close to x such that

$$\tau_2(y) = 0 \text{ and } \text{Im}(d_y \tau_2) + B = \mathbb{R}^n.$$

so that τ intersects B transversely at 0. As a consequence, we can argue completely similar as in the proof of the first bullet, but now with τ_2 instead of σ_2 , and find that locally

$$\tau_2^{-1}(0) = \tau_2^{-1}(B)$$

so that $\tau^{-1}(0)$ is a submanifold of M around x . Its dimension is equal to

$$\begin{aligned} \dim(\tau^{-1}(0)) &= \dim(\ker(d_y \tau_2)) = \dim(M) - \text{rank}(d_y \tau_2) \\ &= \dim(M) - \text{rank}(d_x \sigma_2) = \dim(\sigma^{-1}(0)) \end{aligned}$$

which concludes the proof. □

5.3 Kuranishi models

Lemma 5.3.1. ([CSS14], Lemma 4.6) *Let $f : A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a smooth function such that $f(0) = 0$ and $A \subseteq \mathbb{R}^m$ is open. Then it follows that there exists*

- *a submanifold $Z \subseteq A$ of dimension $\dim(\ker(d_0 f))$,*
- *a vector space $C \subseteq \mathbb{R}^n$ of dimension $n - \dim(\text{Im}(d_0 f))$,*
- *and a smooth map $\tilde{f} : Z \rightarrow C$*

so that $d_0 \tilde{f} = 0$ and the zero set of f and $\tilde{f}$ coincide in a neighborhood of 0.

Proof. Choose a complement C to $\text{Im}(d_0 f)$ in $\mathbb{R}^n$ of dimension $n - \dim(\text{Im}(d_0 f))$, so that then

$$\mathbb{R}^n = \text{Im}(d_0 f) \oplus C.$$

In particular, f is transverse to C at 0. It follows that there exists a neighborhood $U \subseteq A$ of 0 such that f is transverse to C at any point in U : Indeed, define

$$\pi : \mathbb{R}^n \rightarrow \mathbb{R}^n / C = \text{Im}(d_0 f), \tag{5.17}$$

then $\text{rank}(\pi \circ d_0 f) = \dim(\mathbb{R}^n / C)$ is maximal. So by lower semicontinuity of the rank, there exists a neighborhood U of 0 in A so that $\text{rank}(\pi \circ d_y f) = \dim(\mathbb{R}^n / C)$ in U . This means exactly that $\pi \circ d_y f$ is surjective in U , which implies that

$$\text{Im}(d_y f) + C = \mathbb{R}^n \tag{5.18}$$

which means exactly that f is transversal to C in U . Then, theorem 6.30a in [Lee12] says that $Z := f^{-1}(C) \cap U$ is a smooth submanifold of A of dimension $m - \dim(\text{Im}(d_0 f)) = \dim(\ker(d_0 f))$. Therefore, we obtain a map

$$\tilde{f} := f|_Z : Z \rightarrow C.$$

Notice that

$$Z = f^{-1}(C) \cap U = (\pi \circ f)^{-1}(0) \cap U$$

where $\pi \circ f$ is smooth. Furthermore,

$$d_0(\pi \circ f) = \pi \circ d_0f$$

is surjective as $\mathbb{R}^n = C \oplus \text{Im}(d_0f)$, so that 0 is a regular value of $\pi \circ f$. Then

$$T_0Z = \ker(\pi \circ d_0f) = \ker(d_0f) \quad (5.19)$$

and therefore

$$d_0\tilde{f} = d_0f|_{T_0Z} = 0,$$

because for any $v \in T_0Z = \ker(d_0f)$, it follows that $v \in C$ so that per construction

$$d_0\tilde{f}(v) = d_0f(v) = 0.$$

It's also clear that any zero $y \in C$ of f lies in Z by construction, so that

$$\tilde{f}(y) = f(y) = 0. \quad (5.20)$$

Conversely, every zero of $\tilde{f}$ is trivially a zero of f , so that the zero sets of f and $\tilde{f}$ coincide in U . $\square$

Theorem 5.3.2. ([CSS14], Proposition 4.5)(Kuranishi Models) *Let M be a manifold, E and F vector bundles over M , $\sigma : M \rightarrow E$ a section and $\phi : M \rightarrow \text{Hom}(E, F)$ a vector bundle map that satisfies $\phi \circ \sigma = 0$. Furthermore, let G be a Lie group that acts on E such that the zero section is preserved and σ is equivariant.*

$$\begin{array}{ccc} G \circ E & \xrightarrow{\phi} & F \\ \sigma \uparrow \downarrow & \swarrow & \\ G \circ M & & \end{array}$$

Let x denote a zero of σ and $m_x : G \rightarrow M : g \mapsto g \cdot x$. Then there exists

- a submanifold S of M containing x of dimension $\dim(\ker(d_x^{\text{vert}}\sigma)) - \dim(\text{Im}(d_e m_x))$,
- a subbundle $\tilde{E}$ of $E|_S$ of rank $\dim(\ker(\phi_x)) - \dim(\ker(d_x^{\text{vert}}\sigma))$
- and a section $\tilde{\sigma} : S \rightarrow \tilde{E}$

such that

- $d_x^{\text{vert}}\tilde{\sigma} = 0$,
- the zero sets of $\tilde{\sigma}$ and $\sigma|_S$ coincide
- and if x' is a zero of σ sufficiently close to x , then $x' \in G \cdot y$ for some zero y of $\tilde{\sigma}$.

Proof. Take a $\tau \subseteq M$ transversal to m_x at x , so that then

$$T_x\tau \oplus T_x(G \cdot x) = T_xM. \quad (5.21)$$

where we used $\text{Im}(d_e m_x) = T_x(G \cdot x)$.

Step 1:

To prove statement c) later, we first try to find open neighborhoods U_G of e in G and U_τ of x in τ for which $\text{Im}(m|_{U_G \times U_\tau})$ contains an open neighborhood of x in M . Let $m' := m|_{G \times \tau}$, where $m : G \times M \rightarrow M$ is the given action, then its derivative is a map $d_{(e,x)} m' : T_e G \times T_x \tau \rightarrow T_x M$. We claim that this derivative is surjective at (e, x) . As $d_{(e,x)} m'$ is a linear map, it is completely determined by the images of the elements in $T_e G \times \{0\}$ and $\{0\} \times T_x \tau$. Take any $\zeta \in T_e G$ and a curve $g(t) \in G$ for which $g(0) = e$ and $g'(0) = \zeta$, then we have that

$$d_{(e,x)} m'(\zeta, 0) = \frac{d}{dt} \Big|_{t=0} g(t) \cdot x = \frac{d}{dt} \Big|_{t=0} m_x(g(t)) = d_e m_x \zeta$$

so that

$$d_{(e,x)} m'(T_e G \times \{0\}) = \text{Im}(d_e m_x) = T_x(G \cdot x).$$

Additionally, if we take a $\xi \in T_x \tau$ and a curve $y(t) \in \tau$ for which $y(0) = x$ and $y'(0) = \xi$, then it follows that

$$d_{(e,x)} m'(0, \xi) = \frac{d}{dt} \Big|_{t=0} e \cdot y(t) = \frac{d}{dt} \Big|_{t=0} y(t) = \xi$$

from which follows that

$$d_{(e,x)} m'(\{0\} \times T_x \tau) = T_x \tau.$$

As a result, we obtain that

$$\begin{aligned} d_{(e,x)} m'(T_e G \times T_x \tau) &= d_{(e,x)} m'(T_e G \times \{0\}) \oplus d_{(e,x)} m'(\{0\} \times T_x \tau) \\ &= T_x(G \cdot x) \oplus T_x \tau = T_x M \end{aligned}$$

as m_x is transversal to τ at x , which proves the claim that the derivative of m' is surjective at (e, x) . Therefore, m' is submersive at (e, x) and of maximal rank $\dim(M)$. Again, by corollary 5.1.10, we can thus find a neighborhood $U_G \times U_\tau \subseteq G \times \tau$ around (e, x) so that the rank of $d_{(e,x)} m'$ stays maximal in $U_G \times U_\tau$ and thus $m'|_{U_G \times U_\tau}$ has a coordinate representation of the form

$$m'(x_1, \dots, x_n, x_{n+1}, \dots, x_{n+k}) = (x_1, \dots, x_n)$$

where we denoted $k = \dim(G)$. From that, we deduce that $\text{Im}(m'|_{U_G \times U_\tau})$ contains an open neighborhood of x in M . We will use this later to prove c). However, we have yet to construct the desired section $\tilde{\sigma}$.

Step 2:

We will now start the construction of the three objects in the first three dots of the theorem. By shrinking U_τ if necessary, we can take E and F to be trivial:

$$\begin{aligned} E &\cong U_\tau \times V \\ F &\cong U_\tau \times W. \end{aligned}$$

This time, we take abstract vector spaces V and W , rather than writing it as $\mathbb{R}^m$ for some $m \in \mathbb{N}$. Since we are working locally, we can take a chart $\varphi : U_\tau \rightarrow \mathbb{R}^t$ that maps x to 0, where $t := \dim(\tau)$. By identifying the map $\sigma|_\tau$ with the corresponding map in the chart, we get locally

$$\sigma_2|_\tau : \mathbb{R}^t \rightarrow V.$$

Furthermore, locally, we can write $\phi_y = \text{Id}_{U_\tau} \times \phi_y^{(2)}$ for $\phi_y^{(2)} \in \text{Hom}(V, W)$. In that way, $\phi|_\tau$ reduces to a map

$$\phi^{(2)} : U_\tau \rightarrow \text{Hom}(V, W).$$

Let K be a complement to $\text{Im}(\phi_x^{(2)})$ such that

$$K \oplus \text{Im}(\phi_x^{(2)}) = W, \quad (5.22)$$

then clearly $\phi_x^{(2)}$ is transverse to K . Then, by using lower semicontinuity of the rank, we can shrink U_τ if needed so that $\phi_y^{(2)}$ is still transverse to K in U_τ , as we did in lemma 5.3.1. In other words, for all $y \in U_\tau$ holds

$$K + \text{Im}(\phi_y^{(2)}) = W.$$

Now we claim that $\hat{E} := \{(y, (\phi_y^{(2)})^{-1}(K)) | y \in U_\tau\}$ forms a subbundle of $\pi^{-1}(U_\tau) \subseteq E$. Indeed, consider the induced vector bundle map

$$\phi' : E|_{U_\tau} \rightarrow (U_\tau \times W)/K.$$

From equation (5.3) follows that ϕ' has constant rank. As a consequence, we can use theorem 10.34 in [Lee12] so that $\hat{E} = \ker(\phi')$ is a vector subbundle of E .

From the assumption that $\phi \circ \sigma = 0$ follows that

$$\phi_y^{(2)} \circ \sigma_2|_\tau = 0 \text{ for all } y \in U_\tau,$$

which implies

$$\sigma_2(y) \in (\phi_y^{(2)})^{-1}(0) \subseteq (\phi_y^{(2)})^{-1}(K)$$

so that locally, $\sigma|_\tau$ is a section $\sigma|_\tau : U_\tau \rightarrow \hat{E}$. Once more, we can shrink U_τ so that $\hat{E}$ too is trivial: $\hat{E} \cong U_\tau \times \hat{V}$ for a vector space $\hat{V} \subseteq V$. Then $\sigma_2|_\tau$ in the chart becomes a map

$$\sigma_2|_\tau : \mathbb{R}^t \rightarrow \hat{V}$$

such that $\sigma_2|_\tau(0) = 0$, on which we can apply the previous lemma. This provides us with a vector space $\tilde{V} \subseteq \hat{V}$ for which

$$\tilde{V} \oplus \text{Im}(d_0 \sigma_2|_\tau) = \hat{V}, \quad (5.23)$$

and a submanifold S of $\mathbb{R}^t$ containing 0 of dimension $\dim(\ker(d_0 \sigma_2|_\tau))$, defined as

$$S := \sigma_2|_\tau^{-1}(\tilde{V}) \cap U \quad (5.24)$$

for a neighborhood U where $\sigma_2|_\tau$ is transverse to $\tilde{V}$. Additionally, lemma 5.3.1 gives us a map

$$\tilde{\sigma}_2 : S \rightarrow \tilde{V}.$$

for which the differential at 0 vanishes, and the zero set coincides with the zero set of $\sigma_2|_S$ in an open neighborhood. After possibly shrinking, we still call this neighborhood U_τ . We can translate this back to the manifold τ , where now S is a submanifold of τ , and $\tilde{\sigma}_2|_\tau$ and its derivative vanish at x :

$$\tilde{\sigma}_2|_\tau(x) = 0 \quad (5.25)$$

$$d_x \tilde{\sigma}_2|_\tau = d_x^{\text{vert}} \tilde{\sigma} = 0 \quad (5.26)$$

We can turn this locally into a section of $S \times \tilde{V}$:

$$\tilde{\sigma} : S \rightarrow S \times \tilde{V} : p \mapsto (p, \tilde{\sigma}_2(p))$$

In other words, we constructed a section from a submanifold S of τ that contains x , to $S \times \tilde{V}$, a subbundle of $E|_S$. We denote this subbundle by $\tilde{E}$. We have now proven the existence of the three objects in the three bullets. We postpone the proof of the dimension of S and the rank of $\tilde{E}$ until the end of the proof.

Step 3:

Now, statements a) and b) follow almost immediately from the former constructions: Equation (5.26) proves a).

By definition, the zero set of $\sigma|_S$ coincides with the zero set of $\sigma_2|_S$, and the zero set of $\tilde{\sigma}$ coincides with the zero set of $\tilde{\sigma}_2$. Hence by construction, the zero set of $\tilde{\sigma}$ now coincides with the zero set of $\sigma|_S$, so that we're also done proving b).

Step 4:

To prove statement c), we will recall from step 1 that $\text{Im}(m|_{U_G \times U_\tau})$ contains an open neighborhood of x in M , so that also $\text{Im}(m|_{U_G \times U_\tau})$ contains an open neighborhood of x , call it O . Then for any zero x' of σ in O , we can write it as

$$x' = m'(g, y) = g \cdot y \text{ for some } g \in U_G, y \in U_\tau.$$

However, by equivariance of the section holds

$$\sigma_2(y) = \sigma_2(g^{-1} \cdot x') = g^{-1} \cdot \sigma_2(x') = 0$$

which implies that y is also a zero of σ . Since the zero sets of σ_τ and $\tilde{\sigma}$ coincide in U_τ , y is also a zero of $\tilde{\sigma}$. In particular, $y \in S$. This already proves assertion c), but we would like to obtain a formula in terms of sets. The former discussion proves that

$$\sigma_2^{-1}(0) \cap O \subseteq \left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} G \cdot \eta \right) \cap O.$$

Conversely, if we take an element in $\left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} G \cdot \eta \right) \cap O$, then we can write it as $A \cdot \eta$ for some $\eta \in \tilde{\sigma}_2^{-1}(0)$ and $A \in G$. As the zero set of $\sigma_2|_S$ and $\tilde{\sigma}_2$ coincide, η is also a zero of σ_2 . Then it follows by equivariance that

$$\sigma_2(A \cdot \eta) = A \cdot \sigma_2(\eta) = A \cdot 0 = 0.$$

Hence, we also proved the other inclusion, so that we get the equality

$$\sigma_2^{-1}(0) \cap O = \left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} G \cdot \eta \right) \cap O. \quad (5.27)$$

That concludes part c).

Step 5:

Next, we check the dimension of S and the rank of the subbundle $\tilde{E}$. We start with the former. We have the decomposition

$$T_x \tau \oplus \text{Im}(d_e m_x) = T_x M. \quad (5.28)$$

and by equation (5.19) we have

$$T_x S = \ker(d_x \sigma_2|_\tau) = \ker(d_x \sigma_2) \cap T_x \tau. \quad (5.29)$$

Additionally, we have by equivariance that

$$\text{Im}(d_e m_x) \subseteq \ker(d_x \sigma_2). \quad (5.30)$$

Indeed, take some $v \in T_e G$ and consider the curve $\gamma(t) = \exp(tv)$, then it follows from equivariance that

$$\begin{aligned} d_x \sigma_2 \circ d_e m_x(v) &= \frac{d}{dt} \Big|_{t=0} \sigma_2 \circ m_x(\exp(tv)) \\ &= \frac{d}{dt} \Big|_{t=0} \sigma_2(\exp(tv) \cdot x) \\ &= \frac{d}{dt} \Big|_{t=0} \exp(tv) \cdot \sigma_2(x) \\ &= \frac{d}{dt} \Big|_{t=0} \exp(tv) \cdot 0 = 0. \end{aligned}$$

In general holds for finite vector spaces: if $B \subset A$, then $A \cap (B + C) = B + (A \cap C)$, so for

$$\begin{aligned} A &= \ker(d_x \sigma_2) \\ B &= \text{Im}(d_e m_x) \\ C &= T_x \tau, \end{aligned}$$

together with equations (5.28), (5.29) and (5.30), we obtain

$$\begin{aligned} \ker(d_x \sigma_2) &= (\text{Im}(d_e m_x) \cap \ker(d_x \sigma_2)) + (T_x \tau \cap \ker(d_x \sigma_2)) \\ &= \text{Im}(d_e m_x) + T_x S. \end{aligned}$$

As a consequence,

$$T_x S \cong \frac{\ker(d_x \sigma_2)}{\text{Im}(d_e m_x)} = \frac{\ker(d_x^{\text{vert}} \sigma)}{\text{Im}(d_e m_x)} \quad (5.31)$$

from which the dimension formula for S follows.

Lastly, to compute the rank of $\tilde{E}$, we observe that $\tilde{E}_x = \tilde{V}$ and

$$\tilde{V} \oplus \text{Im}(d_x \sigma_2|_\tau) = \hat{V}$$

by equation (5.23). From equation (5.22) and the definition of $\hat{E}$ follows that

$$\hat{E}_x \cong \ker(\phi_x^{(2)}) = \ker(\phi_x) \quad (5.32)$$

so that

$$\tilde{E}_x = \tilde{V} \cong \frac{\hat{V}}{\text{Im}(d_x \sigma_2|_\tau)} \cong \frac{\ker(\phi_x^{(2)})}{\text{Im}(d_x \sigma_2|_\tau)} \quad (5.33)$$

from which the rank of E follows. This concludes the proof. $\square$

Chapter 6

Rigidity and Local Smoothness Results

In this final chapter, we will enjoy the fruits of the labor done in chapter 5, and apply the geometric tools we developed. These results allow us to study rigidity, stability, and the local structure of the deformation spaces. In particular, we formulate cohomological criteria that provide answers to our fundamental problems stated in chapter 1. These results are well known and were already stated in the 1960s by Nijenhuis and Richardson. We first apply the geometric theorems of chapter 5 on the deformation theory of Lie algebras and Lie algebra homomorphisms in an ad hoc manner, following [CSS14]. In these first two sections, we give complete and rigorous proofs of the results, whereas they were merely sketched in [CSS14].

After this, in section 6.3 we will recognize the DGLA theory behind the constructions made in these first two cases. The perspective we make in section 6.3 is our own and as far as we know it is new. With these new insights, we will apply the geometric theorems of chapter 5 on the deformation theory of Lie subalgebras more elegantly with the structures made in chapter 6.3. These last two subsections are our own contributions and thus diverge from [CSS14].

6.1 Lie algebras

Theorem 6.1.1. ([CSS14], Theorem 5.1) *Let $(\mathfrak{g}, \mu)$ be a Lie algebra. Then there exist an open neighborhood U of 0 in $H^2(\mathfrak{g}, \mathfrak{g})$ and a smooth map $\Phi : U \rightarrow H^3(\mathfrak{g}, \mathfrak{g})$ for which*

- a) $\Phi(0) = 0$ and $d_0\Phi = 0$,*
- b) the zeros of Φ are in bijection with a subset of $\mathcal{J}^{-1}(0)$, where 0 corresponds to the original Lie bracket μ*
- c) and in a neighborhood of μ in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, the union of the $\mathrm{GL}(\mathfrak{g})$ -orbits containing Lie brackets that corresponds to a zero of Φ , coincides with the space of all Lie brackets on $\mathfrak{g}$.*

We will apply theorem 5.3.2, Kuranishi models, on the framework we obtained in chapters 2 and 4. We will derive assertions a), b) and c) from the corresponding assertions a), b) and c) of theorem 5.3.2.

Proof. Step 1:

Let $M = \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, $E = M \times \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ and $F = M \times \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}$. Let the section $\sigma := (\text{Id}_M, \sigma_2) : M \rightarrow E$ that is characterized by

$$\sigma_2 = \mathcal{J} : M \rightarrow \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}. \quad (6.1)$$

The section σ has zero $\mu \in M$ because μ is a Lie bracket, so $\mathcal{J}(\mu) = 0$. Let $\phi : M \rightarrow \text{Hom}(E, F) : \eta \mapsto \phi_\eta := (\text{Id}_M, \phi_\eta^{(2)})$ be described by

$$\phi_\eta^{(2)} := \delta_\eta : \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}. \quad (6.2)$$

which is a Chevalley-Eilenberg differential if η is a Lie bracket. Furthermore, the Lie group $\text{GL}(\mathfrak{g})$ acts on $E = (\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}) \times (\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g})$ diagonally by

$$(A \cdot (\eta, \zeta)) = ((A \cdot \eta), (A \cdot \zeta))$$

for $A \in \text{GL}(\mathfrak{g})$, $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, $\zeta \in \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$.

Step 2:

Before we can apply theorem 5.3.2, we have to check whether the conditions are fulfilled. The zero section $Z : M \rightarrow E : \eta \mapsto (\eta, 0)$ is clearly preserved by the Lie group action:

$$A \cdot \text{Im}(Z) = \{(A \cdot \eta, 0) \in E | \eta \in M\} = \text{Im}(Z) \text{ for any } A \in \text{GL}(\mathfrak{g})$$

Additionally, for any $A \in \text{GL}(\mathfrak{g})$ and $\eta \in M$, we have that

$$A \cdot \sigma(\eta) = A \cdot (\text{Id}_M, \mathcal{J})(\eta) = (\text{Id}_M(A \cdot \eta), A \cdot \mathcal{J}(\eta))$$

where for any $u, v, w \in \mathfrak{g}$, we have that

$$\begin{aligned} A \cdot \mathcal{J}(\eta)(u, v, w) &= A\eta(\eta(A^{-1}u, A^{-1}v), A^{-1}w) + \text{c.p.} \\ &= A\eta(A^{-1}A\eta(A^{-1}u, A^{-1}v), A^{-1}w) + \text{c.p.} \\ &= \mathcal{J}(A \cdot \eta)(u, v, w) \end{aligned}$$

and thus

$$A \cdot \sigma(\eta) = (\text{Id}_M(A \cdot \eta), \mathcal{J}(A \cdot \eta)) = \sigma(A \cdot \eta) \quad (6.3)$$

proving that the section σ is equivariant.

Lastly, we claim that $\phi \circ \sigma = 0$. This claim reduces to proving $\phi_\eta^{(2)} \circ \sigma_2(\eta) = 0$ for all $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$. By theorem 3.2.4, we obtain that

$$\delta_\eta = \llbracket \eta, \cdot \rrbracket.$$

Using this, we compute

$$\phi_\eta^{(2)} \circ \sigma_2(\eta) = \delta_\eta \circ \mathcal{J}(\eta)$$

$$\begin{aligned}
&= \delta_\eta \llbracket \eta, \eta \rrbracket \\
&= \llbracket \delta_\eta \eta, \eta \rrbracket - \llbracket \eta, \delta_\eta \eta \rrbracket \\
&= \llbracket \llbracket \eta, \eta \rrbracket, \eta \rrbracket - \llbracket \eta, \llbracket \eta, \eta \rrbracket \rrbracket \\
&= 2 \llbracket \llbracket \eta, \eta \rrbracket, \eta \rrbracket
\end{aligned}$$

by graded skew symmetry. Finally, using the graded Jacobi identity of the Gerstenhaber bracket, we find

$$\phi_\eta^{(2)} \circ \sigma_2(\eta) = 2 \llbracket \llbracket \eta, \eta \rrbracket, \eta \rrbracket = 0 \quad (6.4)$$

This proves the claim. Now that we have checked all of the assumptions needed, it is justified to make use of theorem 5.3.2, Kuranishi models.

Step 3:

Now we will apply theorem 5.3.2, Kuranishi models. We will follow the constructions and adopt the notations in the proof of theorem 5.3.2 to keep track of the relations and conditions between the constructions in the current setup. We start by fixing a τ transversal to the orbit $G \cdot \mu$ in μ , so that

$$T_\mu \tau \oplus T_\mu(G \cdot \mu) = T_\mu M. \quad (6.5)$$

The vector bundles in our setup are already trivial: $E = M \times V$ and $F = M \times W$ where $V = \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ and $W = \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}$. This gives us maps

$$\sigma_2|_\tau : \tau \rightarrow V$$

and

$$\phi^{(2)}|_\tau : \tau \rightarrow \text{Hom}(V, W).$$

Let K be a complement to $\text{Im}(\phi_\mu^{(2)})$ in W , meaning that

$$K \oplus \text{Im}(\phi_\mu^{(2)}) = W.$$

In the proof of 5.3.2, we shrank the neighborhood U_τ of x in τ until all the desired conditions were fulfilled. Hence, denote by U_τ the smallest such open neighborhood. According to that proof, we then find a vector subbundle $\hat{E} \cong U_\tau \times \hat{V}$ of E , where $\hat{V} \cong (\phi_\mu^{(2)})^{-1}(K) = \ker(\phi_\mu^{(2)})$. By taking the chart $\varphi : U_\tau \rightarrow \mathbb{R}^t$, we also obtained $\tilde{V}$ for which

$$\text{Im}(d_0 \sigma_2|_\tau) \oplus \tilde{V} = \hat{V}. \quad (6.6)$$

Moving forward, we acquire a map

$$\tilde{\sigma}_2 : S \rightarrow \tilde{V}$$

from a submanifold S of τ that contains μ , to $\tilde{V}$. For this map holds that

- $\tilde{\sigma}_2(\mu) = 0$,
- a) $d_\mu^{\text{vert}} \tilde{\sigma} = d_\mu \tilde{\sigma}_2 = 0$,
- b) the zero set of $\tilde{\sigma}_2$ coincides with the zero set of $\sigma_2|_\tau = \mathcal{J}|_\tau$,

- c) and if μ' is a zero of $\mathcal{K}$ sufficiently close to μ , then $\mu' \in \text{GL}(\mathfrak{g}) \cdot \eta$ for some zero η of $\tilde{\sigma}_2$.

Step 4:

We will construct Φ using this section $\tilde{\sigma}$. We start by making an identification on the codomain of $\tilde{\sigma}_2$. By equation (6.6), we see that

$$\tilde{V} \cong \frac{\hat{V}}{\text{Im}(d_\mu(\sigma_2|_\tau))} \cong \frac{\ker(\phi_\mu^{(2)})}{\text{Im}(d_\mu^{\text{vert}}(\sigma|_\tau))}.$$

Note that on $G \cdot \mu$, σ is the zero vector at μ by equivariance:

$$\sigma_2(g \cdot \mu) = g \cdot \sigma_2(\mu) = 0$$

so that its derivative vanishes

$$d_\mu(\sigma_2|_{G \cdot \mu}) = 0$$

We could equivalently state that

$$T_\mu(G \cdot \mu) \subseteq \ker(d_\mu^{\text{vert}}(\sigma)) \quad (6.7)$$

It follows from linearity of the derivative and from equation (6.5) that

$$\text{Im}(d_\mu^{\text{vert}}(\sigma|_\tau)) = \text{Im}(d_\mu^{\text{vert}}(\sigma))$$

and thus, if we substitute the maps in our setup

$$\tilde{V} \cong \frac{\ker(\phi_\mu^{(2)})}{\text{Im}(d_\mu^{\text{vert}}\sigma)} = \frac{\ker(\delta_\mu : C^3(\mathfrak{g}, \mathfrak{g}) \rightarrow C^4(\mathfrak{g}, \mathfrak{g}))}{\text{Im}(d_\mu \mathcal{J})} \quad (6.8)$$

$$= \frac{\ker(\delta_\mu : C^3(\mathfrak{g}, \mathfrak{g}) \rightarrow C^4(\mathfrak{g}, \mathfrak{g}))}{\text{Im}(\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g}))} = H^3(\mathfrak{g}, \mathfrak{g}) \quad (6.9)$$

where we used equation (4.2). Denote the linear isomorphism between $H^3(\mathfrak{g}, \mathfrak{g})$ and $\tilde{V}$ by

$$\alpha : \tilde{V} \rightarrow H^3(\mathfrak{g}, \mathfrak{g}) \quad (6.10)$$

that sends $0 \in \tilde{V}$ to $0 \in H^3(\mathfrak{g}, \mathfrak{g})$.

Step 5:

Now we make an identification on the domain of $\tilde{\sigma}_2$. By equation (5.31), it follows that

$$T_\mu S \cong \frac{\ker(d_\mu^{\text{vert}}(\sigma))}{\text{Im}(d_e m_\mu)} = \frac{\ker(d_\mu \mathcal{J})}{\text{Im}(d_e m_\mu)} = \frac{\ker(\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g}))}{\text{Im}(d_e m_\mu)}.$$

We now claim that $d_e m_\mu = -\delta_\mu$, so that in particular the images of $d_e m_\mu$ and δ_μ coincide. To see this, we analyze the maps

$$d_e m_\mu : T_e \text{GL}(\mathfrak{g}) \rightarrow T_\mu(\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g})$$

$$\delta_\mu : \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$$

and remark that $\mathfrak{g}^* \otimes \mathfrak{g}$ is the Lie algebra of the Lie group $GL(\mathfrak{g})$, and $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is a vector space so that it can be identified with its tangent space. Therefore the domain and codomain of the two maps coincide. Additionally, if we take $A \in GL(\mathfrak{g})$ and $u, v \in \mathfrak{g}$, then on the one hand

$$\begin{aligned} d_e m_\mu(A)(u, v) &= \frac{d}{dt} \Big|_{t=0} m_\mu(e^{tA})(u, v) \\ &= \frac{d}{dt} \Big|_{t=0} e^{tA} \cdot \mu(u, v) \\ &= \frac{d}{dt} \Big|_{t=0} e^{tA} \mu(e^{-tA}u, e^{-tA}v) \\ &= A\mu(u, v) + \mu(-Au, v) + \mu(u, -Av) \\ &= A\mu(u, v) - \mu(Au, v) - \mu(u, Av), \end{aligned}$$

and on the other hand

$$\begin{aligned} \delta_\mu(A)(u, v) &= \mu(u, Av) - \mu(v, Au) - A\mu(u, v) \\ &= \mu(u, Av) + \mu(Au, v) - A\mu(u, v). \end{aligned}$$

This proves the claim that

$$d_e m_\mu = -\delta_\mu : C^1(\mathfrak{g}, \mathfrak{g}) \rightarrow C^2(\mathfrak{g}, \mathfrak{g}) \quad (6.11)$$

and thus

$$\text{Im}(d_e m_\mu) = \text{Im}(\delta_\mu : C^1(\mathfrak{g}, \mathfrak{g}) \rightarrow C^2(\mathfrak{g}, \mathfrak{g})).$$

Then the previous identification further reduces to

$$T_\mu S \cong \frac{\ker(\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g}))}{\text{Im}(\delta_\mu : C^1(\mathfrak{g}, \mathfrak{g}) \rightarrow C^2(\mathfrak{g}, \mathfrak{g}))} = H^2(\mathfrak{g}, \mathfrak{g}) \quad (6.12)$$

As S is a manifold, we may choose a local chart

$$\varphi : (S, \mu) \rightarrow (T_\mu S, 0).$$

Then φ is locally a diffeomorphism such that an open neighborhood $U_S \subseteq S$ of μ can be identified with a neighborhood of 0 in $T_\mu S$. Together with the identification $T_\mu S \cong H^2(\mathfrak{g}, \mathfrak{g})$, we obtain a diffeomorphism

$$\beta : (U_S, \mu) \rightarrow (U, 0) \quad (6.13)$$

for some open neighborhood $U \subseteq H^2(\mathfrak{g}, \mathfrak{g})$ of 0.

With the identifications we made on the domain and codomain of $\tilde{\sigma}_2$, we can construct Φ as

$$\Phi := \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1} : U \rightarrow H^3(\mathfrak{g}, \mathfrak{g}) \quad (6.14)$$

Step 6:

Now, we can prove statement a). Firstly, $\beta^{-1}(0) = \mu$ and $\tilde{\sigma}_2(\mu) = 0$ as μ is a zero of $\sigma_2 = \mathcal{J}$, so that

$$\Phi(0) = \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1}(0)$$

$$\begin{aligned}
&= \alpha \circ \tilde{\sigma}_2(\mu) \\
&= \alpha(0) = 0
\end{aligned}$$

because α is a linear isomorphism. Secondly, as $d_\mu \tilde{\sigma}_2 = 0$ by statement a) of theorem 5.3.2, and $\beta^{-1}(0) = \mu$, we have that

$$d_0 \Phi = 0. \quad (6.15)$$

Therefore, we have proven statement a).

Step 7:

For statement b), we can see that for $h \in U$

$$h \text{ is a zero of } \Phi \iff \Phi(h) = 0$$

$$\iff \tilde{\sigma}_2 \circ \beta^{-1}(h) = 0$$

$$\iff \beta^{-1}(h) \text{ is a zero of } \tilde{\sigma}_2$$

However, by statement b) of theorem 5.3.2, a zero of $\tilde{\sigma}_2$ is a zero of $\sigma_2 = \mathcal{J}$. The condition that $\mathcal{J}(\beta^{-1}(h)) = 0$ now means exactly that $\beta^{-1}(h)$ is a Lie bracket. Hence we proved that every zero h of Φ corresponds to a Lie bracket $\beta(h)$. Moreover, $\beta^{-1}(0) = \mu$, so that the zero $h = 0$ corresponds to the original bracket μ .

Step 8:

From statement c) of theorem 5.3.2 follows that we can find an open neighborhood $O \subseteq \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ of μ

$$\mathcal{J}^{-1}(0) \cap O = \left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} \text{GL}(\mathfrak{g}) \cdot \eta \right) \cap O = \left(\bigcup_{h \in \Phi^{-1}(0)} \text{GL}(\mathfrak{g}) \cdot \beta^{-1}(h) \right) \cap O. \quad (6.16)$$

In other words, in a neighborhood of μ in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, the union of $\text{GL}(\mathfrak{g})$ -orbits that contain Lie brackets that correspond to a zero of Φ , coincides with space of all Lie brackets on $\mathfrak{g}$. This is exactly the statement we needed to prove in c). $\square$

The former theorem is theorem B in [NR66].

Remark 6.1.2. By using theorem 6.1.1, we're now able to solve the first bullet of Problem 1 a), Rigidity of Lie algebras. Indeed, if we let $H^2(\mathfrak{g}, \mathfrak{g}) = 0$, then the domain of Φ in the previous theorem reduces to 0. Under the correspondence between zeros of Φ and Lie brackets on $\mathfrak{g}$, the element zero corresponded to μ . Hence, by making use of statement c), then we get that in a neighborhood $O \subseteq \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, the set of Lie brackets on $\mathfrak{g}$ coincides with the $\text{GL}(\mathfrak{g})$ -orbit through μ . In other words, locally holds that

$$\mathcal{J}^{-1}(0) = \text{GL}(\mathfrak{g}) \cdot \mu$$

so that locally, all Lie brackets are isomorphic to μ .

We have thus found an answer to the first bullet of Problem 1 a):

Let $(\mathfrak{g}, \mu)$ be a Lie algebra with $H^2(\mathfrak{g}, \mathfrak{g}) = 0$. Then in a neighborhood of μ in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, all Lie brackets on $\mathfrak{g}$ are isomorphic to μ .

It turns out that the second bullet of Problem 1 a) can be solved with the same condition:

Theorem 6.1.3. ([CSS14], Theorem 5.3) *If $H^2(\mathfrak{g}, \mathfrak{g}) = 0$, then we can find an open neighborhood $U \subseteq \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ of μ and a smooth map $h : U \rightarrow \text{GL}(\mathfrak{g})$ such that $h(\nu) \cdot \mu = \nu$ for all Lie brackets $\nu \in U$.*

Proof. We will apply theorem 5.1.11, Openness of orbits. We work in a similar setting as before: Take $M = \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, $E = M \times \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ and $F = M \times \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}$. Let the section $\sigma := (\text{Id}_M, \sigma_2) : M \rightarrow E$ that is characterized by

$$\sigma_2 = \mathcal{J} : M \rightarrow \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}. \quad (6.17)$$

The section σ has zero $\mu \in M$ because μ is a Lie bracket, so $\mathcal{J}(\mu) = 0$. Furthermore, the Lie group $\text{GL}(\mathfrak{g})$ acts on $E = (\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}) \times (\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g})$ diagonally by

$$(A \cdot (\eta, \zeta)) = ((A \cdot \eta), (A \cdot \zeta))$$

for $A \in \text{GL}(\mathfrak{g})$, $\eta \in \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, $\zeta \in \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$. To apply theorem 5.1.11, we have to check that σ is equivariant and μ is a non-degenerate zero of $\mathcal{J}$. We already proved equivariance of $\mathcal{J}$ in equation (6.3). Hence, the only thing left to prove is non-degeneracy of μ .

If $H^2(\mathfrak{g}, \mathfrak{g}) = 0$, then the sequence

$$C^1(\mathfrak{g}, \mathfrak{g}) \xrightarrow{\delta_\mu} C^2(\mathfrak{g}, \mathfrak{g}) \xrightarrow{\delta_\mu} C^3(\mathfrak{g}, \mathfrak{g})$$

is exact. From equation (6.11), we know that

$$\delta_\mu = -d_e m_\mu : C^1(\mathfrak{g}, \mathfrak{g}) \rightarrow C^2(\mathfrak{g}, \mathfrak{g}).$$

Additionally, from equation (4.2), we see that

$$\delta_\mu = -d_\mu \mathcal{J} = -d_\mu \sigma_2 = -d_\mu^{\text{vert}} \sigma : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g}).$$

If we substitute these maps, together with the explicit expressions for the cochains, into the sequence, we get that

$$\mathfrak{g}^* \otimes \mathfrak{g} \xrightarrow{d_e m_\mu} \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \xrightarrow{d_\mu^{\text{vert}} \sigma} \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$$

is exact. Notice that, as $M = \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is a vector space, we have that

$$T_\mu M \cong M = \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$$

and that $\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ is exactly the fiber of the vector bundle E . Consequently, the exactness of the sequence is just the condition that μ is a non-degenerate zero of σ . Therefore,

we can apply theorem 5.1.11, Openness of orbits, so that we acquire a neighborhood U of μ in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, and map $h : U \rightarrow \text{GL}(\mathfrak{g})$ so that for all zeros $\nu \in U$ of $\sigma = \mathcal{J}$, we have that

$$h(\nu) \cdot \mu = \nu.$$

Since the zeros of $\mathcal{J}$ are exactly the Lie brackets, we have found the desired U and h . $\square$

For problem 2 a), we get the following result:

Theorem 6.1.4. ([CSS14], Theorem 5.4) *If $H^3(\mathfrak{g}, \mathfrak{g}) = 0$, then, as a subspace of $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, the space of Lie brackets on $\mathfrak{g}$ is a manifold in a neighborhood of μ of dimension $\dim(Z^2(\mathfrak{g}, \mathfrak{g}))$.*

Proof. We will apply theorem 5.2.1, Stability of zeros. Before we can apply this theorem, we have to check whether the assumptions are fulfilled. We note that we already proved in the proof of theorem 6.1.1 in equation (6.4) that the condition $\phi \circ \sigma = 0$ is fulfilled. Hence, the only thing left to prove before we can apply theorem 5.2.1 is the exactness of equation (5.16).

If $H^3(\mathfrak{g}, \mathfrak{g}) = 0$, then the following sequence is exact:

$$C^2(\mathfrak{g}, \mathfrak{g}) \xrightarrow{\delta_\mu} C^3(\mathfrak{g}, \mathfrak{g}) \xrightarrow{\delta_\mu} C^4(\mathfrak{g}, \mathfrak{g})$$

Again, by equation (4.2), we recognize $\delta_\mu = -d_\mu \mathcal{J} = d_\mu^{\text{vert}} \sigma : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g})$ and $\delta_\mu = \phi_\mu^{(2)} : C^3(\mathfrak{g}, \mathfrak{g}) \rightarrow C^4(\mathfrak{g}, \mathfrak{g})$, and fill in the Chevalley-Eilenberg cochains. We get that the sequence

$$\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \xrightarrow{d_\mu \mathcal{J}} \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g} \xrightarrow{\phi_\mu^{(2)}} \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}$$

is also exact, where we again identify $T_\mu M \cong \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$, and see that the fibers of E and F are defined as $E_\mu = \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ and $F_\mu = \wedge^4 \mathfrak{g}^* \otimes \mathfrak{g}$. Then we see that the exact sequence is the same as the condition in theorem 5.2.1. From statement 1 of theorem 5.2.1 then follows that $\mathcal{J}^{-1}(0)$ is locally a manifold around μ of dimension

$$\begin{aligned} \dim(\ker(d_\mu^{\text{vert}} \sigma)) &= \dim(\ker(d_\mu \mathcal{J})) \\ &= \dim(\ker(\delta_\mu : C^2(\mathfrak{g}, \mathfrak{g}) \rightarrow C^3(\mathfrak{g}, \mathfrak{g}))) \\ &= \dim(Z^2(\mathfrak{g}, \mathfrak{g})). \end{aligned}$$

This proves the assertion. $\square$

Example 6.1.5. Recall the example we gave in Chapter 2, where $\mathfrak{g} = \mathbb{R}^2$, and take $\mu = 0$. This is the 2-dimensional abelian Lie algebra. As mentioned in example 2.1.26, the moduli space of $(\mathbb{R}^2, 0)$ consists of two elements: the equivalence class of the zero bracket, and the equivalence class of all others. In particular, all Lie brackets on $\mathbb{R}^2$ other than the zero bracket are non-isomorphic to 0, so that $(\mathbb{R}^2, 0)$ cannot be rigid.

On the other hand, we can prove that the deformation space of $(\mathbb{R}^2, 0)$ is locally

smooth. Elements in $\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ take three inputs, and give 0 if these inputs are linearly dependent. However, over $\mathbb{R}^2$, any three vectors are linearly dependent. It follows that $\wedge^3 \mathfrak{g}^* \otimes \mathfrak{g} = \{0\}$, so that the Jacobiator $\mathcal{J} : \wedge^2 \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^3 \mathfrak{g}^* \otimes \mathfrak{g}$ is the zero map. As a consequence, any element in $\wedge^2 \mathfrak{g}^* \otimes \mathfrak{g}$ is a Lie bracket on $\mathfrak{g} = \mathbb{R}^2$. Furthermore, we can compute that for $\mu = 0$, the Chevalley-Eilenberg differential δ_μ vanishes, so that

$$H^k(\mathfrak{g}, \mathfrak{g}) = C^k(\mathfrak{g}, \mathfrak{g}) \text{ for all } k \in \mathbb{N}$$

and in particular $H^k(\mathfrak{g}, \mathfrak{g}) = C^k(\mathfrak{g}, \mathfrak{g}) = 0$ for $k \geq 3$, as every k vectors in $\mathbb{R}^2$ are linearly dependent for $k \geq 3$. This means that we can apply theorem 6.1.3 and conclude that the set of Lie brackets on $\mathbb{R}^2$ is locally a manifold around 0.

By Whitehead's lemma 2.4.8, and theorems 6.1.1 and 6.1.3 all semisimple Lie algebras are rigid. We give two explicit examples of this.

Example 6.1.6. Let $\mathfrak{g} = \mathfrak{sl}(\mathbb{R}^n)$ for any $n \geq 2$ as in example 2.1.7. Then we know from example 2.1.11 that $\mathfrak{sl}(\mathbb{R}^n)$ is semisimple. For any semisimple Lie algebra, we know by Whitehead's lemma 2.4.8 that

$$H^1(\mathfrak{sl}(\mathbb{R}^n), \mathfrak{sl}(\mathbb{R}^n)) = 0 \text{ and } H^2(\mathfrak{sl}(\mathbb{R}^n), \mathfrak{sl}(\mathbb{R}^n)) = 0.$$

By the latter, we can apply theorems 6.1.1 and 6.1.3 and find that $\mathfrak{sl}(\mathbb{R}^n)$ is rigid.

Example 6.1.7. Similarly, if $\mathfrak{g} = \mathfrak{so}(\mathbb{R}^m)$ for any $m \geq 3$ as in example 2.1.8. Then we know by 2.1.12 that $\mathfrak{so}(\mathbb{R}^m)$ is semisimple. Then again by using Whitehead's lemma and theorems 6.1.1 and 6.1.3, we can conclude that $\mathfrak{so}(\mathbb{R}^m)$ is rigid.

6.2 Lie algebra homomorphisms

Theorem 6.2.1. ([CSS14], Theorem 5.7) *Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism. Let G be the simply connected Lie group integrating $\mathfrak{g}$. Then there exist a neighborhood $U \subseteq H^1(\mathfrak{h}, \mathfrak{g})$ around 0 and a smooth map $\Phi : U \rightarrow H^2(\mathfrak{h}, \mathfrak{g})$ such that*

- a) $\Phi(0) = 0$ and $d_0 \Phi = 0$,
- b) *the zeros of Φ are in bijection with a subset of $\mathcal{K}^{-1}(0)$, where 0 corresponds to the original Lie algebra homomorphism ρ*
- c) *and in a neighborhood of ρ in $\mathfrak{h}^* \otimes \mathfrak{g}$, the union of the adjoint G -orbits containing Lie algebra homomorphisms that corresponds to a zero of Φ , coincides with the space of all Lie algebra homomorphisms from $\mathfrak{h}$ to $\mathfrak{g}$.*

We will derive assertions a), b) and c) from the corresponding assertions a), b) and c) of theorem 5.3.2, Kuranishi models.

Proof. Step 1:

We will follow the constructions and adopt the notations in the proof of theorem 5.3.2, Kuranishi models, to keep track of the relations and conditions between the constructions in the current setup. Let $M = \mathfrak{h}^* \otimes \mathfrak{g}$, $E = M \times (\wedge^2 \mathfrak{h}^* \otimes \mathfrak{g})$ and $F = M \times (\wedge^3 \mathfrak{h}^* \otimes \mathfrak{g})$. Let G be the unique simply connected Lie group integrating $\mathfrak{g}$, and consider the adjoint

action by G on M and E . Let the section $\sigma := (\text{Id}_M, \sigma_2) : M \rightarrow E$ that is characterized by $\sigma_2 = \mathcal{K}$, where

$$\mathcal{K}(\varphi)(u, v) = \mu(\varphi(u), \varphi(v)) - \varphi(\nu(u, v)) = \delta_0 \varphi(u, v) + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket(u, v)$$

for the bracket $\llbracket \cdot, \cdot \rrbracket$ we defined on $\wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$. Additionally, define

$$\phi : M \rightarrow \text{Hom}(E, F) : \varphi \mapsto \phi_\varphi := (\text{Id}_M, \phi_\varphi^{(2)})$$

where

$$\phi_\varphi^{(2)} = \delta_\varphi : \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}.$$

We will derive the statements a), b) and c) claimed in this theorem from the corresponding assertions a), b), and c) of theorem 5.3.2.

Step 2:

We first check the assumptions needed to use theorem 5.3.2:

1. G acts on $\mathfrak{h}^* \otimes \mathfrak{g}$ and $\wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$, so that G acts on E
2. As the adjoint action by G on $0 \in \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ is again 0, the zero section $Z : M \rightarrow E$ is preserved by the action.
3. Since for any $g \in G$ and $\eta \in \mathfrak{h}^* \otimes \mathfrak{g}$, the map $\text{Ad}_g \circ \eta$ is a Lie algebra homomorphism, the section $\mathcal{K}$ is equivariant.
4. We know that ρ is a zero of $\mathcal{K}$ because ρ is a Lie algebra homomorphism.

The only thing left to prove before we can use theorem 5.3.2 is that $\phi \circ \sigma = 0$, which is proving that

$$\phi_\varphi^{(2)} \circ \sigma_2(\varphi) = 0$$

for all $\varphi \in \mathfrak{h}^* \otimes \mathfrak{g}$. We can compute that

$$\begin{aligned} \phi_\varphi^{(2)} \circ \sigma_2(\varphi) &= \delta_\varphi \left(\delta_0 \varphi + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket \right) \\ &= (\delta_0 + \llbracket \varphi, \cdot \rrbracket) \left(\delta_0 \varphi + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket \right) \\ &= \delta_0^2 \varphi + \llbracket \varphi, \delta_0 \varphi \rrbracket + \frac{1}{2} \delta_0 \llbracket \varphi, \varphi \rrbracket + \frac{1}{2} \llbracket \varphi, \llbracket \varphi, \varphi \rrbracket \rrbracket \\ &= \llbracket \varphi, \delta_0 \varphi \rrbracket + \frac{1}{2} \delta_0 \llbracket \varphi, \varphi \rrbracket \end{aligned} \tag{6.18}$$

where in the last equation we dropped the first term because δ_0 is a differential and the last term because of the graded Jacobi identity. Lastly, by using that δ_0 is a derivation with respect to $\llbracket \cdot, \cdot \rrbracket$ and graded skew symmetry, we know that the two remaining terms cancel, and thus

$$\phi_\varphi^{(2)} \circ \sigma_2(\varphi) = 0.$$

Step 3: Now we can use theorem 5.3.2, Kuranishi models. We follow the constructions in the proof to get as much information as possible. We start by fixing a transversal τ to the orbit $G \cdot \rho$ in the point ρ , so that then

$$T_\rho \tau \oplus T_\rho(G \cdot \rho) = T_\rho M. \quad (6.19)$$

As the vector bundles E and F are already trivial, we can write $E = M \times V$ and $F = M \times W$ where $V = \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ and $W = \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}$. Then we obtain maps

$$\sigma_2|_\tau : \tau \rightarrow V$$

and

$$\phi^{(2)}|_\tau \rightarrow \text{Hom}(V, W).$$

Choose a complement K to $\text{Im}(\phi_\rho^{(2)})$ in W , so that

$$K \oplus \text{Im}(\phi_\rho^{(2)}) = W.$$

In the proof of 5.3.2, we shrank the neighborhood $U_\tau \subseteq \tau$ of ρ until all the desired conditions were fulfilled. Hence, denote by U_τ the smallest such open neighborhood. Then, by the proof, we find a vector subbundle $\hat{E} \cong U_\tau \times \hat{V}$ of E where $\hat{V} \cong (\phi_\rho^{(2)})^{-1}(K) = \ker(\phi_\rho^{(2)})$. By using a chart $\varphi : U_\tau \rightarrow \mathbb{R}^t$, we also obtain $\tilde{V}$ for which

$$\text{Im}(d_0 \sigma_2|_\tau) \oplus \tilde{V} = \hat{V}. \quad (6.20)$$

The theorem also yields a new map

$$\tilde{\sigma}_2 : S \rightarrow \tilde{V}$$

from a submanifold S of τ that contains ρ , to $\tilde{V}$. For this map holds that

- $\tilde{\sigma}_2(\rho) = 0$,
- a) $d_\rho^{\text{vert}} \tilde{\sigma} = d_\rho \tilde{\sigma}_2 = 0$,
- b) the zero set of $\tilde{\sigma}_2$ coincides with the zero set of $\sigma_2|_\tau = \mathcal{K}|_\tau$,
- c) and if ρ' is a zero of $\mathcal{K}$ sufficiently close to ρ , then $\rho' \in G \cdot \eta$ for some zero η of $\tilde{\sigma}_2$.

Step 4:

We will construct the map Φ out of the section $\tilde{\sigma}$. We start by making an identification on the codomain of $\tilde{\sigma}$. By equation (6.20), we see that

$$\tilde{V} \cong \frac{\hat{V}}{\text{Im}(d_\rho(\sigma_2|_\tau))} \cong \frac{\ker(\phi_\rho^{(2)})}{\text{Im}(d_\rho^{\text{vert}}(\sigma|_\tau))}.$$

As $\sigma_2(G \cdot \rho) = G \cdot \sigma_2(\rho) = 0$, we know that

$$d_\rho(\sigma_2|_{G \cdot \rho}) = 0$$

which means that

$$T_\rho(G \cdot \rho) \subseteq \ker(d_\rho^{\text{vert}}(\sigma)). \quad (6.21)$$

Using this and equation (6.19), we know that

$$\operatorname{Im}(d_\rho^{\text{vert}}(\sigma|_\tau)) = \operatorname{Im}(d_\rho^{\text{vert}}(\sigma)).$$

Substituting this information, together with the actual expressions for the maps σ and $\phi_\rho^{(2)}$, we get that

$$\tilde{V} \cong \frac{\ker(\phi_\rho^{(2)})}{\operatorname{Im}(d_\rho^{\text{vert}}\sigma)} = \frac{\ker(\delta_\rho : C^2(\mathfrak{h}, \mathfrak{g}) \rightarrow C^3(\mathfrak{h}, \mathfrak{g}))}{\operatorname{Im}(d_\rho \mathcal{K})} \quad (6.22)$$

$$= \frac{\ker(\delta_\rho : C^2(\mathfrak{h}, \mathfrak{g}) \rightarrow C^3(\mathfrak{h}, \mathfrak{g}))}{\operatorname{Im}(\delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g}))} = H^2(\mathfrak{h}, \mathfrak{g}). \quad (6.23)$$

where we used the computation in (4.6) which shows that $d_\rho \mathcal{K} = \delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g})$. Denote the linear isomorphism between $\tilde{V}$ and $H^2(\mathfrak{h}, \mathfrak{g})$ by

$$\alpha : \tilde{V} \rightarrow H^2(\mathfrak{h}, \mathfrak{g})$$

that sends $0 \in \tilde{V}$ to $0 \in H^2(\mathfrak{h}, \mathfrak{g})$.

Step 5:

We can also identify the domain of $\tilde{\sigma}_2$. By equation (5.31), we know that

$$T_\rho S \cong \frac{\ker(d_\rho^{\text{vert}}(\sigma))}{\operatorname{Im}(d_e m_\rho)} = \frac{\ker(d_\rho \mathcal{K})}{\operatorname{Im}(d_e m_\rho)} = \frac{\ker(\delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g}))}{\operatorname{Im}(d_e m_\rho)}.$$

We now claim that

$$\delta_\rho = -d_e m_\rho : C^0(\mathfrak{h}, \mathfrak{g}) \rightarrow C^1(\mathfrak{h}, \mathfrak{g}).$$

We know that $\delta_\rho : \mathfrak{g} \rightarrow \mathfrak{h}^* \otimes \mathfrak{g}$ is defined as

$$\delta_\rho v(h) = \mu(\rho(h), v)$$

for $v \in \mathfrak{g}$ and $h \in \mathfrak{h}$. On the other hand, we know that $m_\rho : G \rightarrow \mathfrak{h}^* \otimes \mathfrak{g} : g \mapsto \operatorname{Ad}_g \circ \rho$. Its derivative at e , $d_e m_\rho : \mathfrak{g} \rightarrow \mathfrak{h}^* \otimes \mathfrak{g}$ is then defined for $v \in \mathfrak{g}$ and $h \in \mathfrak{h}$ as

$$d_e m_\rho(v)(h) = (\operatorname{ad}_{\mathfrak{g}}(v) \circ \rho)(h) = \mu(v, \rho(h)) = -\delta_\rho v(h)$$

from which it is clear that

$$\delta_\rho = -d_e m_\rho : C^0(\mathfrak{h}, \mathfrak{g}) \rightarrow C^1(\mathfrak{h}, \mathfrak{g}). \quad (6.24)$$

We can use this, to further identify

$$T_\rho S \cong \frac{\ker(\delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g}))}{\operatorname{Im}(d_e m_\rho)} = \frac{\ker(\delta_\rho : C^1(\mathfrak{h}, \mathfrak{g}) \rightarrow C^2(\mathfrak{h}, \mathfrak{g}))}{\operatorname{Im}(\delta_\rho : C^0(\mathfrak{h}, \mathfrak{g}) \rightarrow C^1(\mathfrak{h}, \mathfrak{g}))} = H^1(\mathfrak{h}, \mathfrak{g}). \quad (6.25)$$

As S is a manifold, we can pick a chart

$$\varphi : (S, \rho) \rightarrow (T_\rho S, 0),$$

which is then locally a diffeomorphism, such that a neighborhood $U_S \subseteq S$ around ρ can be identified with a neighborhood of 0 in $T_\rho S$. Together with the previous identification $T_\rho S \cong H^1(\mathfrak{h}, \mathfrak{g})$, we obtain a diffeomorphism

$$\beta : (U_S, \rho) \rightarrow (U, 0)$$

for some neighborhood $U \subseteq H^1(\mathfrak{h}, \mathfrak{g})$ of 0.

Now that we have constructed isomorphisms on the domain and codomain of $\tilde{\sigma}_2$, we can define the map Φ as

$$\Phi := \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1} : U \rightarrow H^2(\mathfrak{h}, \mathfrak{g}) \quad (6.26)$$

Step 6:

Now we can prove statement a). First, notice that $\beta^{-1}(0) = \rho$ and $\tilde{\sigma}_2(\rho) = 0$ as ρ is a zero of $\sigma_2 = \mathcal{K}$, so that

$$\begin{aligned} \Phi(0) &= \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1}(0) \\ &= \alpha \circ \tilde{\sigma}_2(\rho) \\ &= \alpha(0) = 0 \end{aligned}$$

because α is a linear isomorphism. Secondly, as $d_\rho \tilde{\sigma}_2 = 0$ by statement a) of theorem 5.3.2, and $\beta^{-1}(0) = \rho$, we have that

$$d_0 \Phi = 0, \quad (6.27)$$

which proves statement a).

Step 7:

Now we will prove statement b). We can see that for $h \in U$

$$\begin{aligned} h \text{ is a zero of } \Phi &\iff \Phi(h) = 0 \\ &\iff \tilde{\sigma}_2 \circ \beta^{-1}(h) = 0 \\ &\iff \beta^{-1}(h) \text{ is a zero of } \tilde{\sigma}_2 \end{aligned}$$

Now by statement b) of 5.3.2, a zero of $\tilde{\sigma}_2$ is a zero $\sigma_2 = \mathcal{K}$. Therefore, for any zero h of Φ , we know that $\beta^{-1}(h)$ is a zero of $\mathcal{K}$, meaning that it is a Lie algebra homomorphism from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$. From this we conclude that zeros of Φ correspond to Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$. Moreover, $\beta^{-1}(0) = \rho$, so that the zero $h = 0$ of Φ corresponds the initial Lie algebra homomorphism ρ . This confirms statement b).

Step 8:

Lastly, by statement c) of theorem 5.3.2, we know that we can find an open neighborhood $O \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$ of ρ such that

$$\mathcal{K}^{-1}(0) \cap O = \left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} G \cdot \eta \right) \cap O = \left(\bigcup_{h \in \Phi^{-1}(0)} G \cdot \beta^{-1}(h) \right) \cap O. \quad (6.28)$$

In other words, in a neighborhood of ρ in $\mathfrak{h}^* \otimes \mathfrak{g}$, the union of adjoint G -orbits that contain Lie algebra homomorphisms that correspond to a zero of Φ , coincides with the space of all Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$. This is exactly the statement we wanted in c). $\square$

Theorem 6.2.1 is theorem A of [NR67b].

Remark 6.2.2. By using theorem 6.2.1, we're now able to solve the first bullet of Problem 1 b), Rigidity of Lie algebra homomorphisms. If we let $H^1(\mathfrak{h}, \mathfrak{g}) = 0$, then the domain of Φ in theorem 6.2.1 reduces to 0. Under the correspondence between zeros of Φ and Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$, the element zero corresponded to ρ . Hence, by making use of statement c), then we get that in a neighborhood $O \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$, the set of Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$ coincides with the adjoint G -orbit through ρ . In other words, locally holds that

$$\mathcal{K}^{-1}(0) = G \cdot \rho$$

so that locally, all Lie algebra homomorphisms $\rho' : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ are of the form

$$\rho' = \text{Ad}_g \circ \rho \text{ for some } g \in G.$$

Hence, this leads to the following statement:

Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism with $H^1(\mathfrak{h}, \mathfrak{g}) = 0$. Then in a neighborhood of ρ in $\mathfrak{h}^* \otimes \mathfrak{g}$, all Lie algebra homomorphisms $\rho' : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ are of the form

$$\rho' = \text{Ad}_g \circ \rho \text{ for some } g \in G.$$

Again, the second bullet of problem 1 b) can be solved with the same condition.

Theorem 6.2.3. ([CSS14], Theorem 5.9) Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism. If $H^1(\mathfrak{h}, \mathfrak{g}) = 0$, then we can find an open neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$ of ρ and a smooth map $h : U \rightarrow G$ such that $\rho' = \text{Ad}_{h(\rho')} \circ \rho$ for all Lie algebra homomorphisms $\rho' \in U$.

Proof. We will apply theorem 5.1.11, Openness of orbits. Before we can apply this theorem, it remains to verify the hypotheses. We take the same manifold, vector bundle E , section and Lie group action as in the previous theorem, theorem 6.2.1. Since for any $g \in G$ and $\eta \in \mathfrak{h}^* \otimes \mathfrak{g}$, the map $\text{Ad}_g \circ \eta$ is a Lie algebra homomorphism, the section $\mathcal{K}$ is equivariant. Hence, we only need to prove that ρ is a non-degenerate zero of the section $\mathcal{K}$.

Suppose $H^1(\mathfrak{h}, \mathfrak{g}) = 0$, then the sequence

$$C^0(\mathfrak{h}, \mathfrak{g}) \xrightarrow{\delta_\rho} C^1(\mathfrak{h}, \mathfrak{g}) \xrightarrow{\delta_\rho} C^2(\mathfrak{h}, \mathfrak{g})$$

is exact. We know from equation (6.24) that

$$\delta_\rho = -d_e m_\rho : \mathfrak{g} \rightarrow \mathfrak{h}^* \otimes \mathfrak{g}.$$

Additionally, from equation (4.6) we also know that

$$\delta_\rho = d_\rho \mathcal{K} = d_\rho^{\text{vert}} \sigma : \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}.$$

By substituting these maps, together with the expressions for the cochains into the sequence, we obtain the exact sequence

$$\mathfrak{g} \xrightarrow{d_{em\rho}} \mathfrak{h}^* \otimes \mathfrak{g} \xrightarrow{d_{\rho}^{\text{vert}\sigma}} \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}.$$

As $M = \mathfrak{h}^* \otimes \mathfrak{g}$ is a vector space,

$$T_{\rho}M \cong M = \mathfrak{h}^* \otimes \mathfrak{g}$$

and that $\wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ is the fiber of E . Therefore, the condition that the former sequence is exact, coincides with the condition that ρ is a non-degenerate zero of $\mathcal{K}$. Hence, we can apply theorem 5.1.11 which yields a neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$ of ρ , and a map $h : U \rightarrow G$ so that for all zeros $\rho' \in U$ of $\mathcal{K}$, we have that

$$h(\rho') \cdot \rho = \rho'$$

or equivalently

$$\text{Ad}_{h(\rho')} \circ \rho = \rho'.$$

Since the zeros of $\mathcal{K}$ are exactly the Lie algebra homomorphisms, we found the desired U and h . $\square$

Remark 6.2.4. In Theorem 6.2.3, rigidity is stated with respect to the adjoint action of G on $\mathfrak{h}^* \otimes \mathfrak{g}$ by postcomposition:

$$G \times (\mathfrak{h}^* \otimes \mathfrak{g}) \rightarrow \mathfrak{h}^* \otimes \mathfrak{g} : (g, \eta) \mapsto \text{Ad}_g \circ \eta.$$

However, another natural equivalence relation on Lie algebra homomorphisms from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$ is given by postcomposition with *any* element of $\text{Aut}(\mathfrak{g}, \mu)$, not just an inner automorphisms. Indeed, if $\eta : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ is a Lie algebra homomorphism and $A \in \text{Aut}(\mathfrak{g}, \mu)$, then $A \circ \eta$ is again a Lie algebra homomorphism from $(\mathfrak{h}, \nu)$ to $(\mathfrak{g}, \mu)$, since for all $u, v \in \mathfrak{h}$:

$$A \circ \eta(\nu(u, v)) = A(\mu(\eta(u), \eta(v))) = \mu(A \circ \eta(u), A \circ \eta(v)),$$

where we used that η is a homomorphism in the first equality and that A preserves μ in the second. This defines an action of $\text{Aut}(\mathfrak{g}, \mu)$ on $\mathfrak{h}^* \otimes \mathfrak{g}$ by postcomposition.

The adjoint G -action factors through this action: the action of $g \in G$ on $\mathfrak{h}^* \otimes \mathfrak{g}$ coincides with the $\text{Aut}(\mathfrak{g}, \mu)$ -action of $\text{Ad}_g \in \text{Aut}(\mathfrak{g}, \mu)$. In particular, every G -orbit is contained in an $\text{Aut}(\mathfrak{g}, \mu)$ -orbit, so rigidity with respect to G is a priori a stronger notion than rigidity with respect to $\text{Aut}(\mathfrak{g}, \mu)$.

Theorem 6.2.5. ([CSS14], Theorem 5.10) *Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism. Let $\rho^* : \wedge^k \mathfrak{g}^* \otimes \mathfrak{g} \rightarrow \wedge^k \mathfrak{h}^* \otimes \mathfrak{g}$ be the chain map defined for $\omega \in \wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$ and $v_1, \dots, v_k \in \mathfrak{g}$ as*

$$(\rho^* \omega)(v_1, \dots, v_k) := \omega(\rho(v_1), \dots, \rho(v_k)).$$

If its induced map in cohomology $H^1(\rho^) : H^1(\mathfrak{g}, \mathfrak{g}) \rightarrow H^1(\mathfrak{h}, \mathfrak{g})$ is surjective, we can find a neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$ of ρ and a smooth map $h : U \rightarrow \text{Aut}(\mathfrak{g}, \mu)$ such that*

$$\rho' = h(\rho') \cdot \rho$$

for all Lie algebra homomorphisms $\rho' \in U$.

Proof. To prove this theorem, we apply theorem 5.1.11 where $M = \mathfrak{h}^* \otimes \mathfrak{g}$, $E = M \times \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$, $G = \text{Aut}(\mathfrak{g}, \mu)$ which acts on M and E by postcomposing an element in M or E by an element in $\text{Aut}(\mathfrak{g}, \mu)$. Furthermore, we take $\sigma = \mathcal{K}$ as section of E . We have to prove that $\mathcal{K}$ is equivariant with respect to this action. To this end, take some $\varphi \in \mathfrak{h}^* \otimes \mathfrak{g}$, some $A \in \text{Aut}(\mathfrak{g}, \mu)$ and $u, v \in \mathfrak{h}$. Then it follows that

$$\begin{aligned} \mathcal{K}(A \cdot \varphi)(u, v) &= \mathcal{K}(A \circ \varphi)(u, v) \\ &= \mu(A \circ \varphi(u), A \circ \varphi(v)) - A \circ \varphi(\nu(u, v)) \\ &= A \circ \mu(\varphi(u), \varphi(v)) - A \circ \varphi(\nu(u, v)) \\ &= (A \circ \mathcal{K})(\varphi)(u, v) \\ &= (A \cdot \mathcal{K})(\varphi)(u, v) \end{aligned}$$

where in the third equality, we used that $A \in \text{Aut}(\mathfrak{g}, \mu)$, so that A preserves the bracket. This proves that $\mathcal{K}$ is equivariant with respect to the action by $\text{Aut}(\mathfrak{g}, \mu)$.

We also remark that ρ^* defined for $\omega \in \wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$ and $v_1, \dots, v_k \in \mathfrak{g}$ as

$$(\rho^* \omega)(v_1, \dots, v_k) := \omega(\rho(v_1), \dots, \rho(v_k))$$

is a chain map between $C^\bullet(\mathfrak{g}, \mathfrak{g})$ and $C^\bullet(\mathfrak{h}, \mathfrak{g})$. Indeed,

$$\begin{aligned} ((\rho^* \circ \delta_\mu)(\omega))(v_0, \dots, v_k) &= \delta_\mu \omega(\rho(v_0), \dots, \rho(v_k)) \\ &= \sum_{i=0}^k (-1)^i \mu(\rho(v_i), \omega(\rho(v_0), \dots, \widehat{\rho(v_i)}, \dots, \rho(v_k))) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\mu(\rho(v_i), \rho(v_j)), \rho(v_0), \dots, \widehat{\rho(v_i)}, \dots, \widehat{\rho(v_j)}, \dots, \rho(v_k)) \\ &= \sum_{i=0}^k (-1)^i \mu(\rho(v_i), \omega(\rho(v_0), \dots, \widehat{\rho(v_i)}, \dots, \rho(v_k))) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega(\rho(\nu(v_i, v_j)), \rho(v_0), \dots, \widehat{\rho(v_i)}, \dots, \widehat{\rho(v_j)}, \dots, \rho(v_k)) \\ &= \delta_\rho(\rho^* \omega)(v_0, \dots, v_k) \end{aligned}$$

for all $\omega \in \wedge^k \mathfrak{g}^* \otimes \mathfrak{g}$ and $v_0, \dots, v_k \in \mathfrak{h}$. Here, we used in the third equation that ρ is a Lie algebra homomorphism. This proves that ρ^* is a chain map, so that $\rho^*(Z^1(\mathfrak{g}, \mathfrak{g})) \subseteq Z^1(\mathfrak{h}, \mathfrak{g})$ and $\rho^*(B^1(\mathfrak{g}, \mathfrak{g})) \subseteq B^1(\mathfrak{h}, \mathfrak{g})$, and thus ρ yields a map in cohomology

$$H^k(\rho^*) : H^k(\mathfrak{g}, \mathfrak{g}) \rightarrow H^k(\mathfrak{h}, \mathfrak{g}).$$

We will now prove that the condition that

$$H^1(\rho^*) : H^1(\mathfrak{g}, \mathfrak{g}) \rightarrow H^1(\mathfrak{h}, \mathfrak{g})$$

is surjective implies that ρ is a non-degenerate zero of $\mathcal{K}$. We first prove that this implies that $\rho^* : Z^1(\mathfrak{g}, \mathfrak{g}) \rightarrow Z^1(\mathfrak{h}, \mathfrak{g})$ is surjective. To do so, we need to prove that for

all $\eta \in Z^1(\mathfrak{h}, \mathfrak{g})$, $\eta \in \rho^*(Z^1(\mathfrak{g}, \mathfrak{g}))$. Therefore, take such an $\eta \in Z^1(\mathfrak{h}, \mathfrak{g})$, then η defines a class in $H^1(\mathfrak{h}, \mathfrak{g})$:

$$[\eta] \in H^1(\mathfrak{h}, \mathfrak{g}).$$

As $H^1(\rho^*)$ is surjective, we can write

$$[\eta] = H^1(\rho^*)([\phi]) \text{ for some } \phi \in Z^1(\mathfrak{g}, \mathfrak{g}).$$

This means exactly that

$$[\eta] = [\rho^* \phi]$$

and thus

$$\eta = \rho^* \phi + \delta_\rho g \text{ for some } g \in C^0(\mathfrak{h}, \mathfrak{g}) = \mathfrak{g}$$

Now for $g \in \mathfrak{g}$, consider the map $\text{ad}_g(g) = \delta_\mu(g) : \mathfrak{g} \rightarrow \mathfrak{g}$, which is clearly an element of $Z^1(\mathfrak{g}, \mathfrak{g})$. Then we can see that for all $h \in \mathfrak{h}$

$$\begin{aligned} \delta_\rho g(h) &= \mu(\rho(h), g) \\ &= -\mu(g, \rho(h)) \\ &= -\rho^*(\text{ad}_g(g))(h) \end{aligned}$$

proving that $\delta_\rho g \in \text{Im}(\rho^*)$. In particular

$$\eta = \rho^* \phi - \rho^*(\text{ad}_g(g)) \in \text{Im}(\rho^*)$$

We therefore proved that $\rho^* Z^1(\mathfrak{g}, \mathfrak{g}) \rightarrow Z^1(\mathfrak{h}, \mathfrak{g})$ is surjective. By identifying $Z^1(\mathfrak{h}, \mathfrak{g}) = \ker(\delta_\rho : \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g})$, we can see that this surjectivity is equivalent to the following sequence being exact:

$$Z^1(\mathfrak{g}, \mathfrak{g}) \xrightarrow{\rho^*} \mathfrak{h}^* \otimes \mathfrak{g} \xrightarrow{\delta_\rho} \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g} \quad (6.29)$$

We now want to prove that the exactness of this sequence is equivalent to ρ being a non-degenerate zero of $\mathcal{K}$. To see this, remark that

$$\begin{aligned} \text{Aut}(\mathfrak{g}, \mu) &:= \{A \in \text{GL}(\mathfrak{g}) \mid A \cdot \mu = \mu\} \\ &= \{A \in \text{GL}(\mathfrak{g}) \mid A\mu(u, v) = \mu(Au, Av) \forall u, v \in \mathfrak{g}\}. \end{aligned}$$

To compute its tangent space, take a curve $A_t \in \text{Aut}(\mathfrak{g}, \mu)$ with $A_0 = \text{Id}_\mathfrak{g}$, which then satisfies for all $u, v \in \mathfrak{g}$

$$A_t \mu(u, v) = \mu(A_t u, A_t v).$$

Differentiation with respect to t in $t = 0$ gives

$$\begin{aligned} \dot{A}_0 \mu(u, v) &= \mu(\dot{A}_0 u, A_0 v) + \mu(A_0 u, \dot{A}_0 v) \\ &= \mu(\dot{A}_0 u, v) + \mu(u, \dot{A}_0 v) \end{aligned}$$

which is equivalent to

$$\delta_\mu \dot{A}_0 = 0$$

or

$$\dot{A}_0 \in Z^1(\mathfrak{g}, \mathfrak{g}).$$

We can see that the condition for $A \in \text{GL}(\mathfrak{g})$ to be in $T_e \text{Aut}(\mathfrak{g}, \mu)$ is the same condition as to be in $Z^1(\mathfrak{g}, \mathfrak{g})$, so that

$$T_e \text{Aut}(\mathfrak{g}, \mu) = Z^1(\mathfrak{g}, \mathfrak{g}).$$

After identifying $T_\rho M = \mathfrak{h}^* \otimes \mathfrak{g}$, $d_\rho m_\rho = \rho^*$ with respect to the action of $\text{Aut}(\mathfrak{g}, \mu)$, and $d_\rho^{\text{vert}} \sigma = \delta_\rho$ as proven in theorem 6.2.3, we can see that the exactness of equation (6.29) is equivalent to the exactness of

$$T_e \text{Aut}(\mathfrak{g}, \mu) \xrightarrow{d_\rho m_\rho} T_\rho M \xrightarrow{d_\rho^{\text{vert}} \sigma} E_\rho$$

which is the condition that ρ is a non-degenerate zero of $\mathcal{K}$. We can therefore use theorem 5.1.11 and find a neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{g}$ of ρ and a smooth map $h : U \rightarrow \text{Aut}(\mathfrak{g}, \mu)$ such that

$$\rho' = h(\rho') \cdot \rho$$

for all Lie algebra homomorphisms $\rho' \in U$. □

If we compare theorems 6.2.1 and 6.2.5, we can see that the condition that $H^1(\mathfrak{h}, \mathfrak{g}) = 0$ implies that the map $H^1(\rho^*)$ is surjective. This agrees with remark 6.2.4 where we observed that rigidity with respect to the adjoint G -action is a stronger notion than rigidity with respect to $\text{Aut}(\mathfrak{g}, \mu)$.

The following theorem gives an answer to problems 2 b) and 3, Stability of Lie algebra homomorphisms.

Theorem 6.2.6. ([CSS14], Theorem 5.12) *Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism. If $H^2(\mathfrak{h}, \mathfrak{g}) = 0$, then the space of Lie algebra homomorphisms $\mathfrak{h} \rightarrow \mathfrak{g}$ is a manifold in a neighborhood of ρ , which is of dimension $\dim(Z^1(\mathfrak{h}, \mathfrak{g}))$. Moreover, in that case ρ is stable.*

Proof. We will apply theorem 5.2.1, Stability of zeros. Before we can apply this theorem, we have to check whether the assumptions are fulfilled. We take the same manifold, vector bundles and vector bundle map as in the proof of theorem 6.2.1. Suppose $H^2(\mathfrak{h}, \mathfrak{g}) = 0$, then the sequence

$$C^1(\mathfrak{h}, \mathfrak{g}) \xrightarrow{\delta_\rho} C^2(\mathfrak{h}, \mathfrak{g}) \xrightarrow{\delta_\rho} C^3(\mathfrak{h}, \mathfrak{g})$$

is exact. As shown before in equation (4.6),

$$\delta_\rho = d_\rho \mathcal{K} : \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g},$$

and $\delta_\rho = \phi_\rho^{(2)} : \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}$. Substituting this information, together with the expressions for the cochains, we get that the following sequence is exact:

$$\mathfrak{h}^* \otimes \mathfrak{g} \xrightarrow{d_\rho \mathcal{K}} \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g} \xrightarrow{\phi_\rho^{(2)}} \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}.$$

Here, we identified $T_\rho M \cong \mathfrak{h}^* \otimes \mathfrak{g}$, and $E_\rho = \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g}$ and $F_\rho = \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}$. The exactness of this sequence is the same as the condition in theorem 5.2.1. As we already proved

the condition that $\phi \circ \sigma = 0$ in equation (6.18), it is justified to use theorem 5.2.1. From statement 1 of theorem 5.2.1 then follows that $\mathcal{K}^{-1}(0)$ is a manifold around ρ of dimension $\dim(Z^1(\mathfrak{h}, \mathfrak{g}))$.

Moreover, we claim that from statement 2 of theorem 5.2.1 follows that ρ is stable. Take any Lie bracket μ' on $\mathfrak{g}$. Then instead of considering the graded Lie bracket of definition 3.1.13 with respect to the Lie bracket μ , we can consider this bracket defined with any Lie bracket μ' defined on $\mathfrak{g}$:

$$\llbracket \varphi, \psi \rrbracket_{\mu'}(u_1, \dots, u_{h+k}) := \sum_{\sigma \in S_{h,k}} \text{sgn}(\sigma) \mu'(\varphi(u_{\sigma(1)}, \dots, u_{\sigma(h)}), \psi(u_{\sigma(h+1)}, \dots, u_{\sigma(h+k)}))$$

for $u_1, \dots, u_{h+k} \in \mathfrak{h}$, $\varphi \in C^h(\mathfrak{h}, \mathfrak{g})$ and $\psi \in C^k(\mathfrak{h}, \mathfrak{g})$.

Now we take the section $\tau^{\mu'} = (\text{Id}_M, \tau_2^{\mu'}) : M \rightarrow E$ defined for $\varphi \in \mathfrak{h}^* \otimes \mathfrak{g}$ and $u, v \in \mathfrak{h}$ as

$$\tau_2^{\mu'}(\varphi)(u, v) = \mu'(\varphi(u), \varphi(v)) - \varphi(\nu(u, v)) = \delta_0 \varphi(u, v) + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket_{\mu'}(u, v)$$

and the vector bundle map $\psi^{\mu'}$ where the vertical part in $\varphi \in \mathfrak{h}^* \otimes \mathfrak{g}$ is given by

$$\delta_0 + \llbracket \varphi, \cdot \rrbracket_{\mu'} : \wedge^2 \mathfrak{h}^* \otimes \mathfrak{g} \rightarrow \wedge^3 \mathfrak{h}^* \otimes \mathfrak{g}$$

Then we claim that the composition $\psi^{\mu'} \circ \tau^{\mu'}$ is 0. This comes down to proving for any $\varphi \in \mathfrak{h}^* \otimes \mathfrak{g}$ that

$$(\delta_0 + \llbracket \varphi, \cdot \rrbracket_{\mu'}) (\delta_0 \varphi + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket_{\mu'}) = 0$$

A computation exactly as in equations 6.18, using the properties of the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{g}), \delta_0, \llbracket \cdot, \cdot \rrbracket_{\mu'})$, shows this identity. Hence, by the second bullet of theorem 5.2.1, for any Lie bracket μ' close to μ , there exists a $\rho' \in \mathfrak{h}^* \otimes \mathfrak{g}$ close to ρ such that

$$0 = \tau_2^{\mu'}(\rho')$$

or for all $u, v \in \mathfrak{h}$

$$0 = \tau_2^{\mu'}(\rho')(u, v) = \mu'(\rho'(u), \rho'(v)) - \rho'(\nu(u, v))$$

meaning exactly that $\rho' : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu')$ is a Lie algebra homomorphism. This proves that ρ is stable. $\square$

Example 6.2.7. Let $\rho : (\mathfrak{h}, \nu) \rightarrow (\mathfrak{g}, \mu)$ be a Lie algebra homomorphism for a semisimple Lie algebra $(\mathfrak{h}, \nu)$. Then we know by Whitehead's lemma 2.4.8 that

$$H^1(\mathfrak{h}, \mathfrak{g}) = 0 \text{ and } H^2(\mathfrak{h}, \mathfrak{g}) = 0.$$

Using theorems 6.2.1, 6.2.3 and 6.2.6 we can see that ρ is both rigid and stable.

6.3 The controlling DGLA

The proofs of the results in sections 6.1 and 6.2 make use of the same abstract structure. The observation that the geometric setup of theorems 5.1.11, 5.2.1 and 5.3.2 can be assembled uniformly from the controlling DGLA, as we do in this section, is our own contribution, although the individual ingredients are classical. It is inspired by [DM25] and sections 6.1 and 6.2 of this thesis, which were based on [CSS14]. The presentation here serves two purposes: it clarifies the structure behind the proofs for Lie algebras and Lie algebra homomorphisms given in the preceding sections, and it verifies, once and for all at the level of an arbitrary controlling DGLA, the assumptions required by the geometric theorems of chapter 5, so that we can apply them to the deformation theory of Lie subalgebras in section 6.4, without having to check these assumptions computationally.

In both cases, Lie algebras and Lie algebra homomorphisms, the input data is the controlling DGLA, as discussed in section 4.4. From this DGLA, we can also make the construction needed in the theorem 5.1.11, Openness of orbits, theorem 5.2.1, Stability of zeros, and theorem 5.3.2, Kuranishi models, in chapter 5. Given a controlling DGLA $(L^\bullet, d, [\cdot, \cdot])$, the geometric setup of theorem 5.3.2 is assembled uniformly as follows. We set

$$M := L^1, \quad E := M \times L^2, \quad F := M \times L^3,$$

and G a connected Lie group integrating the Lie algebra $(L^0, [\cdot, \cdot])$, which acts on L^1 by gauge action and on L^2 by adjoint action, as defined in definition 4.4.3. By taking the diagonal action, it then also acts on the trivial vector bundle E . The section $\sigma : M \rightarrow E$ is defined by the Maurer-Cartan equation:

$$\sigma_2(\varphi) := d\varphi + \frac{1}{2}[\varphi, \varphi] \text{ for all } \varphi \in L^1,$$

so that the zeros of σ are exactly the Maurer-Cartan elements of $(L^\bullet, d, [\cdot, \cdot])$, which correspond to the structures we are studying: Lie brackets in the first case, and Lie algebra homomorphisms in the second. The vector bundle map $\phi : M \rightarrow \text{Hom}(E, F)$ is defined by

$$\phi_\varphi^{(2)} := d + [\varphi, \cdot] : L^2 \rightarrow L^3,$$

which is the differential of the DGLA twisted by φ . We obtain a diagram

$$\begin{array}{ccc} G \circ L^1 \times L^2 & \xrightarrow{\phi} & L^1 \times L^3 \\ \sigma \uparrow \downarrow & \swarrow & \\ G \circ L^1 & & \end{array}$$

inspired by the diagram in [DM25] on page 3, and by the diagrams in theorems 5.1.11, 5.2.1 and 5.3.2.

We would like to be able to apply theorems 5.1.11, 5.2.1 and 5.3.2 on this structure. However, to that end, we need to check whether the necessary assumptions

are fulfilled. Propositions 6.3.1 and 6.3.2 will prove exactly that these assumptions are fulfilled.

Proposition 6.3.1. *For a DGLA $(L^\bullet, d, [\cdot, \cdot])$ where the Lie group G integrating the Lie algebra $(L^0, [\cdot, \cdot])$ acts on L^1 and L^2 by gauge action, we have the following statements:*

1. *The Lie group G acts on $E = L^1 \times L^2$*
2. *The Lie group G preserves the zero section $Z : L^1 \rightarrow L^1 \times L^2$.*
3. *We have the relation $\phi \circ \sigma = 0$.*

Proof. We know that G acts on L^1 and L^2 , so that the first statement follows from taking the diagonal action

$$G \times (L^1 \times L^2) \rightarrow (L^1 \times L^2) : (g, (l_1, l_2)) \mapsto (g \cdot l_1, g \cdot l_2).$$

The second statement follows from the fact that the adjoint action of G on L^2 is linear. The condition $\phi \circ \sigma = 0$, required by theorem 5.3.2, immediately follows from the properties of the DGLA. For any $\varphi \in L^1$, it holds that

$$\begin{aligned} (d + [\varphi, \cdot]) (d\varphi + \tfrac{1}{2}[\varphi, \varphi]) &= d^2\varphi + [\varphi, d\varphi] + d[\varphi, \varphi] + \tfrac{1}{2}[\varphi, [\varphi, \varphi]] \\ &= [\varphi, d\varphi] + d[\varphi, \varphi] = 0 \end{aligned}$$

where in the second equality we used that d is a differential and the graded Jacobi identity of $[\cdot, \cdot]$. In the last equality we used the fact that d is a derivation with respect to $[\cdot, \cdot]$, so that the two terms cancel. $\square$

Statement 3 in proposition 6.3.1 is called the Bianchi identity in for example [Baa19] in definition 1.1.6.

Proposition 6.3.2. *Suppose that we have a DGLA $(L^\bullet, d, [\cdot, \cdot])$, with the Lie group G integrating $(L^0, [\cdot, \cdot])$, acting on L^2 by adjoint action and on L^1 by gauge action. Then $\sigma_2(\varphi) = d\varphi + \tfrac{1}{2}[\varphi, \varphi]$ is equivariant.*

Proof. To prove this, we first prove infinitesimal equivariance, meaning for all $x \in L^0$ and $\varphi \in L^1$ the condition that

$$\frac{d}{dt}\bigg|_{t=0} \sigma_2(\exp(tx) \cdot \varphi) = \frac{d}{dt}\bigg|_{t=0} \exp(tx) \cdot \sigma_2(\varphi).$$

or equivalently

$$d_\varphi \sigma_2([x, \varphi] - dx) = [x, d\varphi + \tfrac{1}{2}[\varphi, \varphi]]. \quad (6.30)$$

where $d_\varphi \sigma_2$ is the derivative of σ_2 at φ , which is equal to

$$d_\varphi \sigma_2 \psi = d\psi + [\varphi, \psi].$$

If we compute the left-hand side of equation (6.30), we get

$$\text{LHS} = \frac{d}{dt}\bigg|_{t=0} \sigma_2(\exp(tx) \cdot \varphi)$$

$$\begin{aligned}
&= d_\varphi \sigma_2([x, \varphi] - dx) \\
&= d[x, \varphi] + [\varphi, [x, \varphi]] - [\varphi, dx].
\end{aligned}$$

By using the graded Jacobi identity, graded skew symmetry and the fact that d is a derivation with respect to $[\cdot, \cdot]$, we get that this is equal to

$$\begin{aligned}
&= [dx, \varphi] + [x, d\varphi] + \frac{1}{2}[x, [\varphi, \varphi]] - [dx, \varphi] \\
&= [x, d\varphi] + \frac{1}{2}[x, [\varphi, \varphi]] \\
&= [x, d\varphi + \frac{1}{2}[\varphi, \varphi]] = \text{RHS}
\end{aligned}$$

which is the right-hand side of equation (6.30). Hence, we proved that σ_2 is infinitesimally equivariant. As G is connected, infinitesimal equivariance also implies equivariance of σ_2 . Therefore, the section σ is equivariant. $\square$

6.4 Lie subalgebras

In this chapter, we apply the theorems of chapter 5 on the deformation theory of Lie subalgebras. This time, we use the observations of chapter 6.3, and maximally use the DGLA formalism in the proofs of the theorems. In particular, we know that for a Lie subalgebra $\mathfrak{h} \subseteq (\mathfrak{g}, \mu)$ with complement $\mathfrak{h}^c$ that is also a Lie subalgebra, we had the DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, [\cdot, \cdot])$. On this DGLA, we can immediately apply the constructions made in chapter 6.3 to get the final results on the questions stated in chapter 1.

The results of this section are not new. They are already implied by the more general results in section 5.3 in [CSS14]. However, we give a new way of proving these results. The contribution of this section lies in the presentation and derivation of these results within the DGLA framework developed in the previous chapters. More precisely, we show how the fundamental results in the deformation theory of Lie subalgebras can be recovered from the controlling DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, [\cdot, \cdot])$ and from the general geometric results established in Chapter 5.

Theorem 6.4.1. *Let $(\mathfrak{g}, \mu)$ be a Lie algebra and suppose it has a Lie subalgebra $\mathfrak{h}$ that admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. Take the Lie group K integrating $\mathfrak{h}^c$ that is a subgroup of G , the simply connected Lie group integrating $\mathfrak{g}$. Then there exist an open neighborhood U of 0 in $H^1(\mathfrak{h}, \mathfrak{h}^c)$ and a smooth map $\Phi : U \rightarrow H^2(\mathfrak{h}, \mathfrak{h}^c)$ for which*

- a) $\Phi(0) = 0$ and $d_0\Phi = 0$,
- b) the zeros of Φ are in bijection with a subset of $\mathcal{L}^{-1}(0)$, where $0 \in H^1(\mathfrak{h}, \mathfrak{h}^c)$ corresponds to $0 \in \mathcal{L}^{-1}(0)$.
- c) and in a neighborhood of 0 in $\mathfrak{h}^* \otimes \mathfrak{h}^c$, the union of the K -orbits containing elements $\eta : \mathfrak{h} \rightarrow \mathfrak{h}^c$ that corresponds to a zero of Φ , coincides with the space of all linear maps $\eta : \mathfrak{h} \rightarrow \mathfrak{h}^c$ for which the graph is a Lie subalgebra transverse to $\mathfrak{h}^c$.

We will derive the statements a), b) and c) claimed in this theorem from the corresponding assertions a), b), and c) of theorem 5.3.2, Kuranishi models. This time, we will use the insights we gained in the previous section to prove the conditions needed to apply theorem 5.3.2.

Proof. Step 1:

We consider the controlling DGLA $(C^\bullet(\mathfrak{h}, \mathfrak{h}^c), \delta_{\mathfrak{h}, \mathfrak{h}^c}, \llbracket \cdot, \cdot \rrbracket)$ obtained in proposition 3.2.25 and we set $M = \mathfrak{h}^* \otimes \mathfrak{h}^c$, $E = M \times (\wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c)$ and $F = M \times (\wedge^3 \mathfrak{h}^* \otimes \mathfrak{h}^c)$ to follow the strategy of the previous section. Similarly, we take the Lie group K integrating $(\mathfrak{h}^c, \mu|_{\mathfrak{h}^c})$ that is a subgroup of G , the simply connected Lie group integrating $\mathfrak{g}$. The Lie group K acts on $\mathfrak{h}^* \otimes \mathfrak{h}^c$ by gauge action and on $\wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c$ by adjoint action, as it is given in definition 4.4.3. We also take the section $\sigma : M \rightarrow E$ defined by the Maurer-Cartan equation

$$\sigma_2(\varphi) = \delta_{\mathfrak{h}, \mathfrak{h}^c} \varphi + \frac{1}{2} \llbracket \varphi, \varphi \rrbracket = \mathcal{L}(\varphi)$$

and the vector bundle map $\phi : M \rightarrow \text{Hom}(E, F)$ defined by

$$\phi_\varphi^{(2)} = \delta_{\mathfrak{h}, \mathfrak{h}^c} + \llbracket \varphi, \cdot \rrbracket.$$

Furthermore, we can use proposition 6.3.1 and see that K preserves the zero section $Z : \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow (\mathfrak{h}^* \otimes \mathfrak{h}^c) \times (\wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c)$, and that $\phi \circ \sigma = 0$. We can use proposition 6.3.2 to see that the section $\mathcal{L}$ is equivariant.

Step 2:

Now that we have checked all of the assumptions needed for theorem 5.3.2, Kuranishi models, we can apply it. We follow the constructions in the proof to get as much information as possible. We start by fixing a transversal τ to the orbit $K \cdot 0$ in the point 0, so that then

$$T_0 \tau \oplus T_0(K \cdot 0) = T_0 M. \quad (6.31)$$

As the vector bundles E and F are already trivial, we can write $E = M \times V$ and $F = M \times W$ where $V = \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c$ and $W = \wedge^3 \mathfrak{h}^* \otimes \mathfrak{h}^c$. Then we obtain maps

$$\sigma_2|_\tau : \tau \rightarrow V$$

and

$$\phi^{(2)}|_\tau \rightarrow \text{Hom}(V, W).$$

Choose a complement Q to $\text{Im}(\phi_0^{(2)})$ in W , so that

$$Q \oplus \text{Im}(\phi_0^{(2)}) = W.$$

In the proof of 5.3.2, we shrank the neighborhood $U_\tau \subseteq \tau$ of 0 until all the desired conditions were fulfilled. Therefore, let U_τ be the smallest such open neighborhood. Then, by the proof, we find a vector subbundle $\hat{E} \cong U_\tau \times \hat{V}$ of E where $\hat{V} \cong (\phi_0^{(2)})^{-1}(Q) = \ker(\phi_0^{(2)})$. By using a chart $\varphi : U_\tau \rightarrow \mathbb{R}^t$, we also obtain $\tilde{V}$ for which

$$\text{Im}(d_0 \sigma_2|_\tau) \oplus \tilde{V} = \hat{V}. \quad (6.32)$$

The theorem also yields a new map

$$\tilde{\sigma}_2 : S \rightarrow \tilde{V}$$

from a submanifold S of τ that contains 0, to $\tilde{V}$. For this map holds that

- $\tilde{\sigma}_2(0) = 0$,
- a) $d_0^{\text{vert}} \tilde{\sigma} = d_0 \tilde{\sigma}_2 = 0$,
- b) the zero set of $\tilde{\sigma}_2$ coincides with the zero set of $\sigma_2|_\tau = \mathcal{L}|_\tau$,
- c) and if ξ is a zero of $\mathcal{L}$ sufficiently close to 0, then $\xi \in K \cdot \eta$ for some zero η of $\tilde{\sigma}_2$.

We will construct the map Φ out of the section $\tilde{\sigma}$.

Step 3:

Now we will identify the domain and codomain of this section $\tilde{\sigma}$ to get a new map Φ . We start by identifying the codomain. By equation (6.32), we see that

$$\tilde{V} \cong \frac{\hat{V}}{\text{Im}(d_0(\sigma_2|_\tau))} \cong \frac{\ker(\phi_0^{(2)})}{\text{Im}(d_0^{\text{vert}}(\sigma|_\tau))}.$$

As $\sigma_2(K \cdot 0) = K \cdot \sigma_2(0) = 0$, we know that

$$d_0(\sigma_2|_{K \cdot 0}) = 0$$

which means that

$$T_0(K \cdot 0) \subseteq \ker(d_0^{\text{vert}}(\sigma)). \quad (6.33)$$

Using this and equation (6.31), we know that

$$\text{Im}(d_0^{\text{vert}}(\sigma|_\tau)) = \text{Im}(d_0^{\text{vert}}(\sigma)).$$

Substituting this information, together with the actual expressions for the maps σ and $\phi_0^{(2)}$, we get that

$$\tilde{V} \cong \frac{\ker(\phi_0^{(2)})}{\text{Im}(d_0^{\text{vert}}\sigma)} = \frac{\ker(\delta_0 : C^2(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^3(\mathfrak{h}, \mathfrak{h}^c))}{\text{Im}(d_0\mathcal{L})} \quad (6.34)$$

$$= \frac{\ker(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^2(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^3(\mathfrak{h}, \mathfrak{h}^c))}{\text{Im}(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c))} = H^2(\mathfrak{h}, \mathfrak{h}^c). \quad (6.35)$$

where we used the computation in (4.7) which shows that $d_0\mathcal{L} = \delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c)$. Denote the linear isomorphism between $\tilde{V}$ and $H^2(\mathfrak{h}, \mathfrak{h}^c)$ by

$$\alpha : \tilde{V} \rightarrow H^2(\mathfrak{h}, \mathfrak{h}^c)$$

that sends $0 \in \tilde{V}$ to $0 \in H^2(\mathfrak{h}, \mathfrak{h}^c)$.

Step 4:

Now we will identify the domain of $\tilde{\sigma}$. By equation (4.7), we know that

$$T_0S \cong \frac{\ker(d_0^{\text{vert}}(\sigma))}{\text{Im}(d_e m_0)} = \frac{\ker(d_0\mathcal{L})}{\text{Im}(d_e m_0)} = \frac{\ker(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c))}{\text{Im}(d_e m_0)}.$$

A computation as in equation (4.8) shows that

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} = -d_e m_0 : C^0(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^1(\mathfrak{h}, \mathfrak{h}^c). \quad (6.36)$$

We can use this, to further identify

$$\begin{aligned} T_0 S &\cong \frac{\ker(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c))}{\text{Im}(d_e m_0)} \\ &= \frac{\ker(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^1(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^2(\mathfrak{h}, \mathfrak{h}^c))}{\text{Im}(\delta_{\mathfrak{h}, \mathfrak{h}^c} : C^0(\mathfrak{h}, \mathfrak{h}^c) \rightarrow C^1(\mathfrak{h}, \mathfrak{h}^c))} = H^1(\mathfrak{h}, \mathfrak{h}^c). \end{aligned} \quad (6.37)$$

As S is a manifold, we can pick a chart

$$\varphi : (S, 0) \rightarrow (T_0 S, 0),$$

which is then locally a diffeomorphism, such that a neighborhood $U_S \subseteq S$ around 0 can be identified with a neighborhood of 0 in $T_0 S$. Together with the previous identification $T_0 S \cong H^1(\mathfrak{h}, \mathfrak{h}^c)$, we obtain a diffeomorphism

$$\beta : (U_S, 0) \rightarrow (U, 0)$$

for some neighborhood $U \subseteq H^1(\mathfrak{h}, \mathfrak{h}^c)$ of 0.

Now we can construct Φ as

$$\Phi := \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1} : U \rightarrow H^2(\mathfrak{h}, \mathfrak{h}^c). \quad (6.38)$$

Step 5:

Now that we have made all the constructions, we can start proving the statements a), b) and c). We start with the first one. First, notice that $\beta^{-1}(0) = 0$ and $\tilde{\sigma}_2(0) = 0$ as 0 is a zero of $\sigma_2 = \mathcal{L}$, so that

$$\begin{aligned} \Phi(0) &= \alpha \circ \tilde{\sigma}_2 \circ \beta^{-1}(0) \\ &= \alpha \circ \tilde{\sigma}_2(0) \\ &= \alpha(0) = 0 \end{aligned}$$

because α is a linear isomorphism. Secondly, as $d_0 \tilde{\sigma}_2 = 0$ by statement a) of theorem 5.3.2, and $\beta^{-1}(0) = 0$, we have that

$$d_0 \Phi = 0, \quad (6.39)$$

which proves statement a).

Step 6:

As for statement b), we can see that for $h \in U$

$$\begin{aligned} h \text{ is a zero of } \Phi &\iff \Phi(h) = 0 \\ &\iff \tilde{\sigma}_2 \circ \beta^{-1}(h) = 0 \\ &\iff \beta^{-1}(h) \text{ is a zero of } \tilde{\sigma}_2 \end{aligned}$$

Now by statement b) of 5.3.2, a zero of $\tilde{\sigma}_2$ is a zero $\sigma_2 = \mathcal{L}$. Therefore, for any zero h of Φ , we know that $\beta^{-1}(h)$ is a zero of $\mathcal{L}$, meaning that its graph is a Lie subalgebra of $\mathfrak{g}$ transverse to $\mathfrak{h}^c$. From this we conclude that zeros of Φ correspond to Lie subalgebras

transverse to $\mathfrak{h}^c$. Moreover, $\beta^{-1}(0) = 0$, so that the zero $h = 0$ of Φ corresponds to the initial Lie subalgebra $\mathfrak{h}$. This confirms statement b).

Step 7:

Lastly, by statement c) of theorem 5.3.2, we know that we can find an open neighborhood $O \subseteq \mathfrak{h}^* \otimes \mathfrak{h}^c$ of 0 such that

$$\mathcal{L}^{-1}(0) \cap O = \left(\bigcup_{\eta \in \tilde{\sigma}_2^{-1}(0)} K \cdot \eta \right) \cap O = \left(\bigcup_{h \in \Phi^{-1}(0)} K \cdot \beta^{-1}(h) \right) \cap O. \quad (6.40)$$

In other words, in a neighborhood of 0 in $\mathfrak{h}^* \otimes \mathfrak{h}^c$, the union of adjoint K -orbits that contain linear maps $\mathfrak{h} \rightarrow \mathfrak{h}^c$ whose graph is a Lie subalgebra transverse to $\mathfrak{h}^c$ that correspond to a zero of Φ , coincides with the space of all linear maps $\mathfrak{h} \rightarrow \mathfrak{h}^c$ whose graph is a Lie subalgebra transverse to $\mathfrak{h}^c$. This is exactly the statement we wanted in c). $\square$

Remark 6.4.2. By using theorem 6.4.1, we can now solve the first bullet of Problem 1 c), Rigidity of Lie subalgebras. By letting $H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$, the domain of Φ in theorem 6.4.1 reduces to 0. Under the correspondence between zeros of Φ and linear maps $\mathfrak{h} \rightarrow \mathfrak{h}^c$ whose graph is a Lie subalgebra transverse to $\mathfrak{h}^c$, the element $0 \in U$ corresponds to the 0 map. Hence, by making use of statement c), then we get that in a neighborhood $O \subseteq \mathfrak{h}^* \otimes \mathfrak{h}^c$, the set of Lie subalgebras of $\mathfrak{g}$ transverse to $\mathfrak{h}^c$ coincides with the adjoint K -orbit through 0. In other words, locally holds that

$$\mathcal{L}^{-1}(0) = K \cdot 0.$$

By taking the graph of both sides, this means exactly that locally, all Lie subalgebras $\mathfrak{h}'$ that are transverse to $\mathfrak{h}^c$ are of the form

$$\mathfrak{h}' = \text{Ad}_l(\mathfrak{h}) \text{ for some } l \in K.$$

Therefore, we obtain the following statement.

Let $(\mathfrak{g}, \mu)$ be a Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ a k -dimensional Lie subalgebra that admits a complement $\mathfrak{h}^c$ that is a Lie subalgebra too. If $H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$, then in a neighborhood of $\mathfrak{h}$ in $\text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$, all Lie subalgebras $\mathfrak{h}'$ are of the form

$$\mathfrak{h}' = \text{Ad}_l(\mathfrak{h}) \text{ for some } l \in K.$$

Once again, we see that, with the same condition, we're able to solve the second bullet of problem 1 c).

Theorem 6.4.3. Let $(\mathfrak{g}, \mu)$ be a Lie algebra and suppose it has a Lie subalgebra $\mathfrak{h}$ that admits a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. If $H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$, then we can find an open neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{h}^c$ of 0 and a smooth map $h : U \rightarrow K$ such that $\mathfrak{h}' = \text{Ad}_{h(\xi)}(\mathfrak{h})$ for all Lie subalgebras $\mathfrak{h}' = \Gamma(\xi) \in \text{Tr}_k(\mathfrak{g}, \mathfrak{h}^c)$ for $\xi \in U$.

Proof. We will apply theorem 5.1.11, Openness of orbits. Before we can apply this theorem, it remains to verify the hypotheses. Suppose $H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$, then this means that

$$C^0(\mathfrak{h}, \mathfrak{h}^c) \xrightarrow{\delta_{\mathfrak{h}, \mathfrak{h}^c}} C^1(\mathfrak{h}, \mathfrak{h}^c) \xrightarrow{\delta_{\mathfrak{h}, \mathfrak{h}^c}} C^2(\mathfrak{h}, \mathfrak{h}^c)$$

is an exact sequence. From equation (6.36), we know that

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} = d_e m_0 : \mathfrak{h}^c \rightarrow \mathfrak{h}^* \otimes \mathfrak{h}^c$$

and from equation (4.7) we also know that

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} = d_0 \mathcal{L} = d_0^{\text{vert}} \sigma : \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c.$$

Using this information, we see that the former exact sequence is equivalent to

$$\mathfrak{h}^c \xrightarrow{d_e m_0} \mathfrak{h}^* \otimes \mathfrak{h}^c \xrightarrow{d_0^{\text{vert}} \sigma} \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c$$

being exact. In this sequence, $T_0 M \cong M = \mathfrak{h}^* \otimes \mathfrak{h}^c$ as M is a vector space, and $\wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c = E_0$, the fiber of E at 0. Therefore, we can see that the condition that $H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$ is equivalent to the condition that 0 is a non-degenerate zero of σ . As we know that $\mathcal{L}$ is equivariant by proposition 6.3.2, we can apply theorem 5.1.11 and find a neighborhood $U \subseteq \mathfrak{h}^* \otimes \mathfrak{h}^c$ of 0 and a map $h : U \rightarrow K$ so that for all zeros $\xi \in U$ of $\mathcal{L}$, we have that

$$h(\xi) \cdot 0 = \xi$$

or equivalently

$$\text{Ad}_{h(\xi)}(\mathfrak{h}) = \Gamma(\xi)$$

for all $\xi \in \mathfrak{h}^* \otimes \mathfrak{h}^c$. As the graphs of zeros of $\mathcal{L}$ are exactly Lie subalgebras transverse to $\mathfrak{h}^c$, we have found the desired U and h . $\square$

Lastly, the following theorem gives an answer on problem 2 c).

Theorem 6.4.4. *Let $(\mathfrak{g}, \mu)$ be a Lie algebra and suppose it has a Lie subalgebra $\mathfrak{h}$ that has a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. If $H^2(\mathfrak{h}, \mathfrak{h}^c) = 0$, then the space of Lie subalgebras transverse to $\mathfrak{h}^c$ is a manifold in a neighborhood of $\mathfrak{h}$, which is of dimension $\dim(Z^1(\mathfrak{h}, \mathfrak{h}^c))$.*

Proof. We will apply theorem 5.2.1, Stability of zeros. Before we can apply this theorem, we have to check whether the assumptions are fulfilled. The condition that $H^2(\mathfrak{h}, \mathfrak{h}^c) = 0$ means precisely that

$$C^1(\mathfrak{h}, \mathfrak{h}^c) \xrightarrow{\delta_{\mathfrak{h}, \mathfrak{h}^c}} C^2(\mathfrak{h}, \mathfrak{h}^c) \xrightarrow{\delta_{\mathfrak{h}, \mathfrak{h}^c}} C^3(\mathfrak{h}, \mathfrak{h}^c)$$

is an exact sequence. We know from equation (4.7) that

$$\delta_{\mathfrak{h}, \mathfrak{h}^c} = d_0 \mathcal{L} : \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c,$$

and by definition $\delta_{\mathfrak{h}, \mathfrak{h}^c} = \phi_0^{(2)} : \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c \rightarrow \wedge^3 \mathfrak{h}^* \otimes \mathfrak{h}^c$. By plugging in the former maps in the exact sequence, together with the explicit expressions for the cochains, we get that the sequence

$$\mathfrak{h}^* \otimes \mathfrak{h}^c \xrightarrow{d_0 \mathcal{L}} \wedge^2 \mathfrak{h}^* \otimes \mathfrak{h}^c \xrightarrow{\phi_0^{(2)}} \wedge^3 \mathfrak{h}^* \otimes \mathfrak{h}^c$$

is exact. Here, we made identifications $T_0 M \cong \mathfrak{h}^* \otimes \mathfrak{h}^c$ and $F_0 = \wedge^3 \mathfrak{h}^* \otimes \mathfrak{h}^c$. By the exactness of this sequence, and as we already know from proposition (6.3.1) that the condition $\phi \circ \sigma = 0$ is fulfilled, we are able to use theorem 5.2.1. Statement 1 of theorem 5.2.1 tells us that $\mathcal{L}^{-1}(0)$ is a manifold around 0 of dimension $\dim(Z^1(\mathfrak{h}, \mathfrak{h}^c))$. $\square$

Example 6.4.5. Suppose we have a Lie algebra $(\mathfrak{g}, \mu)$ and a Lie subalgebra $\mathfrak{h}$ that has a complement $\mathfrak{h}^c$ that is also a Lie subalgebra. Additionally, suppose that $(\mathfrak{h}, \mu|_{\mathfrak{h}})$ is a semisimple Lie algebra. Then we know by Whitehead's lemma 2.4.8 that

$$H^1(\mathfrak{h}, \mathfrak{h}^c) = 0$$

so that we can apply theorems 6.4.1 and 6.4.3 to find that $\mathfrak{h}$ is rigid.

Conclusion

The aim of this thesis was to develop, in a single and self-contained framework, the deformation theory of three Lie algebraic objects: Lie algebras, Lie algebra homomorphisms, and Lie subalgebras. Following the viewpoint of Crainic, Schätz and Struchiner [CSS14] and building on the foundational work of Nijenhuis and Richardson, we treated each object in a uniform way. We defined what it means to deform a given structure and when two deformations should be regarded as equivalent, we showed that infinitesimal deformations define elements in a first cohomology group and that equivalent deformations give rise to the same class. Additionally, we introduced in each case a Kuranishi map which detects obstructions to integration. These cohomological descriptions were then the basis for the rigidity, local smoothness, and stability results of chapter 6.

The recurring structure across the three cases is not a coincidence. In chapter 3 and section 4.4 we made explicit the DGLA controlling each deformation problem, in the sense that the objects of interest arise as its Maurer-Cartan elements and the cohomology governing their deformations is the cohomology of this controlling DGLA twisted at the corresponding element. From this point of view, the three concrete theories of sections 4.1, 4.2 and 4.3 become instances of one abstract deformation theory.

For Lie algebras and Lie algebra homomorphisms the results of this thesis are not new. Our contribution there is providing in full the proofs and intermediate steps that the original sources often leave implicit, and organizing the material so that the underlying DGLA theory is clear. For Lie subalgebras we departed from [CSS14] by working under the assumption that the subalgebra admits a complement that is also a Lie subalgebra, to avoid L_∞ -algebra theory. Under this assumption we developed the deformation theory in analogy with the other two cases: we derived the controlling DGLA and obtained the corresponding rigidity and smoothness statements. These results follow from the general theory of Lie subalgebra deformations after the identification $\mathfrak{g}/\mathfrak{h} \cong \mathfrak{h}^c$, but the way of proving these results, has not been presented in this form elsewhere.

Several questions are left open and suggest directions for further work. In chapter 5 we established three geometric theorems, which in chapter 6 we applied to the three deformation problems to obtain rigidity, local smoothness and stability results. In section 6.3, we checked that the assumptions needed for the theorems in chapter 5 are fulfilled for any controlling DGLA. One could actually apply these theorems to an

abstract controlling DGLA. This would yield rigidity and smoothness statements at the level of an arbitrary controlling DGLA, of which our results for Lie algebras, Lie algebra homomorphisms and Lie subalgebras would then be immediate special cases. A second direction is to apply the same analysis as we did to Lie algebras, Lie algebra homomorphisms and Lie subalgebras to other algebraic and geometric structures. Lastly, one could try to generalize the whole framework from differential graded Lie algebras to L_∞ -algebras.

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